

MES RET 100 - Smart Manufacturing Delivery Package/ Procedures

Tuesday, January 9, 2024 9:02 AM

The Smart Manufacturing Delivery RET (RoviSys Execution Template) is our project delivery flow based on our Dx procedures. This template outlines how each of the procedures flow together to define how we execute a comprehensive project within our "Agile for Manufacturing" mindset. This mindset enables the continuous refinement of project scope throughout the lifecycle of a digitalization initiative. It bridges the gaps between purely agile and purely waterfall execution methodologies, addressing key challenges of both approaches.

- **Challenges with pure agile execution:**
 - Lack of overall project goals and clear scope boundaries
 - Absence of a comprehensive roadmap, leading to an unclear definition of completion
 - Increased risk of cost overruns and unmet customer expectations
- **Challenges with pure waterfall execution:**
 - Rigid scope boundaries that limit flexibility and course correction
 - Heavy reliance on upfront functional and design requirements, which can be impractical
 - Risk of weak or incomplete design deliverables that may not be fully understood by the customer at the time of review

By leveraging the strengths of both methodologies with that "Agile for Manufacturing" mindset, our Smart Manufacturing delivery flow provides a more adaptable, goal-oriented approach that drives project success and enhances customer satisfaction.

The RoviSys Smart Manufacturing Delivery brings together the key components from our suite of Dx procedures, all designed to drive a successful digitalization initiative. The package is organized into three core phases: Assessment, Execution, and Deployment. Throughout each phase, we continuously refine the scope - transforming high-level objectives into detailed, actionable technical tasks. Below is a description of each phase:



1. Assessment

The Assessment Package is built from key features of two productized offerings: **DxDiscovery** and **DxArchitecture**. This phase is designed as a short, lightweight engagement that:

- Establishes the current process state
- Defines the rationale behind the digitalization initiative
- Identifies how digitalization will enhance operations
- Develops a strategy to quantify costs and measure success

Additionally, the assessment phase outlines the digitalization tools (platforms, technologies, etc.) to be used and provides an overall deployment roadmap.

This phase often serves as an entry point for RoviSys when engaging with customers. While it may be the first step in a larger digitalization initiative, it is difficult - if not impossible - to fully estimate the project's scope until the assessment is complete. However, if the customer has already conducted initial groundwork, that progress can be leveraged to streamline the assessment process.

Please refer to the [Assessment RET](#) for additional details on executing this step.

2. Project Execution

The DxExecution productized offering converts business process scenarios into specific technical tasks, which are then implemented and deployed using the selected technology stack.

During this phase:

- User stories are created based on business process scenarios.
- Technical development tasks are defined and built according to these user stories.
- Frequent customer review cycles ensure that deliverables align with expectations.
- Business process alignment guarantees that every technical feature has a clear purpose and measurable ROI.

Additionally, a detailed design specification is generated as part of each sprint, ensuring an "as-built" technical document is available for future development and technical support teams.

Please refer to the [DxExecution RAP](#) for additional details on executing this step.

3. Project Deployment

The DxDeployment productized offering ensures full user adoption across the site(s) while promoting customer ownership and long-term system maintenance.

Key aspects of this phase include:

- Detailed deployment planning to coordinate activities and personnel efficiently.
- Risk minimization strategies to prevent downtime and product quality issues.
- User training and adoption to ensure seamless integration into daily operations.

Please refer to the [DxDeployment RAP](#) for additional details on executing this step.

MES RET 102 - OEE Implementation

Friday, February 14, 2025 3:53 PM

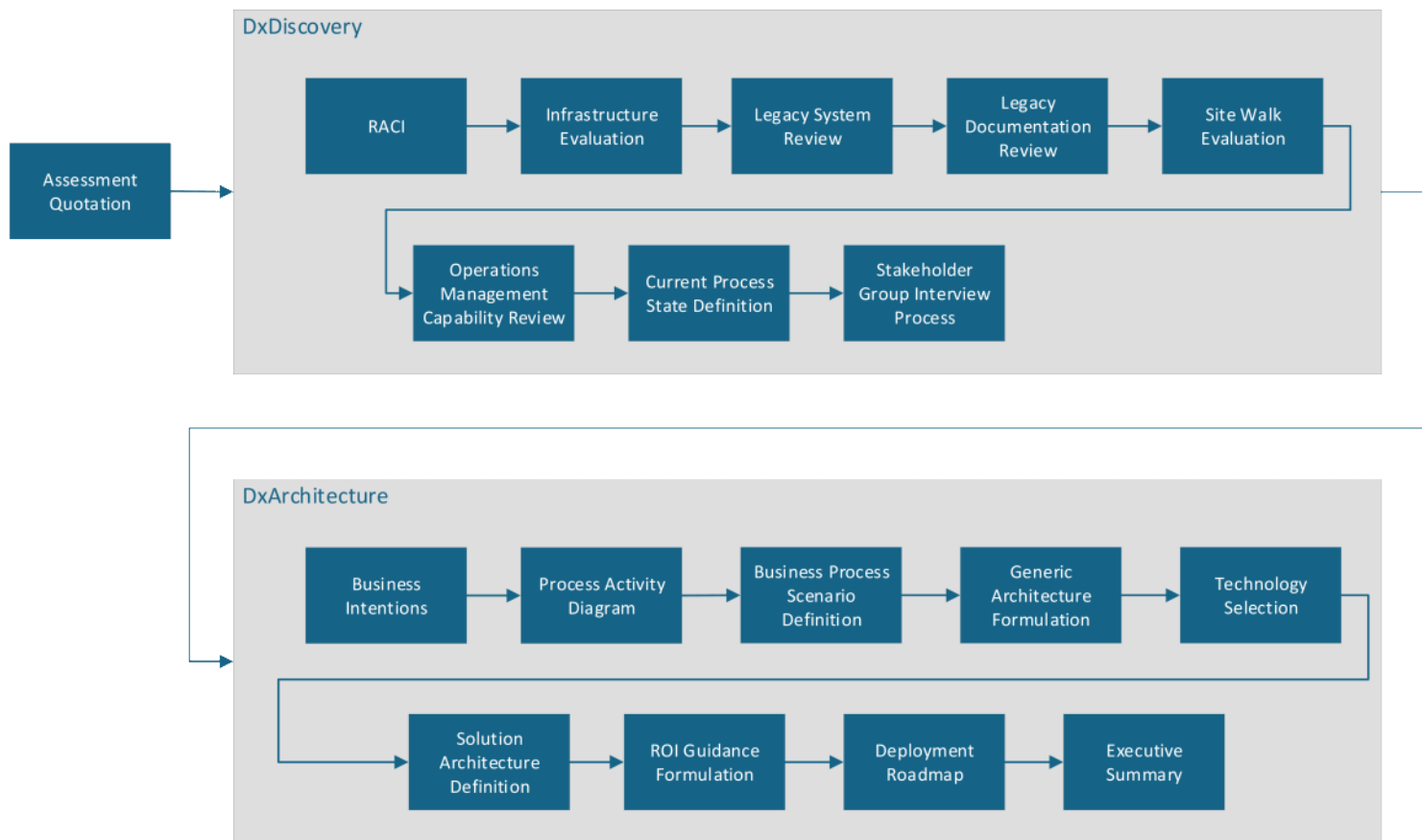
Define the sequence, how to scale the RAPs to deliver.
Can be applied to the proposal effort.

MES RET 101 - Assessment Package Flow

Tuesday, January 9, 2024 9:02 AM

The assessment is meant to be a relatively lightweight engagement that establishes the current process state, the rationale behind the digitalization initiative, the manner in which digitalization will be leveraged to provide positive impact to operations, and a strategy for quantifying the success of the initiative. Additionally, the assessment identifies the digitalization tools (platforms, technologies, etc.) that will be leveraged, as well as an overall deployment roadmap for these technologies. The assessment commonly serves as an 'entry point' for RoviSys when working with a customer who is interested in an MES. It is difficult if not impossible to accurately estimate the overall scope of the initiative until the assessment has been completed. However, it is possible that the customer has already completed initial work that can be leveraged to shorten the assessment effort.

The following chart details the sequence of events (features) typically executed as part of a holistic assessment effort. Note that some features irrelevant to a holistic digitalization effort have been omitted. These features may be required if the DxDiscovery product is being executed by itself, and/or in the case of a more control systems / automation-centric assessment.



MES RIS 100 - Assessment Package Quotation

Tuesday, March 12, 2024 8:18 AM

A proposal must be formulated for the customer that provides the execution strategy for the assessment effort, as well as a cost estimate. RoviSys may be inclined to perform the assessment as a cost of sales activity. However, assessment execution is not only an income stream for RoviSys, but also shows commitment on the part of the customer. The customer is more likely to commit time and resources to the effort if their money is on the line.

Key aspects of interest to the customer that should be detailed within the assessment proposal include the following:

- RoviSys background, why are we well suited to perform the assessment? This includes a focus on our process overview, digitalization delivery centers of excellence (CoEs), and example success stories
- What are the goals of the assessment? For example, the customer may be looking to replace legacy systems, improve efficiency/quality/etc., support a greenfield plant expansion, reduce labor costs, or a host of other 'levers' that can be pulled via usage of digitalization
- Schedule - what activities are envisioned and what is the general duration of each. A sample schedule is provided in the assessment proposal template provided later in this MES RAP.
- Deliverables - what is provided to the customer as collateral at the conclusion of the assessment effort?
- RoviSys team - what roles from RoviSys will be supporting the assessment effort?
- Customer time considerations - what information will be needed from the customer? Who is expected to participate on behalf of the customer and what time commitments will be required of these individuals?
- Pricing - What will the assessment cost, including services as well as travel & living expenses
- Assumptions - Clarifications concerning assumptions for the assessment effort

Typical deliverables provided to the customer as part of the assessment effort include the following:

- Any stakeholder interview recordings captured, as well as sanitized stakeholder interview notes captured by RoviSys
- Process background description or corresponding 'current state' SIPOC register
- Legacy system review notes and legacy application register (as applicable)
- Business intention listing
- 'To-be' business process scenarios (SIPOC register)
- Generic architecture
- Technology option background
- Completed technology option comparison matrix
- Technology-specific architecture
- Formulated roadmap
- Execution summary

Typical assumptions provided within the assessment proposal include the following

- No software development or configuration will be provided as part of the assessment proposal
- The customer will make available all required stakeholders needed to support the assessment effort based upon the timing required by the effort
- Any background documentation possessed by the customer to support the assessment effort will be provided

- Assumptions concerning the site(s) at which assessment activities will take place, as well as activities envisioned to require an onsite presence vs. activities that will be completed remotely within RoviSys offices.
- Expectations concerning when RoviSys can start the effort after project award

The assessment is meant to be a lightweight effort, so the schedule impact of the assessment should be minimized where possible. However, adequate time must be allocated to perform the assessment activities, as well as to support customer review. An example of a common assessment schedule to formulate a holistic MES roadmap for between one and three similar sites is as follows:

Time	Activity
Week 0	Project Award, Kickoff, and Planning Activities
Week 1	Onsite Workshop
Week 2	Business Intentions and Process Scenario Definition
Week 3	Business Intentions and Process Scenario Definition
Week 4	Technology Assessment and Roadmap
Week 5	Technology Assessment and Roadmap
Week 6	Deliverable Review
Week 7	Executive Summary and Wrap Up

As a rough estimate, allow for the creation of one business process scenario per hour. This allows the duration of time required for this activity to be roughly estimated based on the number of facilities and the complexity of the business processes. A someone simple, single site process may only require one week, while multiple business units across multiple sites may require four weeks or more.

Due to the nature of the assessment effort, tight review cycles are required and the customer should be limited in the number of reviews permitted.

Party	Activity	Typical Permitted Duration
RoviSys	Document Draft	Per deliverable estimates
Customer	Deliverable Review – Cycle 1	2 days
RoviSys	Review, Address and Incorporate comments – Submit for Cycle 2	2 days
Customer	Deliverable Review – Cycle 2	2 days
RoviSys	Review, Address and Incorporate comments – Submit for Approval	1 days
Customer	Review of mitigation and deliverable revisions based on existing comments/feedback	1 days

Resourcing:

The assessment effort will require a varied resource mix based upon the scope and effort. Key RoviSys resources required for the assessment include the following:

- Project Manager (Works with customer team on scheduling and other logistics, facilitates periodic status meetings, invoices for the effort)
- Lead Functional Consultant (facilitates stakeholder interviews, leads business process scenario documentation effort, leads ROI analysis strategy formulation, leads creation of execution

- roadmap)
- Functional Analyst Assist (helps take notes during interview workshops, aids in the creation of documentation)
- Solution Architect (generates technical architectures, identifies shortlisted technology platform options, performs technology comparison, provides advisement during execution roadmap formulation)

Assessment Budgetary Formulation:

Formulation of the assessment quotation is based upon application of the envisioned project team against the formulated schedule. The lead functional consultant will be heavily involved (80%+) across the entire initiative. The functional analyst assist will be heavily involved during the onsite workshops, and often around 50% involved throughout the remainder of the engagement. The PM likely will be involved at between 10 and 20%, although involvement could be higher if the PM participates in project workshop activities and travels with the team. Higher PM involvement will be required in the initial week to aid in planning activities. The solution architect will be most heavily involved during the technology assessment schedule periods.

Typically, an assessment aligning to the above schedule falls around \$75,000 (based upon 2024 standard rates).

Assessment Discount Strategy:

There are strategies RoviSys can suggest if a customer is unwilling to pay for the assessment effort. RoviSys can suggest that a discount equal to some percentage (such as 50%) of the assessment effort will be applied to the implementation if the customer chooses RoviSys for the implementation scope. One time discounts can also be applied to give the customer a more favorable rate.

Please confirm with your GM or applicable vertical/horizontal director that such discount strategies are warranted before proceeding with such strategies.

We should also point out the deliverables that will be provided to the customer after the assessment as a value statement.

Example Assessment Proposal:

An example assessment proposal in PowerPoint format is provided at the following link:

[P24XXX01a - MES Assessment and Roadmap.pptx](#)

MES RAP 101 - DxDiscovery

Tuesday, January 9, 2024 9:02 AM

The ultimate goal of the DxDiscovery product offering is to gather information that will shape the remainder of the engagement. This includes a recommendation concerning the path forward, which may include execution of the DxArchitecture and/or DxLean productized services offerings.

The discovery effort does not provide a detailed roadmap, design for the solution, or allow for the build and deployment of any software component. These expectations must be managed with the customer.

The following is a breakdown of the general activities conducted as part of the DxDiscovery effort. The execution of each process is further described in corresponding RIS documents.

Discovery Quotation:

The customer must be given a proposal for the discovery effort. This proposal should clarify the duration of the effort, execution strategy, required meetings, deliverables, and associated costs.

RACI Formulation:

RACI formulation aids in the definition of the various participants in the effort, and their responsibilities, both from RoviSys and customer perspectives. This includes not just the project team itself, but also various stakeholders who will need to participate in current process state explanations, interviews, and deliverable review.

Infrastructure Evaluation:

An evaluation of the infrastructure in place across the enterprise is often required to evaluate its ability to support a digitalization initiative, identifying aspects that may need to be bolstered as part of the effort. Infrastructure includes both process control and enterprise plan networks, cloud connectivity, provisioned cloud infrastructure, integration layers and associated integration applications in place (such as OPC servers, MQTT brokers, etc.), cybersecurity resilience, and data analytics layers (data warehouses, data lakes, etc.).

Legacy System Review:

Often a manufacturing enterprise is already leveraging legacy systems / applications as part of their existing digitalization footprint. It is important that these legacy applications are properly understood, including the role of each, functionality provided, and any issues - including both useability and maintenance concerns. This will help to ensure no key piece of legacy functionality is overlooked and that ultimately a holistic architecture is created detailing which applications will be replaced and which will be integrated into the future solution.

Site Walk Evaluation:

A site walkdown includes a tour of the facility, showing the RoviSys team how product flows, how individuals interact with the process, and the environment in which individuals perform their daily work. This provides a first-hand view of the manufacturing process, including identification of physical and logistical constraints and opportunities that can be factored into the recommended path forward. A site walk also provides an opportunity to ask questions directly to shop-floor personnel that can give valuable perspective not able to be obtained from other customer stakeholders. This activity enables consideration of the sizing of the facility into future quotation efforts. Potential interaction points and

associated devices, such as tablets, terminals, scanners, readers, etc. can be envisioned and the practicality of leveraging such devices can be determined as well.

Panel Walk Down:

This feature consists of inventorying, assessing, and inspecting the control system panels within the facility to determine their location, serviced machinery, and hardware age, as well as to evaluate the layout, organization, and clarity of wiring and component identification in each cabinet. This enables RoviSys to recommend improvements to the control system infrastructure, including hardware upgrades or improved organization. Such improvements can be properly phased, leading to minimal production impact.

Legacy Documentation Review:

A review of any legacy documentation possessed by the enterprise, including legacy system design documents, user guides, training guides, process flow diagrams, etc. can help give RoviSys a better understanding of the legacy systems and processes in place within the facility. This allows RoviSys to ask more thoughtful, relevant questions during other DxDiscovery activities such as site walks and stakeholder interviews rather than spend time asking high-level questions simply to obtain sufficient background.

Operations Management Capability Review:

In many cases, a customer may be unfamiliar with the capabilities provided by an MES or MOM application, and there may be confusion as to the scope of the initiative as a result. Even customers familiar with the digitalization of operations management functions may lack a deep understanding of capabilities provided by the different layers in the ISA-95 model, considering such systems as ERP, MES/MOM, SCADA, etc. During this activity, RoviSys provides an overview of the capabilities provided as part of a digitalized ISA-95 compliant solution stack, focusing particularly on Level 3 capabilities most often provided by Layer 3 applications.

Current Process State Definition:

This effort helps to baseline the current state of the facility or facilities under the scope of the assessment. This exercise gives RoviSys background concerning current shop floor activities and system-level processes that will help to frame the rest of the assessment effort.

Stakeholder Group Interview Process:

During this activity, RoviSys learns more from key personnel in the enterprise about the processes and activities observed during current state process definition. Probing questions are asked that allows RoviSys to draw conclusions concerning what processes work well and what processes could be enhanced by a greater degree of digitalization.

Perceived Benefit: (Next Steps Value Identification)

During this activity, the perceived benefit of pursuing a next step is documented and a justification formulated. While a formal ROI analysis is not typically conducted at this stage, distillment of the benefits of moving forward will aid the customer in justifying a larger initiative and provide the customer a more specific focus for constrained budgets and resources. For example, RoviSys may recommend a DxArchitecture engagement, DxLean engagement, pilot DxExecution project, or a combination of all these productized services offerings.

Executive Summary:

The executive summary presentation distills the activities, key findings, and recommendations into a format that can be digested by broader customer stakeholders including an executive leadership team or steering committee. It should present a compelling plan that highlights the benefits of moving forward with a deeper effort and keys to success.

(Published) MES RIS 101.1 - DxDiscovery Quotation

Tuesday, September 3, 2024 6:53 AM

Please reference:

[DxDiscovery-Quotation.aspx](#)

A proposal must be formulated for the customer that provides the execution strategy for the DxDiscovery effort, as well as a cost estimate. Often, discovery is done as the initial phase in an assessment, requiring the assessment package to be quoted in its entirety, but a customer needing high-level direction with minimal cost impact may choose to execute a DxDiscovery effort in a standalone manner.

RoviSys may be inclined to perform the discovery scope as a cost of sales activity. However, DxDiscovery execution is not only an income stream for RoviSys, but also shows commitment on the part of the customer. The customer is more likely to commit time and resources to the effort if their money is on the line.

Key aspects of interest to the customer that should be detailed within the DxDiscovery proposal include the following:

- RoviSys background, why are we well suited to perform the activity? This includes a focus on our process overview, digitalization delivery centers of excellence (CoEs), and example success stories
- What are the goals of the DxDiscovery effort? For example, the customer may be looking to replace legacy systems, improve efficiency/quality/etc., support a greenfield plant expansion, reduce labor costs, or a host of other 'levers' that can be pulled via usage of digitalization
- Schedule - what activities are envisioned and what is the general duration of each.
- Deliverables - what is provided to the customer as collateral at the conclusion of the DxDiscovery effort?
- RoviSys team - what roles from RoviSys will be supporting the DxDiscovery effort?
- Customer time considerations - what information will be needed from the customer? Who is expected to participate on behalf of the customer and what time commitments will be required of these individuals?
- Pricing - What will the DxDiscovery effort cost, including services as well as travel & living expenses
- Assumptions - Clarifications concerning assumptions for the DxDiscovery effort

Typical deliverables provided to the customer as part of the DxDiscovery effort include the following:

- Any stakeholder interview recordings captured, as well as sanitized stakeholder interview notes captured by RoviSys
- Process background description or corresponding 'current state' SIPOC register
- Legacy system review notes and legacy application register (as applicable)
- Perceived benefit of moving forward with a broader effort, and the direction in which RoviSys feels the customer should move forward based upon our findings.
- Execution summary

Typical assumptions provided within the DxDiscovery proposal include the following

- No software development or configuration will be provided as part of the scope

- The customer will make available all required stakeholders needed to support the DxDiscovery effort based upon the timing required by the effort
- Any background documentation possessed by the customer to support the DxDiscovery effort will be provided
- Assumptions concerning the site(s) at which the DxDiscovery will take place, as well as activities envisioned to require an onsite presence vs. activities that will be completed remotely within RoviSys offices.
- Expectations concerning when RoviSys can start the effort after project award

The DxDiscovery engagement is meant to be a lightweight effort, so the schedule impact of the initiative should be minimized where possible. However, adequate time must be allocated to perform the activities, as well as to support customer review. An example of a common DxDiscovery schedule to formulate a recommended path forward is as follows:

Time	Activity
Week 0	Project Award, Kickoff, and Planning Activities
Week 1	Onsite Workshop (one site - add additional weeks for additional remote sites) - Site tour - Legacy application review - Stakeholder interviews
Week 2	Notes Compilation Operations Management Capability Review
Week 3	Summarization of Findings and Recommendation

Due to the nature of the DxDiscovery effort, tight review cycles are required and the customer should be limited in the number of reviews permitted.

Party	Activity	Typical Permitted Duration
RoviSys	Document Draft	Per deliverable estimates
Customer	Deliverable Review – Cycle 1	2 days
RoviSys	Review, Address and Incorporate comments – Submit for Cycle 2	2 days
Customer	Deliverable Review – Cycle 2	2 days
RoviSys	Review, Address and Incorporate comments – Submit for Approval	1 days
Customer	Review of mitigation and deliverable revisions based on existing comments/feedback	1 days

Resourcing:

The DxDiscovery effort will require a varied resource mix based upon the scope and effort. Key RoviSys resources required for the effort include the following:

- Project Manager (Works with customer team on scheduling and other logistics, facilitates periodic status meetings, invoices for the effort)
- Lead Functional Consultant (facilitates stakeholder interviews and legacy application review)
- Functional Analyst Assist (helps take notes during interview workshops, aids in the creation of documentation)

DxDiscovery Budgetary Formulation:

Formulation of the DxDiscovery quotation is based upon application of the envisioned project team against the formulated schedule. The lead functional consultant will be heavily involved (80%+) across the entire initiative. The functional analyst assist will be heavily involved during the onsite workshops, and often around 50% involved throughout the remainder of the engagement. The PM likely will be involved at between 10 and 20%, although involvement could be higher if the PM participates in project workshop activities and travels with the team. Higher PM involvement will be required in the initial week to aid in planning activities.

Typically, an DxDiscovery effort aligning to the above schedule falls around \$15,000 - 20,000 (based upon 2024 standard rates).

DxDiscovery Discount Strategy:

There are strategies RoviSys can suggest if a customer is unwilling to pay for the DxDiscovery effort. RoviSys can suggest that a discount equal to some percentage (such as 50%) of the effort will be applied to the implementation if the customer chooses RoviSys for the implementation scope. One time discounts can also be applied to give the customer a more favorable rate.

Please confirm with your GM or applicable vertical/horizontal director that such discount strategies are warranted before proceeding with such strategies.

We should also point out the deliverables that will be provided to the customer after the DxDiscovery as a value statement.

Example Assessment Proposal:

An example assessment proposal in PowerPoint format is provided at the following link. This proposal can be condensed to focus on only the DxDiscovery aspects.

[P24XXX01a - MES Assessment and Roadmap.pptx](#)

Expectation Setting:

To avoid confusion at the end of the discovery effort, it should be made clear to the customer the goals of the discovery effort, benefits, and deliverables. Some key points to highlight include the following:

- No software development, configuration, or pilot deployment will be performed as part of the effort
- The goal of the effort is to make a high-level recommendation concerning a path forward. Further analysis will likely be required to hone in on a specific platform to be deployed and roadmap for software deployment. This can be provided via a RoviSys DxArchitecture engagement.
- Benefits to the customer in addition to a path forward recommendation include:
 - o Distillment of documentation
 - o Distillment of feedback from operations concerning challenges, pain points, and opportunities for improvement
 - o A manufacturing capability map aligned to the ISA-95 standard detailing relevant capabilities of important to the organization and a prioritization of capabilities of focus. This can be leveraged to ensure all stakeholders 'speak the same language' regarding capabilities of focus

(Published) MES RIS 101.2 - DxDiscovery RACI

Wednesday, February 7, 2024 12:55 PM

Please Reference:

[DxDiscovery RACI](#)

RACI (or RASCI) formulation aids in the definition of the various participants in the effort, and their responsibilities, both from RoviSys and customer perspectives. This includes not just the project team itself, but also various stakeholders who will need to participate in current process state explanations, interviews, and deliverable review.

Each DxDiscovery activity and deliverable should be broken out on the RACI template based on the envisioned schedule, the scope of the formulated discovery interviews, and the resulting deliverables.

General items to be captured within the RACI or RASCI and assigned include the following:

- Participation in regular status meetings
- Providing prerequisite discovery effort information, such as legacy documentation. Each prerequisite item desired should be broken out as a separate line item.
- Scheduling workshops and other meetings
- Execution of the site tour
- Each workshop meeting session
- Every DxDiscovery deliverable - these deliverables should be broken up into granular components as different stakeholders will have interest in different sections.
 - o Current process state
 - o Stakeholder group interview notes
 - o Perceived benefit of further digitalization initiative
 - o Executive summary
 - o Etc.
- Executive summary presentation participants

This example is based upon the following generic RACI template:

<https://rovi.sharepoint.com/:x:/r/sites/mes/Shared%20Documents/3.Strategy/PMO/Document%20Templates/Submitted/PMO%20Template%20010%20-%20Customer%20Name.Project%20No.PD%23%23%23.Responsibility%20Matrix%20-%20RACI.xlsx?d=wadcb74d3c9154f3d8fa454f2083d6c5c&csf=1&web=1&e=uyby2d>

The RASCI model specifies definition of stakeholders as follows:

- Responsible for ensuring completion of the activity. The responsible person is the one who does the work to complete the task or create the deliverable. Typically, a RoviSys resource is named as responsible for the completion of the various Assessment deliverables, although the customer may be responsible for providing prerequisite information.
- Accountable for reviewing and confirming the completion of the activity. The accountable party approves the deliverable provided as a result of a given assessment activity.
 - o Typically, a customer resource is named as the accountable party, ensuring the provided deliverable is accurate and soliciting feedback from other consulted parties. Rather than requiring the document be signed off by many reviewers, one accountable party helped to streamline review and approval cycles.
 - o In the case of stakeholder interview RACI items, the accountable party may be the representative providing input during the interview session, but may be required to solicit

- input from other consulted individuals who are not physically present in the session.
- Not every customer stakeholder may be required to review the assessment in its entirety. A production manager may desire to review business process scenarios specific to their area, but need not review the technical architecture envisioned for example.
- Supporting roles may be other RoviSys engineers or individuals assisting the responsible party in completing the task at hand in the case of deliverables provided by RoviSys (such as documentation). This may include other RoviSys engineers / functional architects.
- Consulted during information gathering or deliverable review. Consulted individuals may be additional customer reviewers or individuals from which information is captured by the accountable party.
- Informed throughout the activity, but not required for review or information gathering purposes. These individuals are typically customer roles who need to be kept abreast of the progress of the activity by being provided status updates, or being included in status update meetings. They might include heads or directors of affected customer teams and senior customer leadership.

The RoviSys templated assessment agenda and RACI/RASCI template will both contain generic stakeholder roles. Translating these generic roles into specific roles possessed by an organization is crucial for operational efficiency. Generic roles, while useful for defining broad responsibilities, may not align perfectly with the unique structure, capabilities, and expertise within an organization. By mapping these roles to specific positions, it is ensured that tasks are assigned to the most suitable individuals. This specificity enhances accountability, as it is clear who is responsible for what, and improves communication by aligning expectations with actual job functions within the organization. Moreover, it allows for more accurate resource planning and utilization, reduces redundancy, and helps in identifying any gaps in coverage.

As a result, RoviSys should discuss with the customer project manager and relevant project team members to ensure that the generic roles for which communication is requested is translated to the specific roles and responsibilities within the customer's organization while working to identify the key individuals for which involvement is requested, rather than simply pass the generic templates to the customer and ask for these to be filled in.

(Published) MES RIS 101.3 - Infrastructure Evaluation

Thursday, August 22, 2024 4:22 PM

Please Reference:

[Infrastructure Evaluation](#)

The degree of infrastructure evaluation will depend based upon the ultimate goal of the organization. An MES/MOM-centric effort requires a high-level review of the underlying shop floor infrastructure, considering factors of importance when selecting and deploying such an application. Factors to consider during such an evaluation include the following:

- General network architecture, including shop floor and enterprise network delineation
- Network coverage on the shop floor, including support for wired and wireless devices
- Ability to send/receive data from the shop floor, including existing OPC or MQTT pipelines
- Existence of a historian layer
- Ability to integrate the shop floor and enterprise network layers to cloud-hosted applications
- Existence of terminals / clients on the shop floor that may be reused

An infrastructure-specific discovery effort that would be commonly completed by the RoviSys BIIT team will require a deeper, more methodological analysis.

(Published) MES RIS 101.4 - Legacy System Review

Thursday, August 22, 2024 4:22 PM

Please Reference:

[Legacy System Review](#)

As part of the discovery effort, a review of any legacy systems leveraged as part of manufacturing will aid in an understanding of current plant processes, constraints, and opportunities for replacement or augmentation. Legacy system replacement is often a key aspect of the business case / ROI for a digitalization initiative, so it is important to perform a thorough review to be in a position to make recommendations concerning the path forward.

Prior to any workshop or sessions, an inventory of legacy applications should be created by working with key customer IT / OT subject matter experts. The following is a listing of general information to be captured as part of this inventory:

- Application Name
- Site
- Plant Area
- Age
- Vendor (or home-grown)
- General Function
- Criticality (mission critical to production, helpful to production, unnecessary to production, etc.)
- Primary User Interface (web client, thick client, mobile, hosted within Office application such as Access or Excel, etc.)
- Primary SME (who can assist with demos and other questions)
- Support Status (actively supported by vendor/creator, vendor/creator support available, no support available)
- External interfaces / interactions with other applications
- External interfaces / interactions with underlying automation layer
- Master data management / configuration considerations (how is the application configured, who maintains it, what processes exist concerning master/configuration data management?)

Any documentation available from existing applications should be obtained as part of legacy documentation review (MES RIS 101.7). This may include user manuals, design documentation, troubleshooting guides, and other resources. If possible, these documents should be qualified by the customer to confirm how up-to-date and accurate they are felt to be. This documentation should be reviewed prior to any interactive demo.

Any questions on unclear or poorly documented functionality can be captured from this documentation review and raised to the customer during the interactive demonstration.

RoviSys can then arrange a demonstration of each application with the primary SME or delegate. It will not be feasible during such a demonstration to document every single function, and this may not even be useful as the goal of a digitalization initiative is typically not to provide a one for one replacement of the legacy application stack. Rather, general application functionality and usability should be captured, including the following key aspects:

- How do users access and log into the application?
- Speed and responsiveness of the application, to baseline expectations of any new toolset
- Business logic / functions implemented within the application

- Data obtained from the application, including any reports, dashboards, and metrics
- Historical data maintained within the application
- Devices utilized to input data (touch screen, keyboard, scanner, etc.)
- Means of troubleshooting application issues (log files, alarm banners, etc.)
- Any antidotal feedback provided by the demonstrator. Particular bugs, challenges using the application, or worthwhile features that would be undesirable to lose.

The demonstrator may not be able to answer detailed technical questions concerning the application and its architecture. In these cases a technical subject matter expert with application experience may need to be consulted.

With the information gathered, this should put RoviSys in a position to provide an initial recommendation concerning future state - are there logical reasons to keep the application, integrate it into a broader landscape, or replace it with a better alternative.

(Published) MES RIS 101.5 - Legacy Documentation Review

Thursday, August 22, 2024 4:27 PM

Please Reference:

[Legacy Documentation Review](#)

A review of relevant documentation concerning the process, legacy systems, or paper leveraged as part of manufacturing can aid in an overall understanding of the current operational processes that will in turn allow RoviSys to ask more intelligent questions throughout the discovery effort and make more educated recommendations at the conclusion of the activity.

What to Request:

There are a variety of documents to be considered, detailed as follows:

- Plant layout, flow, and systems architecture:
 - o Plant layout diagrams
 - o Plant process flow diagrams
 - o System swim lane diagrams
 - o Network architecture/layout diagrams
 - o Legacy software system inventory listing
- Shop floor processes:
 - o Standard operating procedures
 - o Examples of paper forms / checklists leveraged on the floor
 - o Examples of paper or electronic schedules leveraged
- Legacy software system operation:
 - o Legacy application functional or design specifications
 - o User manuals
 - o Training documentation
- Product Traceability:
 - o Examples of paper travelers or other production-specific paperwork
- Reporting:
 - o Example paper reports
 - o Compiled Excel sheets
 - o Exports of digital reports

Documentation Qualification:

As documentation is provided, an attempt should be made by RoviSys to qualify the documentation with our contacts. Some documentation may be directionally correct but could be out-of-date. Where possible, identifying the date of the document will also aid in this qualification. Documents that are many years old may not be entirely accurate.

When to Request:

Requests for legacy documentation should be made prior to the discovery workshop, allowing sufficient time for the customer to compile such information and for RoviSys to perform a cursory review prior to interactive sessions.

Our early-stage project contacts may not have the ability to capture everything in the limited time allocated. Additional references will likely come up during the discovery workshop discussions, which may require post workshop review and follow-up.

Order of Review:

The condensed timeline of a DxDiscovery effort may not permit time for detailed review of everything provided. Therefore, typically the most digestible documents should be reviewed first, and lengthy detailed documentation digested as time permits. A general strategy is as follows:

- Review process flows, layouts, and architectures to become acquainted with the plant layout, process flow, and infrastructure constraints
- Review production paperwork and schedule examples, as these will likely lead to questions and clarifications that will need to be discussed during the workshop sessions
- Review reporting examples, as these are also likely to lead to questions and clarifications that will need to be discussed during the workshop sessions
- User manuals and training documents will provide insight as to application functionality and user work processes
- Lastly, functional and design documents may be reviewed as time permits, but these documents are likely to take the most time to digest and are most likely to contain inaccuracies as they are often not updated as changes and enhancements are made.

(Published) MES RIS 101.6 - Site Walk Evaluation

Thursday, August 22, 2024 4:23 PM

Please Reference:
[Site Walk Evaluation](#)

A plant tour / site walk is often conducted early during the assessment workshop. The plant tour helps give the assessment team perspective concerning the actual constraints and processes conducted on the shop floor. It is important to maximize the value of the site walk, as opportunities to spend time on the shop floor are typically few.

Safety Considerations:

First and foremost, it is important to conduct site walks in a safe manner. There are inherent risks with being on the factory floor in an unfamiliar environment. Some key considerations include the following:

- Verify personal protective equipment (PPE) requirements prior to arriving onsite. Safety shoes should be purchased prior and brought to site. A hard hat, safety vest, safety glasses, and hearing protection are other typical requirements. These can be brought to site but are usually made available by the customer. Some sites may require long-sleeve shirts or may provide protective sleeves to wear. Sites maintaining a certain sanitation level may require wearing gowns, hair nets, and beard nets.
- It is likely that jewelry (watches, rings, earrings, etc.) will need to be removed prior to the site walk.
- When walking the site, stay alert and close to the tour guide.
 - o Always yield to fork truck drivers and make eye contact / acknowledge the driver before crossing in the path of a fork truck
 - o Yield to automated guides vehicles
 - o Stay within designated pedestrian walkways
- Watch for light curtains and physical e-stop gates. Do not cross these, as it may put you at risk as well as shut down a line or piece of equipment.
- Avoid loose-fitting clothing and take care to ensure clothing or hair do not come in close contact with moving equipment.
- Do not touch equipment or products unless explicitly given permission by your tour guide. In-process product can be hot, sharp, or caustic. Touching may also contaminate the product.
- Watch for trip hazards and take care on stairs and overhead walkways.
- Note taking can be difficult while moving through the facility. If an important note needs to be taken, ask the tour guide to pause / stop while this is recorded.

Pre Site Walk:

- It is useful to go over the process in-person or virtually prior to the site walk to better understand the process steps, production flow, line and equipment configuration, quality-centric activities, and other factors. This information can be difficult to convey in full while on the tour and a review prior to the site walk can help give focus to specific activities of interest.
 - o Any process flow or plant layout documents that may be available can aid in this review. Ask if the plant has any such documents available.
- Ask that the site walk / plant tour be done sequentially based on the order of production rather than based on the physical plant layout or convenience. Conducting the site walk in a sequential manner will lessen confusion.

During the Site Walk:

- Be sure to ask or note questions. It can be difficult to speak / discuss while on the plant floor due to noise. Have a means of noting follow-up questions (such as a small notebook) and raise these in a quieter location or back in the office area if required.
- It is useful to talk to personnel on the floor, such as operators, quality technicians, maintenance workers, etc. about their role. Ask your tour guide if there are any opportunities to talk to these resources prior to the tour so that the tour guide can flag down individuals. Keep in mind that these individuals likely have time-sensitive work to attend to, so try to keep the conversation short (a few minutes).
- Tour guides and personnel will often focus the discussion on the manufacturing process itself, including technical details as to how equipment / machines operate. This can be useful, but more important than this is a discussion of 'people processes' and 'system processes, including the following:
 - o Who does what
 - o How do individuals know what to do next
 - o What electronic systems are leveraged on the line
 - o What common issues occur on the line and how are these issues resolved
- Ensure explicit permission has been granted to take pictures or even have cell phones on the floor. Some sites prohibit the presence of electronic devices on the floor and picture taking due to intellectual property concerns. In some cases, the tour guide or another member of the customer team may be able to take pictures on our behalf as instructed.
- Do not take product from the shop floor unless explicit permission is given, including product perceived as scrap/waste.

Site Walk Checklist:

The following are common things to consider during a site walk:

- Process overview - allowing the ability to relate previous and subsequent discussions to what you saw
 - o How does material flow and how is it moved between locations. How are raw materials staged, where do they come from, and who moves them?
- Scale, size of the operation
 - o A solution may or may not make sense based on the size, logistics involved in performing the operation
 - An operator cannot walk a half-mile back and forth to verify something 20 times a day for example
- Operational constraints that may guide the system user experience
 - o A gloved operator may not be able to interact easily with a keyboard for example
 - o Certain mobile devices may be impractical due to the shop floor environment
- A chance to observe potential issues on the shop floor firsthand and ask why these issues occur
 - o Why is that machine backed up or not running?
 - o What is that terminal used for?
 - o What is that operator doing right now?
 - o Where does that material go next?
- Current displays and communication means. These may all be able to be replaced or augmented by the new system.
 - o Whiteboards, pegboards
 - o Paperwork, work instructions
 - o Legacy systems/applications in use
 - o Overhead displays
- The manner in which material/WIP is identified

- Temporary labels, permanent labels, paper travelers, stickers, etc.
- These can be important when considering how a digital system may fit into or replace current identification processes
- How material is organized on the shop floor
 - Often, there are situational scrap, rework, or quarantine locations on the floor that will need to be tracked by the system. Why is material sitting in a given location and what is its status.

Post Site Walk:

- Ensure that time is allocated to the workshop schedule post site walk to ask follow-up questions based upon observations and notes in a quieter, conference room environment.
- The customer will likely ask for feedback concerning the process, including our thoughts on complexity, uniqueness, or specific concerns / constraints. Throughout the plant tour, be prepared for this question and ensure thoughtful feedback can be provided concerning ways in which digitalization may be able to add substantial value based on observations, or specific concerns that could impact the success of the initiative and will have to be accommodated as a result.
- Handwritten notes should be reviewed, organized, and transposed into an electronic format as soon after the site walk as possible, while thoughts are still fresh. Notes taken onsite are usually abridged and with poor handwriting due to the environment and pace.

(Published) MES RIS 101.7 - Panel Walk Down

Thursday, August 22, 2024 4:26 PM

Please Reference:

[Panel Walk Down](#)

(Published) MES RIS 101.8 - Operations Management Capability Review

Wednesday, February 7, 2024 12:56 PM

Please Reference:

[Operations Management Capability Review](#)

In many cases, a customer may be unfamiliar with the capabilities provided by an MES or MOM application, and there may be confusion as to the scope of the initiative as a result. Even customers familiar with the digitalization of operations management functions may lack a deep understanding of capabilities provided by the different layers in the ISA-95 model, considering such systems as ERP, MES/MOM, SCADA, etc. During this activity, RoviSys provides an overview of the capabilities provided as part of a digitalized ISA-95 compliant solution stack, focusing particularly on Level 3 capabilities most often provided by Layer 3 applications.

To facilitate this exercise, a standard capabilities map has been defined, which can be found at the following link:

[RoviSys S95 capability example.pptx](#)

This can be reviewed with the customer to aid in understanding what capabilities are often found within the various levels. The focus of a digitalization assessment is largely on Level 3, although other RoviSys services as a product (SaaS) offerings may also cover levels 1 & 2.

Using the generic RoviSys capability map as a guideline, the following associated exercises can be conducted with the customer:

- The customer can be educated as to the core capabilities typically provided by a digitalization stack, specifically Level 3 operations management capabilities relevant to the assessment effort. Misconceptions as to what should be contained within the layers and associated systems such as SCADA, ERP, and MES can also be resolved during this review.
- The capabilities can be reviewed with the customer to ensure alignment of naming. The customer may require the capability in question, but refer to it by a different name. Therefore, it is important to discuss each capability with the customer rather than simply present the various terms, to eliminate any misunderstandings.
- Any legacy systems leveraged by the customer to provide each capability can be captured, allowing for the creation of a legacy application map that will help to guide the remainder of the assessment.
- RoviSys can work with the customer to prioritize certain capabilities of focus based upon this high-level information. For example, struggles with a legacy system or lack of an automated system to provide a certain capability may cause a certain capability to be prioritized over others.

Based on this discussion, a revised capability map can be provided specific to the customer, ensuring that relevant capabilities are shown, capabilities are named according to customer vernacular, legacy systems providing the capabilities are overlaid, and specific prioritizations are highlighted.

A customer unfamiliar with MES / MOM may benefit from viewing a demo of a typical MES / MOM solution. There is some risk in providing a demo to the customer that the demo environment will not look and feel like the customer's operation, causing confusion or loss of interest. As a result, RoviSys should strive to draw parallels to how the demonstrated functionality could be applied to the customer's enterprise, and should work with the demo presenter to make the demo as realistic and topical as possible.

(Published) MES RIS 101.9 - Current Process State Definition

Wednesday, February 7, 2024 12:56 PM

Please Reference:

[Current Process State Definition](#)

This effort helps to baseline the current state of the facility or facilities under the scope of the assessment. This exercise gives RoviSys background concerning current shop floor activities and system-level processes that will help to frame the rest of the assessment effort, including specific topics to discuss and questions to ask during stakeholder interviews.

The following are typical means by which RoviSys gathers current process state information:

Documentation Review:

The customer may possess legacy documentation that can be reviewed in an offline manner. This includes user manuals, SOPs, piping and instrumentation diagrams (P&IDs), swimlane diagrams, process flows, etc. Early in the assessment initiative RoviSys should formulate an ask for such documentation. However, care must be taken not to spend too much time reviewing legacy documentation that may be incorrect or may be cumbersome / confusing when a more direct means of obtaining the same information (site tours, stakeholder reviews, etc.) may be more efficient.

Conference Room Overview:

During this session, the customer walks RoviSys through the manufacturing process, including key user and/or system touchpoints. It can be advantageous to discuss how the site / enterprise operates in a conference room setting prior to the site tour as it is easier to communicate and ask questions in a meeting setting than on the floor with noise and other distractions. This discussion provides a baseline for the site tour that allows more useful details to be gathered during the tour.

Site Tour:

The primary goal of a site tour is not to understand the customer's manufacturing and supporting business processes, as this information can be more easily obtained offline. Rather, the site tour helps to establish perspective that may weigh into the subsequent business intentions and business process scenarios. Some examples include the following:

- The speed at which production moves on the shop floor
- Locations at which manual operations occur or areas where key data may be captured in an automated manner
- The size and scale of the facility, which may necessitate or eliminate certain solution options (such as mobile devices, vision systems, etc.)
- The scope of an operator's tasks
- Opportunities for overhead visualizations, stack lights, paging, and other notification conduits
- Locations where whiteboards or bulletin boards may exist to capture or convey information that could be replaced with a digital system
- Usage of paper work instructions
- Methods of identifying raw materials, WIP product, and finished goods (labels, barcodes, etc.)

The site tour can also help to highlight concerns that can be discussed during the stakeholder interview process, including the following examples:

- Large buildups of staged WIP product between stations
- Lines / machines not running
- Alarms
- Large amounts of scrap or suspect product
- Paper-based operations (tickets, travelers, labels, etc.)
- Inefficient work processes (Operator has to walk large distances to complete tasks, perform multiple disconnected operations in sequence, etc.)

Lastly, the site tour can permit access to speak to operators, supervisors, quality techs, etc. first hand to gain insights concerning how these individuals perform their daily work, concerns, or ideas they may have for improvement. The tour will not allow every scenario to be captured but allows us to hear what top of mind for these individuals. Additionally, it helps to build early relationships and impressions that we are at the site to help, not hurt operations.

If possible ask the customer to execute the site tour based upon the product flow sequence, rather than convenience of other factors. This reduces confusion and helps us relate the tour to the process overview previously discussed.

Legacy System Review:

Often, legacy systems can be shown in a conference room or web conference setting. The customer can provide a high-level overview of the various functions supported by the legacy systems, information provided, and can highlight specific challenges, bugs, or gaps. The time taken to load or save data can be captured as a rough baseline. Note that the goal of a digitalization initiative is rarely to duplicate a legacy system function-for-function, but a legacy system review can help highlight what works well and what does not.

Current Process State Documentation:

Conference room and web-based reviews should be recorded and transcribed if at all possible. Seek the customer's permission prior to recording, and provide any recordings captured to the customer after the sessions.

Taking pictures of the shop floor may be permitted, but ***always*** seek the customer's permission before taking photos. In some cases cameras and other recording devices are not permitted on the shop floor or explicit authorization must be secured prior. In other cases, the customer may be able to take photos and provide them.

If permitted by the customer, bring a lightweight notebook and writing utensil on the plant tour to jot down key notes or ideas.

A designated notetaker is useful during conference room walkthroughs and legacy application reviews to ensure that key facts are not missed as the leader of the meeting is focused on driving the conversation and asking questions. Notes should be cleaned up and provided to the customer after the sessions.

Executing legacy application reviews over web conference allow screen captures or videos of legacy applications to be taken.

It can be useful in the assessment to provide a concise summarization of learnings, including a process overview, the role of legacy applications, and major concerns / pain points captured during observations. The customer knows their process, so this is not the primary benefit/focus of the

assessment. Rather, this documentation shows the customer that RoviSys understands their process, including concerns and constraints that were communicated to RoviSys. This baseline documentation can also be of use to the customer if a goal of the assessment document is to form the basis of a bid package for solution implementation that will be provided to multiple vendors.

The customer may also ask RoviSys to present back details concerning the customer's operation and initial observations / recommendations to prove competence and understanding. Be prepared for this ask if it occurs.

(Published) MES RIS 101.10 - Stakeholder Group Interview Process

Wednesday, February 7, 2024 12:56 PM

Please Reference:

[Stakeholder Group Interview Process](#)

Purpose:

The stakeholder interview process is important to gain insights facilitating formulation of the 'to-be' business process scenarios. The goal of this exercise is to better understand the role of each stakeholder in the manufacturing process, then apply our knowledge of digitalization / MES to recommend altered work processes that are likely to provide tangible benefits. The goal of the exercise is NOT for the stakeholder to tell us what those future processes should be. Stakeholders may have opinions concerning various aspects of their daily work, data they wish was more accessible, systems that they wish had better user interfaces, etc., and it is important to capture that feedback. However, stakeholders often will not realize that there is a better way to do what they are doing, and the customer is paying RoviSys to help identify those insights as part of the assessment.

The stakeholder group interviews should minimize the time required of both the stakeholders and ourselves. Therefore, these interviews must be properly planned, efficiently executed, and effectively documented.

Current Process State Background:

Ideally, preliminary conversations with the customer project team concerning the goals of the initiative, the workflows within the organization, and various plant personnel should provide sufficient background to give us an understanding of who we want to talk to and what we want to find out more about. Use this exercise to formulate the scope of the stakeholder interviews. The size and scope of the assessment effort will also come into play. For example, if we are focusing on a maintenance initiative in a certain area it may not be as worthwhile to talk to quality personnel in an unrelated area.

Stakeholder Group Identification:

The various groups we desire to interview must first be identified. Customers often have different names for different groups and roles within the organization, so we must coordinate with our project contacts to understand these groups / roles. The following is a generic list of the various groups we generally desire to interview:

- Plant Manager
- Production Managers
- Controls Engineers
- IT / Manufacturing IT
- Finance / Business Analysts
- Schedulers
- Receiving
- Area Managers / Leads
- Warehouse Managers
- Quality Managers
- Process Engineers
- Maintenance Managers

- Utilities Managers
- HR Managers

Stakeholder Identification:

Depending on the role, it may be impractical to talk to all individuals in the organization. Therefore, we must collaborate with our project contacts who must in turn collaborate with others in the organization to identify the key individuals to be interviewed, based upon factors such as seniority, knowledge, and communication skills. Identifying a stakeholder that can speak to multiple areas can lead to time efficiencies.

It is often beneficial to interview several individuals in small groups to avoid redundancy and save time during the process. We should typically avoid interview meetings with large groups of unrelated individuals though. The larger the group setting, the less likely some individuals will be to provide insights, while others may be more likely to posture or overstate certain aspects.

Additional Site Inclusion:

Even if the scope of the project is limited to one site in a multi-site corporation, consideration should be given to visiting / interviewing other sites to ensure that a global solution is defined, rather than one that only meets the needs of the site in question.

Agenda Formulation:

We want to be as minimally disruptive as possible to the day-to-day jobs of the individuals we identify to interview. Therefore, sessions should be scheduled based on availability and kept as short as possible. Typically 1-2 hours is the maximum amount of time that can be expected for a specific stakeholder / stakeholder group interview session.

A typical DxDiscovery stakeholder interview agenda for a specific manufacturing site is around 2-3 days in duration. The following document reference contains a sample agenda:

[Sample DxDiscovery Agenda Questions.xlsx](#)

Our customer project contact should be able to look at calendars and help find specific times during which specific stakeholder groups are available.

The above document reference also contains sample interview questions for each stakeholder group that can be tailored based on the customer, industry, and other factors, and included in the meeting invites.

Meeting Invites:

We do not want customer stakeholders to feel like they need to perform a large degree of preparation prior to the interview sessions. However, it is useful to include sample interview questions in the invites, along with any other specific requests. For example, we may want a participant to demonstrate a legacy system, bring log sheets or work instruction samples, or perform other prework. This prework should be clearly stated in the invites so time is not wasted obtaining this information during the session or waiting for follow-ups to be performed.

Interview Execution:

It is okay for the conversation to drift off script in certain cases, and this can be useful to gain insights

concerning associated processes and systems that may not be under the scope of the initiative if brought up by the interviewee. However, tactfully steer things back to the topics at hand if the interviewee veers too far off course. A 'parking lot' listing can be used to capture complex topics that will require additional follow-up, action items, and low-priority tangents.

Workshop Status Reporting:

A daily status report should be completed for the purpose of updating internal RoviSys MES division management (director, technical strategy GM, MoC lead, etc.) on the progress of the stakeholder interview workshop. The status report should be completed daily, maintained within OneNote, and a link provided daily to the management team via email or Teams message. This team currently includes the following individuals:

Justin Vaught
John Vargo
Keith Beaubien

Other individuals with interest in the outcomes of the session should also be included at the discretion of the RoviSys lead.

The following is a link to the daily status report template:

[MES RIS 101.5.1 - Stakeholder Workshop Status Report Template](#)

Workshop Wrap-Up:

At the end of each workshop day, the next day's agenda should be reviewed with the customer team, including any session or time changes due to either customer resource availability or prior sessions that ran long / short, and the expected attendees to confirm their availability. Open actions / parking lot items should also be reviewed, including prioritization, follow-up date, and assigned responsible party. A responsible party should be assigned if an action requires follow-up and one was not assigned during the capture of the item. This may require customer project team input or follow-up.

At the end of the workshop, next steps should be reviewed, including the need to schedule follow-ups, notes / documentation to be provided, and RoviSys deliverables to be provided based upon the workshop findings.

Interview Documentation:

At a minimum, the RoviSys team should consist of the interviewer, who is focused on asking prompting questions and driving forward the session, and a note taker, who is focused on the capture of information. Additionally, it is recommended that sessions be recorded so that unclear discussions can be replayed during our subsequent review. In an in-person setting, an omnidirectional microphone can facilitate recording in an inobtrusive manner.

Post interview, the notes from the sessions should be read through to ensure understanding and sanitized. Some customers will explicitly ask for the notes from the sessions but it is good practice to send them to the customer, even if not explicitly requested. Additionally, customers should be provided with any recordings that were taken.

(Published) MES RIS 101.10.1 - Stakeholder Workshop Status Report Template

Wednesday, August 14, 2024 7:24 AM

Please Reference:

[Stakeholder Workshop Status Report Template](#)

Session Date:

RoviSys Team Members Attending:

Sessions Held:

Schedule Adherence (are we on track or drifting off track):

Key Takeaways from Sessions:

Key Follow-Ups / Actions:

Link to Notes Taken During Sessions:

Any Needs from MES Division Management?:

(This may include clarification on topics raised that caused confusion, resourcing to assist with follow-ups, a need to connect to potential vendors, backchanneling issues to our customer contacts, etc.)

(Published) MES RIS 101.11 - Perceived Benefit

Wednesday, February 7, 2024 12:58 PM

Please Reference:
[Perceived Benefit](#)

The goal of the DxDiscovery effort is to gather sufficient information to aid the customer in making a high-level recommendation as to the best path forward to facilitate improvement. Several typical options are as follows:

- Execution of a DxArchitecture effort to better define the manner in which digitalization can be applied to improve plant processes, efficiency, quality
- Execution of a DxArchitecture effort to better define the manner in which digitalization can be applied to replace obsolete or disconnected systems
- Execution of a DxLean effort to identify plant process improvements that can increase efficiency and reduce waste, independent of a digitalization effort
- Problems and potential outcomes may possess sufficient clarity that a pilot implementation initiative can be proposed
- Proposal of a full digitalization initiative, including architecture, implementation, and deployment
- Execution of a deeper controls / automation front-end engineering design (FEED) effort to define the logistics of an automation initiative
- Determination that additional discovery is required, for example visiting other facilities or considering other business units

In making a recommendation, specific findings captured as part of the DxDiscovery effort should be cited as justification for the path forward. Specific observations should be distilled and benefits described at a high level.

(Published) MES RIS 101.12 - Executive Summary

Wednesday, September 4, 2024 1:54 PM

Please Reference:

[Executive Summary](#)

The executive summary presentation distills the activities, key findings, and recommendations into a format that can be digested by broader customer stakeholders including an executive leadership team or steering committee. It should present a compelling plan that highlights the benefits of the initiative, roadmap, and keys to success.

Presentation Components:

The following are the topics typical of an executive summary presentation

Agenda:

The agenda should be presented at the beginning of the presentation so that the group at hand understands what will be covered during the discussion.

Introduction:

Key RoviSys assessment team members, their roles, and credentials should be captured and presented.

RoviSys Background:

Customer stakeholders unfamiliar with RoviSys will likely benefit from a brief overview of our size, locations, and capabilities. This introduction should be quick and to the point, allowing for ample time to cover the logistics and findings within the assessment itself.

DxDiscovery Activity Overview:

This section should present an overview of the discovery activities and overall timeline involved.

Key Individuals and Sites:

This section should present the key individuals from which information was solicited or who reviewed and approved various aspects of the assessment findings. In the case of a global enterprise-centric assessment, the representative site(s) of focus and reasons for selection of those sites as focuses should be discussed.

Capabilities of Focus:

Summarization of a standard capability map and presentation of the capabilities of focus is useful to lay the foundation for the focus of the assessment, including functional capabilities in scope and out of scope.

Key Findings:

The customer stakeholders for whom the presented is targeted likely understands the manufacturing and associated operational processes at a high-level. However, key findings should be summarized, including gaps, concerns, and challenges pertaining to current operational workflows.

Next Steps:

The expected next steps should be clearly articulated. Next steps may include a proposal for architecture formulation, a Lean effort, or other paths forward as detailed in [MES RIS 101.11 - Perceived Benefit](#). RoviSys should coordinate with the project team to recommend next steps that are believed to

be agreeable to the presentation audience prior to giving the presentation, so that there is little chance of surprise.

Questions:

Time should be allocated at the end of the presentation for questions and follow-up actions.

Presentation Roles:

In addition to the primary presenter, a note taker should be assigned who can document any questions, key points raised, or follow-up actions. The note taker should be prepared to summarize these points verbally at the end of the presentation session, so that it can be assured that RoviSys captured all follow-ups and nothing was misinterpreted.

Recording:

Permission should be requested at the beginning of the presentation to record the session. This is useful so that the presentation can be shared with any stakeholders who were unable to attend, as well as so that RoviSys can review any points made that require clarification or re-examination.

Pre-Review:

A pre-review of the executive summary presentation should be conducted with the customer project champion or a small subset of the customer stakeholder team prior to wider distribution of the presentation and conducting the presentation in front of a larger customer team. This will ensure alignment with the core team and reduce the chance of any surprises concerning our findings and recommendations. The customer project champion or project team may also be able to call out sensitive topics or better ways of communicating our findings and recommendations to ensure the messaging is better received by the broader customer stakeholder community.

(Published) MES RAP 102 - DxArchitecture

Thursday, August 22, 2024 4:13 PM

Please Reference:

[DxArchitecture.aspx](#)

RACI Formulation:

RACI formulation aids in the definition of the various participants in the effort, and their responsibilities, both from RoviSys and customer perspectives. This includes not just the project team itself, but also various stakeholders who will need to participate in deliverable review.

Business Intention Definition:

Business intentions are high-level goals and overriding philosophies that will shape the remainder of the initiative. These intentions should be aligned to corresponding returns on investment and/or customer organizational standards to which the initiative must adhere.

Process Activity Diagram:

The process activity diagram documents the various production processes in a 'swimlane' format, showing the system responsible for each activity, as well as system-to-system interactions or user-to-system interactions. This activity is especially useful when a customer possesses existing legacy systems and is struggling with picturing how the MES / MOM initiative will fit into the overall systems landscape.

Business Process Scenario Definition:

Business process scenarios capture high level details concerning the various activities (processes) envisioned to be executed within the manufacturing enterprise, once digitalization is applied. Key aspects of business process scenario definition include documentation of the specific process steps, as well as identification of the specific inputs to the process (incoming data), suppliers of the inputs, outputs from the process (outbound data), and consumers of that data.

Generic Architecture Formulation:

The generic architecture shows the various interactions envisioned between systems as well as touchpoints between people and systems (clients, scanners, instruments, etc.), agnostic of a specific digitalization platform. This documentation aids in understanding how the MES/MOM platform stack will fit into the overall enterprise systems landscape, and can help better visualize architectural constraints that may need to be considered when performing technology selection activities.

Technology Selection Process:

The result of this activity is clarity concerning the solution stack that will be leveraged to deliver on the envisioned process scenarios and corresponding ROI. A variety of measurable criteria are evaluated to ensure that there is a solid justification for the technology selection(s) made.

Solution Architecture Definition:

The solution architecture offers additional clarity not provided in the generic architecture concerning the selected digitalization solution stack and how this fits into the overall customer enterprise

landscape. Specific modules to be implemented in the selected platform(s) as well as interface points are highlighted.

ROI Guidance Formulation:

During this activity, the customer is provided guidance concerning how to calculate the expected return on investment (ROI) of the solution stack. Various opportunities for ROI are considered during this exercise, including reductions in legacy system infrastructure and maintenance, tangible ROI, and calamity avoidance ROI.

Deployment Roadmap Formulation:

The deployment roadmap details how the solution stack is envisioned to be deployed across the area, site, enterprise based on the overall scope of the initiative. This plan helps to establish the required implementation project phases upon which the quotation effort can be based. As a result, this activity naturally progresses to the Spend Allocation Phase.

Executive Summary:

The executive summary presentation distills the activities, key findings, and recommendations into a format that can be digested by broader customer stakeholders including an executive leadership team or steering committee. It should present a compelling plan that highlights the benefits of the initiative, roadmap, and keys to success.

(Published) MES RIS 102.1 - DxArchitecture RACI

Wednesday, February 7, 2024 12:55 PM

Please Reference:

[DxArchitecture RACI](#)

RACI (or RASCI) formulation aids in the definition of the various participants in the effort, and their responsibilities, both from RoviSys and customer perspectives. This includes not just the project team itself, but also various stakeholders who will need to participate in current process state explanations, interviews, and deliverable review.

Each DxArchitecture activity and deliverable should be broken out on the RACI template based on the envisioned assessment schedule, the scope of the formulated assessment interviews, and the resulting deliverables.

General items to be captured within the RACI or RASCI and assigned include the following:

- Participation in regular status meetings
- Every DxArchitecture deliverable - these deliverables should be broken up into granular components as different stakeholder will have interest in different sections.
 - o Business intentions
 - o Business process scenarios (by plant area / function)
 - o Generic architecture
 - o Technology assessment
 - o System architecture
 - o ROI guidance
 - o Etc...
- Executive summary presentation participants

This example is based upon the following generic RACI template:

<https://rovi.sharepoint.com/:x:/r/sites/mes/Shared%20Documents/3.Strategy/PMO/Document%20Templates/Submitted/PMO%20Template%20010%20-%20Customer%20Name.Project%20No.PD%23%23%23.Responsibility%20Matrix%20-%20RACI.xlsx?d=wadcb74d3c9154f3d8fa454f2083d6c5c&csf=1&web=1&e=uyby2d>

The RASCI model specifies definition of stakeholders as follows:

- Responsible for ensuring completion of the activity. The responsible person is the one who does the work to complete the task or create the deliverable. Typically, a RoviSys resource is named as responsible for the completion of the various Assessment deliverables, although the customer may be responsible for providing prerequisite information.
- Accountable for reviewing and confirming the completion of the activity. The accountable party approves the deliverable provided as a result of a given assessment activity.
 - o Typically, a customer resource is named as the accountable party, ensuring the provided deliverable is accurate and soliciting feedback from other consulted parties. Rather than requiring the document be signed off by many reviewers, one accountable party helped to streamline review and approval cycles.
 - o In the case of stakeholder interview RACI items, the accountable party may be the representative providing input during the interview session, but may be required to solicit input from other consulted individuals who are not physically present in the session.
 - o Not every customer stakeholder may be required to review the assessment in its entirety. A

production manager may desire to review business process scenarios specific to their area, but need not review the technical architecture envisioned for example.

- Supporting roles may be other RoviSys engineers or individuals assisting the responsible party in completing the task at hand in the case of deliverables provided by RoviSys (such as documentation). This may include other RoviSys engineers / functional architects.
- Consulted during information gathering or deliverable review. Consulted individuals may be additional customer reviewers or individuals from which information is captured by the accountable party.
- Informed throughout the activity, but not required for review or information gathering purposes. These individuals are typically customer roles who need to be kept abreast of the progress of the activity by being provided status updates, or being included in status update meetings. They might include heads or directors of affected customer teams and senior customer leadership.

The RoviSys templated assessment agenda and RACI template will both contain generic stakeholder roles. Translating these generic roles into specific roles possessed by an organization is crucial for operational efficiency. Generic roles, while useful for defining broad responsibilities, may not align perfectly with the unique structure, capabilities, and expertise within an organization. By mapping these roles to specific positions, it is ensured that tasks are assigned to the most suitable individuals. This specificity enhances accountability, as it is clear who is responsible for what, and improves communication by aligning expectations with actual job functions within the organization. Moreover, it allows for more accurate resource planning and utilization, reduces redundancy, and helps in identifying any gaps in coverage.

As a result, RoviSys should discuss with the customer project manager and relevant project team members to ensure that the generic roles for which communication is requested is translated to the specific roles and responsibilities within the customer's organization while working to identify the key individuals for which involvement is requested, rather than simply pass the generic templates to the customer and ask for these to be filled in.

(Published) MES RIS 102.2 - Business Intention Definition

Tuesday, January 9, 2024 9:02 AM

Please Reference:

[Business Intention Definition](#)

Business intentions are high-level goals, guiding principles, and overriding philosophies that will shape the remainder of the initiative. These intentions should be aligned to corresponding returns on investment and/or customer organizational standards to which the initiative must adhere. Additionally, they should permit corresponding business process scenarios and/or solution architecture aspects to be tied to them. Business intentions should allow individuals across the customer enterprise to understand the reasons to execute the initiative and standards to which the initiative will adhere without reading the entirety of the assessment. Different categories of business intentions include the following:

Organizational Standards:

These business intentions establish constraints to which the solution must adhere to be deemed viable. Often these intentions are based upon corporate standards imposed by IT/OT organizations. Some examples include the following:

- Usage of cloud infrastructure for hosting or data repositories.
- User interface requirements, such as usage of mobile devices, web interfaces, 'single pane of glass' concepts, thin clients, etc.
- Corporate communications standards (protocols such as OPC or MQTT, middleware application layers, etc.)
- Corporate standard software solutions that must be leveraged (data repositories, reporting, analytics, etc.)
- Adherence to specific security standards (network segmentation, user authentication, etc.)

Compliance:

The system may need to comply with certain industry or regulatory standards. Examples of this include:

- CFR 21 Part 11 (life sciences)
- ITAR (government / defense)
- ISA-95 (enterprise organization modeling)
- ISA-88 (batch system modeling)

High-Level Goals:

These business intentions state the perceived benefits of the solution against which success will be measured. The goals may or may not be quantifiable. Some examples of goals include the following:

- Increase OEE by X percentage points
- Increase first-pass quality by X
- Reduce scrap production by X
- Reduce on-hand inventory by X
- Reduce unplanned downtime by X
- Replacement of specific legacy systems
- Reduction in manual paperwork, dual data entry, and other time consuming efforts
- Automate specific manual work processes

- Ease of maintainability by site or corporate teams
- Replacement of certain legacy systems

These high-level goals often have specific ties to ROI formulation. As a result, they can be easily championed within the customer organization. These goals are also most easily tied to corresponding business process scenarios that help to detail how each goal will be achieved. Note that a business process scenario may be able to be tied to one or more business intentions.

Clarifying Intentions:

These business intentions express what the initiative will *not* provide to address misconceptions and reduce confusion. For example, the initiative may not replace certain legacy applications, provide certain functionality, be deployed to all sites/areas, etc.

(Published) MES RIS 102.3 - Process Activity Diagram

Thursday, September 19, 2024 8:09 AM

Please Reference:

[Process Activity Diagram](#)

The process activity diagram documents the various production processes in a 'swim lane' format, showing the system responsible for each activity, as well as system-to-system interactions or user-to-system interactions. This activity is especially useful to facilitate a technical team's understanding of the role of various systems and how they fit together. The process activity diagram highlights areas where detailed system interface design will be required, as well as cases where user interfaces will need to be made available. These diagrams can be particularly useful when a customer possesses existing legacy systems and is struggling to picture how the MES / MOM initiative will fit into the overall systems landscape.

The first step in creation of a process activity diagram is identification of the distinct systems and roles to be shown. Common systems to be portrayed include the following:

- MES / MOM
- Enterprise Resource Planning System (ERP)
- Laboratory Information Management System (LIMS)
- Laboratory Execution System (LES)
- Warehouse Management System (WMS)
- Computerized Maintenance Management System (CMMS)
- Process Control System (PLCs, DCS, SCADA, etc.)
- Historian
- Advanced Production Scheduling Application (APS)
- User roles (Operator, Scheduler, Lab Technician, Maintenance, etc.)

Usually the diagram does not need to go to the level of showing specific modules within systems. This may make the diagram too complex and hinder understanding, although more detailed variants could be built as required by certain stakeholder groups.

It is useful to arrange systems in the order of the ISA-95 systems hierarchy from left to right, with process control systems coming first, then historian, MES, and lastly enterprise systems.

To the extreme left is a representation of the sub-process within the overall process being represented. These sub-processes are labeled and shown as circles, with arrows between indicating the flow as the process represented is executed. In some cases a sub-process may branch to multiple sub-processes based upon a decision point, or a sub-process may be skipped if some variants of the process do not require it.

Individual steps and interactions are then broken out into rectangles, placed under the system or person performing the step in question. Interactions between systems are then shown as arrows, indicating the source and destination of the interaction or data transfer in question. Arrows and boxes may be color-coded with a legend to draw attention to specific steps or interactions of interest. For example, if the goal of the diagram is to highlight all MES <-> ERP interaction, these can be color-coded so that they are easily viewed.

A process activity diagram could be generated for a general sub-process, and then referenced throughout other process activity diagrams rather than re-generated in detail on each. For example,

multiple different batch processes may possess a weigh and dispense sub-process, which could be documented in detail and then referenced in various process activity diagrams rather than laid out in detail within each.

Detailed inputs, outputs, suppliers, and consumers need not be shown on the process activity diagram. Rather, these details are shown in the SIPOC diagram created for a specific process.

The following is a reference to a process activity diagram template that can be leveraged as a basis for creation of process activity diagrams:

[DxArchitecture Process Activity Diagram Template.vsd](#)

The following is an example process activity diagram that can be shown to customers / prospective customers as examples or used as inspiration for creating process activity diagrams specific to a customer project:

[EX001-MES-FRS04 - Assembly System Interactions.vsd](#)

[EX002-MES-FRS07 - Packaging System Interactions.vsd](#)

(Published) MES RIS 102.4 - Business Process Scenario Definition (SIPOC)

Tuesday, January 9, 2024 8:53 AM

Please Reference:

[Business Process Scenario Definition \(SIPOC\)](#)

Purpose:

Business process scenarios capture high level details concerning the various activities (processes) envisioned to be executed within the manufacturing enterprise, once digitalization is applied.

The goals of the defined business process scenarios include the following:

- Be easily understood by non-technical stakeholders
- Be documented in an efficient, repeatable manner as part of the assessment process
- Be technology agnostic. The business process scenario should refer to various business systems at a high level (MES, ERP, LIMS, control system, etc.), but not delve into the nuances of applying a specific platform or technical solution
- Establish 'guardrails' concerning the solution to be provided, so that the intended processes to be facilitated by the solution are clear, reducing the possibility of future scope creep
 - o Both typical (sunny day) and exception (rainy day) scenarios should be captured
- Provide a basis upon which more detailed user stories can be built. User stories should be matrixed to business process scenarios so that the goal and reasoning for each user story is clear
- Highlight required user-to-system and system-to-system interactions, which can be leveraged when building more detailed technical documentation (architecture diagrams, value stream maps, etc.)

The functionality basis established by the business process scenarios assists not only with up-front understanding of the scope of the solution and value to be provided, but also serves as a basis for future implementation and deployment activities. For example, properly-defined business process scenarios can serve as a foundation for testing plans (integration testing, FAT, UAT, etc.) and creation of role-specific training materials such as SOPs. Therefore, it is important to keep defined business process scenarios up-to-date throughout the initiative so that they can be utilized later in the project lifecycle, further enhancing their value.

Methodology:

The MES division has adopted the 'SIPOC' methodology for the definition of business process scenarios. This is an industry standard methodology that has been adapted to better serve the needs of our MES project initiatives.

S = Process suppliers

I = Process inputs

P = Process

O = Process outputs

C = Process consumers

The anatomy of a SIPOC includes the following:

Header: A short description of the activity (business process scenario) being represented by the SIPOC

Inputs: A table containing all envisioned inputs to the process. This table contains three columns, including the following:

- Type = a categorization indicating the type of input. Valid types include:
 - o User
 - o System
- Supplier = the user or system role expected to provide the input
- Input Detail = A description of the input itself

Process: A table outlining the various steps required to execute the process. This table contains three columns, including the following:

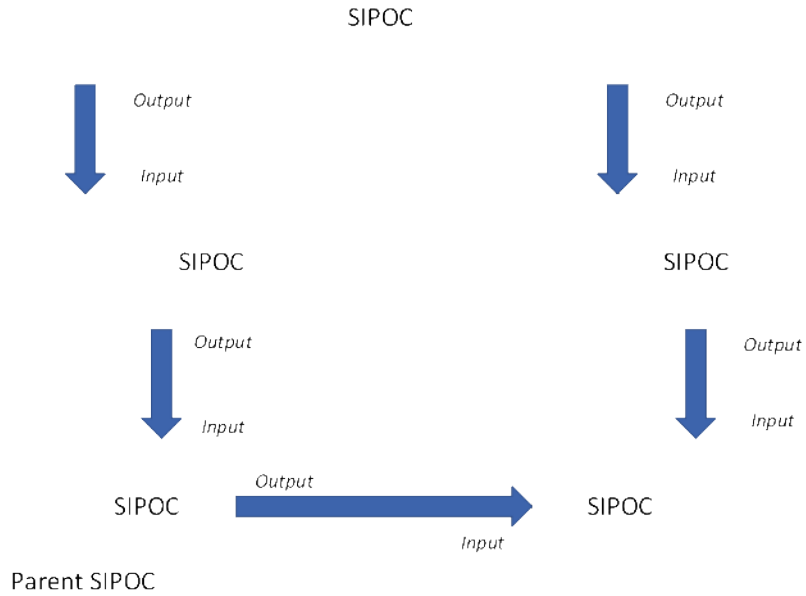
- Who = a reference to the actor performing the process step. An actor may be a user or a specific system
- What = a description of the process step
- External = A flag indicating that a supporting process step must occur that is envisioned to live outside the scope of the system (typically MES solution) being implemented

Outputs: A table containing all envisioned outputs of the process. This table contains three columns, including the following:

- Type = a categorization indicating the type of output. Valid types include user and system
- Consumer = the recipient of the output. The consumer will typically leverage the output as an input to a subsequent activity
- Output Detail = A description of the output itself

It is important that SIPOCs be hierarchical in nature, and support drill-down into more specific activities. The highest-level SIPOC in a manufacturing facility represents the overall conversion of raw materials as input to finished goods as output. Various sub-processes needed to facilitate this, such as scheduling, raw material receiving, quality control, warehousing, etc. all derive from this highest-level SIPOC in some manner. SIPOCs must also be able to be chained together, as the outputs of a given SIPOC are often inputs to another. An example of the hierarchical nature of SIPOCs is shown in the following diagram:

Parent SIPOC



Other guidelines concerning the definition of each SIPOC include the following:

- The inputs and outputs should capture any person or system that interacts with the MES for the scenario
- The process should include all inputs and outputs captured, in addition to any MES operators involved directly in the process
- The process should be captured in a way that is clearly the basis for a series of User Stories. Each process step in the lowest-level SIPOC diagrams are likely individual user stories.
 - o Any inputs / outputs that are of type User would require a UI component.
 - o Any inputs / outputs that are of type System would require some interface to the MES that would be portrayed on the architecture diagram

SIPOC Authoring

A preferred template has been created to facilitate SIPOC definition. Other authoring tools may also be appropriate for a specific situation, but the preferred template is recommended as a starting point.

The preferred template makes use of Microsoft PowerPoint. This template provides a means of defining the SIPOC title, process steps, inputs, and outputs. Additionally, a PowerPoint link can be added to a process step, allowing that step to link to a child SIPOC diagram. This allows for support of hierarchical 'drill-down' capabilities.

At the bottom of a SIPOC diagram, a placeholder allows a reference to a parent SIPOC to be defined. This allows 'roll-up' capabilities, so that the viewer can quickly navigate to the parent of a specific SIPOC.

Lastly, related SIPOCs can be grouped using PowerPoint sections.

The SIPOC template can be found in the following location:

DxArchitecture SIPOC Template.pptx

Examples of SIPOC diagrams can be found in the following locations:

[DxArchitecture SIPOC Template - Batch Execution Example.pptx](#)

[DxArchitecture SIPOC Template - Customer-specific example.pptx](#)

Business Process Scenario Reuse

The MES division will genericize core manufacturing business process scenarios, including creation and maintenance of baseline SIPOC diagrams that can be leveraged as a starting point for customer engagements. A register of business process scenarios and corresponding SIPOCs, based upon manufacturing type (batch, continuous, and discrete) will be maintained as an Excel sheet within the business process scenario repository. This register will also contain linkages to baseline SIPOC diagrams that can be utilized as reference.

When commencing a new assessment effort for which SIPOC definition will occur, the assessment resource should first consult the SIPOC register to capture all relevant SIPOCs that may be applicable to the process in question. These SIPOCs can then be extended and new SIPOCs added as required based on the specific customer work processes.

At the end of an assessment engagement, an effort should be made to identify any scenarios / SIPOCs that could be of use to other assessment engagements in the future. These should be genericized where possible, including the removal of customer-specific terms and intellectual property, then added to the SIPOC reuse repository. Please consult the MES leadership team if assistance is needed with this effort.

The business process scenario register is located in the following location:

[DxArchitecture Business Process Scenario Register.xlsx](#)

This register possesses two tabs - one tab contains a reference of SIPOCs for general MES drivers. The second contains a reference of SIPOCs facilitating decision making processes common in manufacturing enterprises.

(Published) MES RIS 102.4.1 - Usage of SIPOC to Capture Decision Points

Tuesday, January 28, 2025 8:10 AM

Please Reference:

[Usage of SIPOC to Capture Decision Points](#)

Background:

A key consideration when documenting business process scenarios is to properly distill decision points throughout the process leveraging the SIPOC format, including all inputs needed to support the decision, the process entailed in making the decision, and resulting decision output.

This is particularly applicable to life sciences digitalization deployments, as the primary driver for the initiative is ultimately to support product release by exception. This necessitates exception scenarios / anomalies to be identified and resolved throughout the process, then reviewed subsequent to making a final decision concerning the ability of the product to be released to the consumer. However, a goal of digitalization in general is to enable decision makers, and thus it is important for these decision points to be properly distilled and agreed upon as part of business process scenario definition.

Longer-term, if these decision points are properly captured and clearly defined, agentic systems may be applied to support decision making, reducing human intervention. These systems can determine specifically when human feedback is required, present targeted information supporting the decision making process so that the human need not spend time mining data from disparate data sources, and may ultimately be able to make certain decisions autonomously. Therefore, proper documentation of all decision points is essential to future support for such agentic systems.

SIPOC Hierarchy:

- Decision points may warrant creation of dedicated child SIPOC diagrams that are related to higher-level parents. For example, there may be an overall SIPOC driving product release upon production completion. This SIPOC may be fed by the output of numerous child SIPOCs, representing smaller decision points that feed into the overall product release decision. For example, exception scenarios may arise throughout production warranting scrapping a batch, reworking an assembly, or moving forward due to the high likelihood that the product will still be compliant. These decisions must be reviewed and determined to be of no risk to product safety prior to the distribution of the product to the consumer.

Input and Decision Considerations:

- The SIPOC author must ensure that all inputs needed to support the decision in question are captured. This may include both immediate data points created in the context of the batch / assembly in question, as well as contextual and historical data from other products / batches / production runs that may be needed by a person or agent to properly make a decision. Some examples include the following:
 - o If a dimensional check fails, measuring equipment calibration records may need to be consulted to confirm that the measuring device was properly calibrated within the proper time thresholds. The act of reviewing these records and making a decision should be captured as part of the SIPOC documentation process, and the decision made (recalibrate

- and retest the part, fail the part, pass the part, etc.) captured as an output.
- If a part or material is leveraged that was not indicated on the BOM, supplier records, including the CoA, may need to be consulted. Perhaps a substitute part was sent by the supplier that possesses the same quality criteria and is an acceptable substitute? Or, an inferior part may have been provided that will require the assembly to be rejected?
- A part may be sourced from multiple different suppliers, but historical data indicates that parts from a certain provider often lead to downstream quality issues. Usage of the part from the suspect supplier may warrant additional quality verifications (additional dimensional checks, electrical checks, visual inspections, etc.) before release of the assembly.
- In the event that a batch exceeds some CTQ (critical to quality) parameter, such as a pH, temperature, etc., historical trend data may need to be consulted to look at the value over time. Was the exceeded value an anomaly that was quickly brought back into conformance, or has the historical data point been trending in the wrong direction for some time? The latter is indicative of a larger issue that should be resolved and may necessitate scrapping the batch, while the former may be acceptable should subsequent testing prove that the batch ultimately meets all established quality criteria.
- An improper amount of a certain raw material may have been added, which requires consultation with a staff chemist to determine if the raw material addition error will lead to a potency issue or presents a risk to patient safety. The chemist may be required to perform additional lab tests or consult historical data from patient trials to make such determination.

Denotation:

- Consideration should be given to breaking out decision making processes, particularly those requiring multiple inputs and/or sequential decision making steps, into their own SIPOC diagrams for ease of review and understanding.
- All inputs (direct or indirect) required to support the decision should be captured, including the source / supplier of the input. Consideration must be given not only to input values that are immediately accessible as part of the specific product's production, but also indirect or historical inputs that may need to be obtained by the decision maker to utilize in making a judgement call concerning product quality.
- An output that is either a supporting decision point that feeds a higher-level SIPOC or ultimately a decision point concerning the quality of a finished product should be clearly denoted as such.

Summary:

Effective documentation of business process scenarios requires clearly defining decision points using the SIPOC format, ensuring all necessary inputs, decision-making processes, and outputs are captured. This is especially crucial in life sciences digitalization efforts, where exception handling is key to product release. Properly structured decision points may enable future automation and reduce human intervention, supporting agentic systems that assist or autonomously make decisions.

(Published) MES RIS 102.5 - Generic Architecture

Wednesday, February 7, 2024 12:57 PM

Please Reference:

[Generic Architecture](#)

The generic architecture shows the various interactions envisioned between systems as well as touchpoints between people and systems (clients, scanners, instruments, etc.), agnostic of a specific digitalization platform. This documentation aids in understanding how the MES/MOM platform stack will fit into the overall enterprise systems landscape, and can help better visualize architectural constraints that may need to be considered when performing technology selection activities.

There are two common processes leveraged to document a generic architecture. One or both of these may be of use to generate and provide to the customer during the assessment phase.

Block Diagram:

A block diagram shows specific systems or system placeholders and associated interconnections between them. The block diagram is often organized according to the ISA-95 hierarchy:

- ERP
- MES / MOM
- SCADA / HMI
- Controls

Legacy systems and applications expected to remain within the solution stack can be explicitly called out, while placeholders for new applications can be referenced generically. Additionally, it can be useful to document 'before and after' block diagrams, showing the existing systems landscape and to-be systems landscape after legacy applications are decommissioned and/or new functionality is incorporated.

The block diagram may also convey boundaries between on-premises, cloud, and corporate applications, network delineations (controls network, site business network, corporate network), and other aspects related to infrastructure (like user interface types and locations).

A complex architecture and associated block diagram is not always better. Rather, steps should be taken to simplify the architecture wherever possible, and the generic architecture exercise can help to highlight cases where the architecture envisioned is overly complex. For example, a large number of crisscrossed point-to-point interfaces could be replaced by a centralized data pipeline / universal namespace (UNS) that follows more of a hub and spoke architecture.

IT Infrastructure Constraints:

Architectural constraints due to IT infrastructure aspects should also be considered and noted as part of the generic architecture definition, including the following:

- Network security requirements
- Long-term data storage and retrieval considerations
- Networking requirements and network latency
- Data historization
- Shop floor data connectivity needs

- User tracking and authentication

The MES solution selected must fit within any pre-defined IT policies or associated constraints that are discovered as part of the assessment.

(Published) MES RIS 102.6 - Technology Selection Process

Wednesday, February 7, 2024 12:57 PM

Please Reference:

[Technology Selection Process](#)

The result of the technology selection activity is clarity concerning the solution stack that will be leveraged to deliver on the envisioned process scenarios and corresponding ROI. A variety of measurable criteria are evaluated to ensure that there is a solid justification for the technology selection(s) made.

Technology Platform Shortlist:

The first step in executing the technology assessment is shortlisting the platforms to be evaluated. If RoviSys was contracted directly by the customer to perform an independent analysis, we have the opportunity to select and rate various competing platform solutions. However, if RoviSys was brought into the opportunity by a vendor, we may not be afforded this opportunity. In this case, the technology assessment should focus on how to best utilize the vendor platform and where weaknesses in the vendor platform may require supplemental components that RoviSys can recommend due to our independence.

Given the hundreds of platforms and niche solutions on the market, it is impractical to evaluate every option. Rather, our experience is applied to first recommend a set of short-listed candidates. Typically, the shortlist is comprised of the following:

- MES priority platforms that we feel may have merit as the basis for the solution. This is based upon industry, functional requirements, manufacturing type (batch, continuous, discrete), and required interfaces to external systems. The MES Platform Highlights register is a valuable resource when determining applicable shortlisted platforms:
<https://rovi.sharepoint.com/sites/mes-dsc-project-execution/Lists/MES%20Platform%20Comparison/AllItems.aspx>
- Platforms that may already be owned / leveraged by the customer
- Platforms that the customer may desire to be included in the comparison
- A custom development alternative may also warrant inclusion in some cases. Particularly, the customer may desire a custom-developed solution to be considered, or the scope of the deployment may yield benefits that outweigh the higher development and maintenance costs of a custom option. For example, a customer possessing 100 sites may see a custom option with no software licensing costs as an attractive option, but this is very likely impractical for a customer with a handful of sites.

Some solutions may inherently be non-starters due to certain aspects. For example, a cloud or SaaS solution may not be permitted by the customer, or a customer may exclusively want a SaaS solution. The formulated business intentions concerning infrastructure and platform constraints may automatically exclude or include certain options, without a full technology comparison being required.

Technology Option Background:

It is useful to provide a brief background for each technology option selected to provide the customer with additional information. This includes the following:

- Website

- Location of headquarters and global office reach
- Year of founding
- Approximate company size
- Key market differentiators
 - Typical markets / manufacturing types supported
- Gartner report recognition (if applicable)
- General technology stack overview (components, hosting model, etc.)

Criteria Definition:

The criteria to be utilized include a mix of general solution capabilities, vendor stability and reliability, and aspects specific to the functionality required to fulfill the various business process scenarios. Criteria should be grouped into overall categories.

Generic categories to be considered include the following:

Business Fit:

- These criteria rank the selected options based upon their applicability to the customer's manufacturing type, vendor and integrator presence in regions / continents aligning to the customer manufacturing plant footprint, evidence that the vendor solution has been applied in situations of similar scale / complexity, and vendor stability (ongoing investment in solution, financial stability, etc.)

User Experience Fit:

- These criteria rank the user experiences provided by the selected options, including required form factors (web, mobile, etc.), robustness of the out-of-the-box user experiences provided by the platform, and ability to extend the user interfaces to provide visualizations unique to the customer.

Functional Capability Fit:

- These criteria rank each solution according to its ability to deliver needed functionality based upon the defined business process scenarios. A platform may possess screens, workflows, and event models that correlate well to the needed functionality, or provide a solid foundation upon which the functionality can be built. It is important to think about how the platform can be leveraged specific to the business process scenario. For example, many platforms generically support schedule execution, but a certain specific platform may be well suited to a scenario in which multiple orders possessing the same container type can be consolidated and treated as one larger order at a filler. The advice of RoviSys solution SMEs or solution architects may be warranted to assist with this ranking exercise.

External Systems Integration Fit:

- These criteria rank each solution according to its ability to integrate to external systems according to the envisioned generic architecture. A certain solution may provide a specific interface as-is, or provide a solid framework for the creation of the interface.

Technical Implementation Fit:

- These criteria rank each solution in terms of the ease of maintenance and deployment. Availability of vendor technical support conduits, product documentation, an established technical support community, as well as support for modern development, troubleshooting, and testing methodologies are all considerations in this section.

Solution Deployment Flexibility Fit:

- These criteria rank ease of deployment for each shortlisted option. This includes solution

alignment to the customer's preferred hosting model (on-premises, cloud, SaaS, etc.) the magnitude of infrastructure required and associated cost, and other factors such as the ability of the solution to host multiple site implementations within a single instance. Other deployment considerations include ease of solution installation and patching, disaster recovery and redundancy functionality, as well as support for complementary data warehousing or data lake solutions.

Solution Cost:

- These criteria rank ease solution in terms of cost. This includes simplicity of the vendor costing model, scalability of the cost model, anticipated licensing cost relative to its peers, anticipated maintenance cost relative to its peers, and anticipated services cost relative to its peers. For example, a low-cost solution with limited out-of-the-box capabilities may prove to possess a higher cost in the long run than a high-cost solution with rich out-of-the-box capabilities. Solution cost is one area where a scaling factor larger than 0-5 may be justified. For example, a solution in a different order of magnitude ballpark may warrant scaling at the high end of a 0-10 spectrum to better highlight the pricing differential.

Scaling Factor Definition:

A scaling factor is then established against which each technology option is ranked. For example, a rating of 5 may indicate an option is an excellent fit, while 0 indicates no fit (function not achievable within the platform, even with customization). A scale of 1-5 is generally appropriate for the purposes of the ranking. A larger scale such as 1-10 can be unwieldy, as it can be difficult to quantify subtle differences (such as between a 6 and a 7).

Weighting Factor Definition:

Each category is assigned a weighting factor based upon the importance of a given area from both RoviSys' and the customer's perspectives. Functional fit is typically assigned the highest weighting, as the most important aspect of the solution is its ability to fulfill the envisioned business process scenarios. However, an extremely cost-conscious customer may weight cost more heavily than user experience fit. External system integration may not be as important to a customer with a limited external solution stack.

Criteria within a category may also be weighted and a weighted average calculated to arrive at an overall average score for the category. Otherwise, a straight average of the scores within the category is taken to arrive at an overall score for the category.

A weighted average of all category scores can then be taken to arrive at an overall score for each option.

Freeform Comments:

It is useful to provide comments and explanations describing why certain criteria were ranked in a given manner. This can be useful for reference or when reviewing the scorecard with the customer.

Example:

Below is a link to an example MES vendor scorecard:

[MES Vendor Scorecard Example.xlsx](#)

Non-Disclosure:

A given vendor within the scorecard may disagree with our assessment or be offended based upon the ranking assigned. Based on this, it is important not to disclose the scorecard in its entirety to vendors,

and to ask the customer not to disclose the rankings to competing vendors or outside parties.

Summary:

A summary must be drafted that provides the following aspects:

- A high-level overview of the selected options to be evaluated
- A summary of the comparison criteria leveraged
- The resulting selection and highlighted differentiators that led to its selection

(Published) MES RIS 102.7 - Solution Architecture Definition

Wednesday, February 7, 2024 12:58 PM

Please Reference:

[Solution Architecture Definition](#)

The solution architecture offers additional clarity not provided in the generic architecture concerning the selected digitalization solution stack and how this fits into the overall customer enterprise landscape. Specific modules to be implemented in the selected platform(s) as well as specific interface points are highlighted.

Additionally, specific architectural aspects needed to support the deployments (servers, cloud infrastructure, networking) are depicted.

The solution architecture is typically of interest to IT / OT groups and related technical teams within the customer enterprise so that they understand how the solution will fit into the landscape and what they will need to provide to ensure deployment.

Definition concerning infrastructure requirements, such as networking, server sizing, and other constraints should be provided with the solution architecture to allow IT / OT organizations to begin preparing for installation and deployment of the solution stack. Long procurement lead times and the need to plan infrastructure installation / configuration activities often necessitate some level of infrastructure specification during the assessment phase. Solution scaling will continue to be refined during technical design and master data definition throughout the dxExecute activity.

The solution architecture may also highlight CI/CD conduits, development -> QA -> production environment progressions, automated unit testing frameworks, and other aspects related to the development of the solution rather than the end-state production environment.

The following is an example of a platform-specific system architecture:

[Sample System Architecture Map.vsd](#)

Other architectural aspects should also be captured and denoted within the solution architecture, including the following:

- Network security
- Long-term data storage and retrieval considerations
- Networking requirements and network latency
- Data historization
- Shop floor data connectivity conduits
- User tracking and authentication

(Published) MES RIS 102.8 - ROI Guidance Formulation

Wednesday, February 7, 2024 12:58 PM

Please Reference:

[ROI Guidance Formulation](#)

It is important to provide our customers guidance concerning how the solutions we propose will provide a return on investment (ROI). It is generally beyond our capabilities to calculate the ROI for our customers in a vacuum, as we lack specific knowledge on various costs including raw materials, finished goods, labor, etc. However, we can provide guidelines and examples that our customers can leverage, allowing them to plug in financial numbers to determine expected ROI.

Note that ROI calculations are approximations. We should err on being conservative with our estimations so that customer expectations are managed throughout the initiative.

There are three primary types of ROI that should be considered:

- Legacy system infrastructure and maintenance cost reduction
- Tangible ROI
- Calamity avoidance ROI

These three considerations are broken down in the following sections.

Legacy System Infrastructure and Maintenance:

Often, MES initiatives are justified due to infrastructure and IT support savings. Some potential considerations include the following:

- Reduction in licensing and recurring vendor support costs due to legacy application elimination or consolidation
- Reduction in physical infrastructure cost due to consolidation of legacy servers, migration to the cloud, removal of costly mainframe systems, or transition from thick clients (physical PCs) to web or thin clients
- Reduction in IT / OT support staff due to reduction in the application portfolio requiring upkeep and troubleshooting
- Removal of unstable systems requiring frequent intervention to keep running
- Elimination of physical interfaces requiring specialized network hardware, wiring, etc.

Tangible ROI:

Tangible ROI is based upon direct cost savings due to implementation of new processes enabled by the MES platform. Note that installing the MES platform itself does not provide a tangible ROI. Rather, the efficiencies gained by the new work processes facilitated by the MES are what provide the benefit. Therefore, it is important to properly define and manage these work process changes alongside the deployment of the MES itself.

Some examples of tangible ROI include the following:

- Reduction in time spent inputting data into the system or running reports to find insights.
- Direct interfaces to external systems such as ERP allowing data to be transferred in real-time rather than relying upon manual data transposition

- Direct interfaces to shop floor devices, ensuring that product and recipe information is conveyed in real-time with less potential for data entry error
- Access to work instructions, changeover procedures, or other information allowing manual tasks to be completed more efficiently or accurately, leading to greater production uptime or increased first-pass yield
- Optimization algorithms allowing greater throughput due to shop floor schedule optimization and resulting increased capacity
- Greater insight into poorly performing lines or work cells, allowing work processes to be enacted to more quickly achieve an ideal production state, increasing the overall equipment effectiveness (OEE) of the facility
- Reduction in safety stock, in-flight WIP, or raw material inventory, in turn increasing working capital
- Standardization of processes, reducing site- or area-specific training demands and ensuring all personnel complete a task in the same manner

Some aspects of tangible ROI may have a greater impact than others, depending on the customer, industry, and even region where the plant is located. Reduction of manual processes may have a dramatic impact in a region where labor is expensive or difficult to secure, or the task is highly complex in nature, but may be negligible in an area where manpower is cheap and plentiful. Raw material inventory reductions may lead to substantial savings if the cost of such raw materials is high, but if raw materials are cheap and easily obtained this may not be a factor.

Engagement of the plant financial team is crucial during the ROI formulation exercise to ensure factors are taken into account that may not be known to the project team. The finance team will have insights into labor, capital, and material costs that will allow for more accurate ROI formulation.

Again, it is important to think of tangible costs savings from a work process perspective. For example, deployment of an OEE dashboard to the facility will not lead to an automatic increase in OEE. Rather, the enhanced visibility provided by the dashboards should trigger the execution of work processes that more quickly return the facility to producing good quality product at the target ideal rate.

Manual OEE calculations or time studies may be required to be performed over a time period to determine where the plant currently operates and then cost savings projected based on this data. For example, if the equipment is down two hours a day and it is felt that with proper visibility and triggering of associated downtime resolution work processes this downtime can be decreased to 1 hour a day, the additional output able to be realized in the extra hour of uptime is the tangible ROI.

Calamity Avoidance ROI:

Calamity avoidance ROI is the most difficult to calculate because the impact is vague in nature, but often provides the greatest justification for the initiative. This type of ROI is realized via avoidance of an incident that could severely impact the company, requiring money to be spent to resolve and/or reducing the company's sales. There is a chance the incident may or may not occur even if the MES is put in place, but the goal of the MES platform is to act as an insurance policy of sorts, making the likelihood of occurrence far less than it otherwise would be.

Some examples of calamity avoidance ROI include the following:

- Prevention of an equipment failure that leads to a facility shutdown and substantial lost production
- Reduction in the likelihood of a recall scenario due to enactment of consistent quality procedures, such as periodic sample and test operations. A recall scenario not only impacts cost of scrapped product that must be removed from the value stream and destroyed, but also impacts the

company's reputation and could lead to consumer avoidance of the company's products depending upon the severity.

- Containment of a quality issue or recall scenario due to availability of material genealogy data. For example, perhaps a specific raw material lot was the source of a quality issue. The MES allows data to be quickly obtained concerning all finished goods leveraging that raw material lot.
- Enforcement of operations that promote safety, such as preventing operators from performing tasks for which they lack qualifications or training. Avoidance of operator injury or death prevents lawsuits as well as subsequent loss of reputation.
- Reduction in the likelihood of a cybersecurity issue, infrastructure obsolescence, or a support issue that could make legacy systems inoperable and halt operations for an indeterminate time period.

Calamity avoidance ROI is especially critical when dealing with life sciences MES initiatives. These regulated industries require a full audit trail of both system and operator actions, an electronic record of batch production, and other aspects promoting traceability. Lack of this information may make the enterprise more susceptible to FDA audit, potentially requiring costly remediations to be enacted in a quick manner, and forcing the company to halt production until such actions are taken.

The customer may be able to cite previous incidents and the associated impact when assigning tangible dollars to calamity avoidance ROI. Or, the customer may have already performed simulated studies concerning the impact of various recall or safety scenarios. The customer may also need to use statistics and historical data to predict the likelihood of an incident over any given period. For example, the customer may predict that a major recall is likely to occur every 10 years, with a 10 percent chance of a recall in any given year.

Note that even a conservatively low estimate of the potential cost savings by incident avoidance is often enough to justify the money spent on the MES initiative.

ROI Documentation Practices:

ROI should be tied back to specific business process scenarios and business intentions. If a group of business process scenarios must be implemented in parallel to lead to the ROI, this should be documented so that the customer understands that they must implement all of the related business process scenarios before the ROI will be obtained. A description of the recommended calculation should be provided, allowing the customer to define specific numbers based upon their business. In some cases the customer may not want to disclose the specific numbers to RoviSys, which is okay. What is of utmost importance is that the customer feels confident that the solution proposed will positively impact their bottom line to the extent that proceeding with the MES initiative is justified.

ROI Payback Period:

The payback period is the duration in which the MES installation and associated services will be paid for, and after which all cost savings result in profit increase. When prioritizing business process scenarios to be implemented, the ROI payback period should be considered to identify functionality that provides the greatest benefit in the shortest timeframe. Finance should again be consulted to determine the company's priorities. For example, it may be more desirable to implement functionality that yields an immediate \$500K savings than one that will save the company \$1MM after three years of runtime.

The MES software licensing model must also be considered when determining ROI. A perpetual licensing model will require a greater up-front expenditure, but once 'paid off' by the ROI over some timespan, all additional savings are pure profit. A subscription-based system must provide cost savings greater than the subscription cost over the subscription period in question to be financially viable, and uphold this savings over the lifespan of the MES.

Reference Documentation:

The following spreadsheet contains a reference of typical drivers for an MES implementation, along with associated tangible and calamity avoidance ROI examples.

<https://rovi.sharepoint.com/:x:/r/sites/mes-dsc-project-execution/Shared%20Documents/General%20Knowledge%20Resources/LMS%20Training/Assessment%20Process/MES%20Driver%20Matrix.xlsx?d=w0f8a80e91f674a3294cf7aeb0cf11196&csf=1&web=1&e=VP4nKu>

(Published) MES RIS 102.8.1 - Functionality Baseline

Monday, March 11, 2024 11:16 AM

Please Reference:
[Functionality Baseline](#)

This stage of quotation helps categorize what is provided 'out-of-the-box' with minimal investment (including utilization of solution accelerators that may already be possessed by RoviSys), in contrast to deltas that will require more extensive development. This exercise establishes a 'menu' for the customer so that they understand the tradeoffs inherent in providing a minimum viable product vs. a solution that will provide a fuller return on investment (ROI) or otherwise will be tailored to fit their specific desires.

In some cases, the customer may have explicit interest in understanding where customizations and additional development effort will be required and what specific impacts these aspects have on project cost. In these scenarios it is helpful to show the customer the functionality baseline quotation and the optional extensions / customizations as cost adders. In other cases it may not be appropriate to show this information to the customer (for example if usage of a baseline solution is not practical or if RoviSys is not being given the opportunity to contrast the two options in an RFP response scenario). In these cases it is still worthwhile to leverage the baseline quotation as a starting point, then add required customizations onto the baseline quotation cost.

A repository of generic baseline quotations is required to draw from to establish the baseline for the situation in question. Each generic baseline quotation will capture the platform in question, any accelerators or reuse components to be leveraged, a description of the functionality provided, a listing of activities to be done configure the functionality and scale out the solution, and approximate time requirements as the magnitude of the deployment scale increases. For example, deployment of the Ignition OEE accelerator will necessitate configuration of downtime reason trees, which will require a varying time duration based upon the number of units / machines present in the facility.

The current baseline functionality quotation template is located within the following location:

[Base Functionality Cost Template.xlsx](#)

The repository of pre-populated templates are located in the following directory:

[Baseline Functionality Quotations](#)

Templates should be moved into the 'Submitted' subdirectory when created, and then moved to the 'Approved' subdirectory when approved by a member of MES management.

This template workbook contains the following tabs/sections:

Overview

- This section provides background concerning the baseline functionality quotation, including the revision date, MES platform in question (such as GE Proficy, TrakSys, etc.), the platform version, a description of the functionality, and references to pre-built accelerators that may be required, including a description as to how they will be leveraged.
- Additionally, this section provides the ability to track a revision history, as well as detailed concerning the author, approver, and date approved. All base functionality templates should be

approved by a member of MES division management and the approval date captured. An approval date that is older than one year from the current date should be re-approved by a member of the MES division management team before being leveraged.

Platform Licensing

- This section provides details concerning any platform licensing that will be required to support the solution in question, including a description of the various modules and units required. In this context a 'unit' is a generic representation of any licensing aspect that will need to scale. This could include user counts, production lines / equipment, production events, etc. The estimated cost per unit is presented, rolling up into the total cost for the module in question. These numbers can be used as budgetary guidance but should always be verified with the appropriate current vendor pricing lists or sales personnel.

Base Accelerator Cost

- In some cases RoviSys may choose to sell a base accelerator module at a given cost to recoup initial development and ongoing maintenance expenses. Any such costs are captured in this tab, including a description of the accelerator module in question, the number of units, cost per unit, and resulting total cost per module. Similar to the above, a 'unit' is a generic representation of any licensing aspect that will need to scale.

Services Cost

- This section captures all services that would be expected to realize deployment of the functionality to a manufacturing facility. The rate structure should be updated based upon current RoviSys rates or the rates agreed to with the customer.
- Hours are captured by role on a 'per unit' basis, including architect, lead, and test-level resources, as well as corresponding project management time. Similar to the above, a 'unit' is a generic representation of any services aspect that will need to scale.
- Tasks are split according to several overall buckets, including configuration activities (such as master data input, CI/CD pipeline management, etc.), testing activities (such as FAT, UAT, integration testing, etc.) and deployment activities (such as deployment plan definition and solution deployment to the facility), and associated organizational change management (OCM) activities, such as training support, workstation configuration, etc.
- Typically project management time is determined by applying a multiplier to the overall project hours. This sheet allows the multiplier to be input, and project management time will be calculated accordingly.
- The sheet will multiply the per unit estimates by the number of units input to arrive at a total hour count and associated cost estimate for the baseline functionality realization.

Assumptions

- This section lists any assumptions concerning the functionality that should be considered, including supported functionality, functionality not supported, prerequisite activities required, and other aspects that should be considered by the RoviSys sales individual when applying the base functionality cost template to a specific situation.

(Published) MES RIS 102.8.2 - Budgetary Establishment

Monday, March 11, 2024 11:16 AM

Please Reference:

[Budgetary Establishment](#)

During this phase, the budgetary constraints concerning execution of some portion of the envisioned roadmap is established. A small-scale initiative can likely be quoted via a detailed task breakdown, as the scope items are clear and the implementation effort can be envisioned by a solution architect.

Framework Agreement:

However, more commonly, there will still be unknowns concerning the effort as a detailed design has not been completed. Further, a desire to maintain agility means that there must be some room to flex scope throughout the effort as user stories are defined in detail and technical design aspects are clarified based on the user stories. Lastly, a customer may be hesitant to award RoviSys the full scope for a multi-year and/or multi-site initiative up-front due to the size of the effort and lack of scope clarity.

Based on these aspects, it is recommended that we work with the customer to establish a framework agreement, consisting of an overall budget, milestones to be achieved throughout roadmap execution, and a budgetary estimate for each phase of the roadmap. The framework agreement represents a commitment by the customer to proceed with the initiative in full, utilizing RoviSys as the implementation partner throughout the effort, provided the formulated business intentions are being met and the initiative is generally meeting the schedule and budget established.

Using this information, as well as the results of any ROI analysis performed jointly by RoviSys and the customer, the customer is able to establish a budget for the initiative and seek approval of this budget from higher-level leadership.

The following aspects should be included within the framework agreement proposal:

- Initiative success criteria
 - Reference to the business intentions established during the assessment effort
 - Reference to the roadmap established during the assessment effort
 - *Note that if a formal assessment effort was not conducted prior to creation of the framework agreement, business intentions and a roadmap must still be formulated, leveraging information provided by the customer or informal findings captured during project discussions*
- Details concerning the extent of roadmap execution covered by the framework agreement
 - Reference to the business process scenarios to be completed during each phase of the roadmap, based upon the scenarios and roadmap formulated during the assessment effort
 - *Note that if a formal assessment effort was not conducted prior to creation of the framework agreement, business process scenarios must still be formulated, leveraging information provided by the customer or informal findings captured during project discussions*
- Estimated budget for completion of each phase of the roadmap, including the following aspects
 - User story formulation and resulting technical design
 - Implementation
 - Any existing functional baselines applicable to this scope can be applied and built upon, per [MES RAP 102.1](#)

- Internal and customer acceptance testing phases
- Deployment
- Hypercare and post-deployment support
- Associated expenses
 - If specific durations onsite are unknown, often a percentage of the total scope can be calculated for expenses. 15% is a common number but this should be adjusted based upon the relative amount of time envisioned to be onsite.
 - Make it known to the customer that expenses are always invoiced at RoviSys cost and expense costs are rough estimates at this time.
- OPE
 - Typical OPE for a digitalization initiative will consist of platform licensing costs
 - Consider if the pricing model is perpetual or subscription-based and factor cost accordingly. Assume that a subscription-based cost will likely increase by some factor (such as 5%) throughout a multi-year engagement
 - Include any recurring support fees throughout the lifespan of the initiative, particularly in the case of a perpetual pricing model
 - Include RoviSys margin if the licensing is to be procured by RoviSys. Typical OPE margin is 15%.
 - Consider any discounts available to RoviSys if RoviSys is procuring the licensing. For example, if RoviSys is provided a 10% discount when purchasing from the vendor, it is appropriate to add 5% for our OPE margin when quoting a cost to the customer.
- External resource needs
 - RoviSys may desire to contract in an external resource, such as a vendor solution architect, to provide guidance / oversight / review or to help bolster the internal project team.
- Consider that when estimating a multi-year initiative, RoviSys resource and external vendor rates will likely increase each year.
- Process for consumption of funds from this budget by RoviSys
 - Description of this workflow, including detailed change request proposal to be authored by RoviSys, required customer approvals and corresponding approval time requirements, and PO issuance
- Project status meeting and governance meeting cadences
 - Project status reporting deliverables to be provided and reviewed during these meetings
 - General agendas for these meetings
- Delineation between RoviSys and customer roles, including roles RoviSys will make available and customer role expectations

Establishment of budgetary numbers for the framework agreement often requires formulating a resource loading against the proposed project timeline. This strategy is detailed in [MES RAP 102.3](#).

It is not expected that the customer will issue a PO to RoviSys for the full amount budgeted in the framework agreement. However, the customer should seek internal approval for the amount indicated. Signing the framework agreement constitutes an agreement by the customer to proceed with utilization of RoviSys to move forward with the digitalization initiative, although this signature does not constitute a legally binding agreement to execute all aspects of the initiative nor does it require the customer to pay RoviSys for the amount in full. It should be clear in the framework agreement that the figures provided are budgetary estimates that should be used solely by the customer for internal budgeting purposes.

Detailed Proposal:

A detailed proposal will need to be issued under two scenarios:

- The project consists of a well-defined scope with minimal project phasing requirements
- A specific scope is being 'carved out' of the framework agreement for execution

The detailed proposal should consist of either a fixed-price or time and materials (not to exceed) estimate to achieve the scope. It must be clear in the proposal what scope is being executed by RoviSys, and any assumptions or items not in scope must also be clarified.

Creation of a work breakdown structure (WBS) is often required to properly visualize all tasks that will be required. This should consider not only implementation, but also the following general aspects:

- Infrastructure establishment, as well as supporting CI/CD processes (Dev / QA / Production instances, deployment pipelines, source control repositories, automated testing frameworks, etc.)
- User story definition and resulting technical design
- Configuration
- Development
- Master data definition, verification, and loading cycles
- Unit testing
- Organizational change management activities, such as training plans / materials creation
- Internal holistic testing (regression, performance)
- Customer-facing testing (FAT, UAT, integration testing)
- Deployment planning creation
- Deployment
- Post-deployment 'hypercare' support
- Technical team handoff
- Project meeting cadences (status meetings, design and build progress reviews, governance meetings, etc.)

Any existing functional baselines applicable to this scope can be applied and built upon, per [MES RAP 102.1](#).

Project management should also be incorporated. A typical PM time allocation is 10% of the project scope, although a higher PM engagement percentage may be required in the case of complex scope, need for coordination across multiple internal / external teams, or a demanding customer. A project coordinator role may also be appropriate based on the size of the initiative.

Project expenses must be included based upon envisioned onsite travel. Remember that in the case of multiple resources being onsite, expenses must be included for each resource. For example, during a period of 24x7 onsite coverage, one resource cannot cover 24 hours / day. As a result, multiple resources and associated travel expenses for those resources must be allocated. Also, consider whether resources will be traveling together or separately, flying or driving, etc. The RoviSys proposal template expenses estimation tab (available via Excel in the RoviSys Sales templates folder) is useful in estimating expenses. Be sure to edit the top of this tab based upon travel distance, airfare costs, hotel costs in the area in question, and other factors.

Any OPE required should be finalized and quoted in a more concrete manner based upon needed licensing. In some cases base licensing may have already been purchased, but the specific project phase may necessitate expansion of the licensing to support additional features or expansion across the facility or into subsequent facilities.

Lastly, external resource needs (such as vendor architect support or external contractors) that can be more exactly envisioned and estimated as part of the detailed proposal phase should be included.

Of equal if not greater importance to project pricing is inclusion of the project execution strategy into the proposal. This is often RoviSys' differentiator when executing MES engagements. Key aspects of the project execution strategy to be included include the following:

- Envisioned project schedule, including key milestones/sprints, deliverables, and onsite periods
- Required RoviSys and customer resources, including envisioned loading throughout the duration
- Prerequisite needs from the customer (such as infrastructure, training team support, etc.)
- Key risks and envisioned mitigations
- Other assumptions factoring into the estimate

A proposal checklist has been created that should be leveraged when creating a detailed proposal to ensure that various considerations have been factored into the estimate. This checklist is located in the following location:

[A4M Project Estimation Checklist.xlsx](#)

Ensure that the detailed proposal is provided in the customer far enough in advance to ensure that customer review and approval cycles can occur without overall program disruption, specifically in the case where multiple proposals are being 'carved out' as execution throughout a multiple phase initiative progresses. This ensures that project momentum is maintained and the project team can be kept intact.

(Published) MES RIS 102.8.3 - Resource-Based Quotation Methodology

Monday, March 11, 2024 11:19 AM

Please Reference:
[Resource-Based Quotation Methodology](#)

In some cases, a rough schedule and a listing of desired scope items may exist, but there is not enough task granularity to formulate a task-based quotation. It may not be feasible to generate all the various tasks due to lack of clarity and/or lack of time. In these cases, a strategy to give the customer an approximate budgetary estimate is to formulate a quotation based upon resource loading.

Even if a task-based estimate is being created, formulation of a resource-based model can be a useful 'cross-check' to ensure that the task estimates are realistic. Task-based estimates tend to be light when one considers the full team that will be required to support the project over the course of the project's lifecycle. A resource-based model may highlight areas that need to be bolstered.

To formulate this style of quotation, first the calendar schedule is laid out (often within an Excel sheet). Calendar time may be represented generically (month one, month two, etc.) if the start date is unclear, or a starting date established and the calendar built upon this.

Next, the calendar time within the schedule is split up according to envisioned project phases. Key phases that may be represented include the following examples:

- Assessment
- Project planning
- User story definition
- Sprint execution (x number of sprints envisioned)
- Sprint testing (x number of sprints envisioned)
- Sprint deployment
- Hypercare support

The duration of a specific phase is approximated based upon the customer's schedule constraints and our experience in the time required to complete given tasks. The customer may have a target go-live date aligned to a site startup, supporting system (such as ERP) cutover, or other factors. In these cases, one can work backward from the go-live date to slot in the various activities above.

Some timeline approximations are as follows:

- A full assessment effort, including site visit, business process scenario definition, technology comparison, architecture formulation, roadmap formulation, and executive summary, typically requires between 6 and 8 weeks, although this time period can be compressed.
- Detailed project planning is often a 2-week effort
- User story definition may typically take between 1 and 2 weeks prior to the kickoff of each sprints
 - Definition of initial user stories must be done before the commencement of the first sprint. Thereafter, user story definition for the next sprint can overlap with execution and testing of the current sprint. However, in the event of this overlap, sufficient architect and lead-level resources must be available to drive both the current and next sprint.
 - There may be a desire to sequence user story development and sprint build/test in a sequential manner with no overlap. However, consideration must be given in this scenario to ensure that the project developers have value-add tasks during user story development. Otherwise, it is impractical to ramp the development team on and off the project to keep them busy during user story definition.
- Sprint execution is often 4-6 weeks in duration, although this is dependent upon the complexity of the sprint. It is at the estimator's discretion based on project scope and experience whether a larger number of shorter sprints (2 weeks for example) or a smaller number of longer sprints are better.
- Sprint testing is often 2 weeks in duration.
- A tight schedule may require sprints to overlap with multiple workstreams involved. This will in turn require additional resources to be allocated to support the overlapping sprints.
- For a large-scale MES rollout, execution of the FAT and UAT activities for a full MES deployment, including remediation, may last between four and six weeks. For example, one week for FAT, one week to resolve issues from FAT, one week for UAT, and one week to resolve issues from UAT.
- Other testing periods may also be required, including regression testing, integration testing, performance testing, simulation testing, and other aspects.
- Ample time should be allocated for deployment preparation and deployment activities
- Go-live and post deployment hypercare support periods must be accounted for.
 - A typical hypercare period may be somewhere between three and four weeks, with a gradual taper from around the clock onsite coverage to a remote as-needed model.
 - Not only should the customer-facing onsite support resources be allocated, but also offsite resources who can help aid the support team in the resolution of issues, implementation of code fixes or enhancements, and other matters.
- Concerning a multi-site deployment, in theory the first site to be deployed should align with development of a global template that encompasses a large degree of functionality that will be required for subsequent sites. Thereafter, subsequent site implementations should be shorter in duration, largely requiring delta functionality implementation that is subsequently rolled into the global template, master data definition / import, and deployment activities.

Once the project execution activities, phases, and approximate durations are laid out, a resource loading can be formulated. First, the resources who must be involved in the effort must be identified. This typically goes beyond technical leads and developers, depending on the project magnitude. However, it is likely that not every resource should be staffed 100% throughout the full duration of the project. Rather, resource loading will rise and fall based upon the project phase and even specific period within the phase.

Resources often allocated and relative periods of high or low loading include the following:

- Project Manager
 - Most heavily loaded at the beginning of the engagement to aid in project organization, creation of the project execution and quality plans, schedule formulation, and other initial project planning activities. Also prepares kickoff meeting agenda and leads kickoff meetings.
 - Also often heavily involved during deployment and 'hyper care' support periods, aiding in the capture and clarification of observations / user issues, triaging of these issues, generation of change requests, etc.
- Project Coordinator

- The project coordinator may assist the project manager, particularly on large engagements, with schedule management, budgetary tracking, issue tracking, etc.
- **Solution Architect**
 - The solution architect is typically most heavily engaged toward the beginning of the project as overall architectures, interface specifications, and other high-level design activities are completed. The solution architect then typically takes a smaller supporting role.
 - On smaller-scale initiatives the solution architect and technical lead may be fulfilled by the same role, or a shared solution architect may be staffed on the project in a limited capacity to facilitate oversight and verification of design / architecture.
- **Functional Analyst**
 - The functional analyst is involved in the beginning of the project to facilitate definition of business process scenarios. The functional analyst is also heavily involved in user story refinement at the beginning of each sprint. In between, the functional analyst typically acts in a supporting role. This resource will also likely be heavily involved with post-deployment user support activities.
- **Technical Lead**
 - This resource is heavily involved throughout the project lifecycle to aid in translation of user stories to technical design and guiding / supporting the development team. Multiple technical leads may be required to support an aggressive schedule or multiple workstreams in the case of complex initiatives.
- **Developer**
 - Developers may not be able to fully engage in the project until the implementation portion of each sprint. Developers are often engaged 100% through these periods, although a developer could be partially allocated in the case of scope that may not require a full-time resource. Developers often transition to full-time support resources during hyper care support periods. Additionally, it is often useful to maintain some level of developer staffing throughout testing periods (FAT, UAT, etc.) so that any issues or changes identified can be quickly resolved.
- **Master Data Lead**
 - The master data lead may be engaged early in the project and throughout sprint user story development activities to define master data needs, collection strategies, and templates of value. The master data lead is also often engaged during deployment activities to perform master data loads, and during post-deployment support periods to aid in master data corrections and supplemental loads. This role is most applicable to large-scale initiatives that span large manufacturing facilities.
 - On smaller-scale initiatives the technical lead may also serve as the master data lead / resource.
- **Master Data Resource**
 - The master data resource may be engaged throughout configuration periods and during deployment to help collect, cleanse / verify, and import master data.
- **Organizational Change Management (OCM) Lead**
 - This resource aids in the definition and coordination of organizational change management activities. This resource may be involved with test strategy formulation, training strategy formulation and training material creation, training of the customer's training team (typically RoviSys tries to avoid being required to train end users), and determining needed hardware / user experiences on the shop floor (tablets, thin clients, PCs, scanners, etc.)
 - The resource is often engaged early in the project to aid in formulation of OCM plans / strategies, throughout the tail end of each sprint to verify testing completion, and prior to deployment to aid in training documentation creation and 'train the trainers' activity execution.
 - On smaller-scale initiatives the functional analyst may also serve the OCM lead / resource.
- **OCM Resource**
 - On larger-scale initiatives, additional OCM resource(s) may be required to assist the OCM lead in documentation and other activities.
- **Deployment Lead (particularly in cases where deployments are frequent and require substantial coordination)**
 - An initiative may require a dedicated deployment lead to coordinate deployment activities, formulate required deployment plans, and ensure that deployments are completed successfully. This resource may assist early in the project to formulate deployment strategies, at the tail end of each sprint to aid in the deployment of functionality to a QA or testing environment, and prior to go-live to coordinate production deployment readiness verifications and production deployment activities.
 - This resource may also aid in the rollback of delta functionality implemented for subsequent sites as part of a global rollout to the global template and coordination of updates to sites that may have already gone live, ensuring that all sites possess the same code base.
- **Process Lead (similar to a functional analyst in scope, a process lead can serve as an extension of the customer to aid in the technical team's understanding of shop floor processes and workflows)**
 - This role is most prevalent on large-scale initiatives, and/or when the customer's manufacturing processes are extremely complex.

Once resource allocations have been made, the total resource duration can be captured by summing the allocations. This sum can be multiplied by the rate being used for that resource to come up with a total cost for that resource over the duration. Each cost can then be summed to calculate the total services cost.

Expenses calculation:

Expenses that will be incurred throughout the project's lifecycle must also be allocated. One strategy to perform this estimation is to capture the number of resources and associated days anticipated to be onsite under the resource loading matrix. Expense cost per day can be calculated based on hotel, food, and other incidental expense estimations. Airfare and travel time can also be captured and calculated based on a sum of the number of trips anticipated. It is best to assume some expense buffer as expenses are typically invoiced as incurred.

Key periods of travel / onsite presence will depend on the nature of the engagement. Periods of travel often allocated include the following:

- Onsite customer kickoff meeting
- Site tour and assessment workshop
- Master data definition workshop
- Key milestone build review sessions (for example, 25% review, 50% review, etc.)
- Project governance reviews
- Operational readiness planning activities (equipment placement and installation coordination, etc.)
- Factory acceptance testing (FAT)
- User acceptance testing (UAT)
- Deployment readiness checklist execution
- Deployment execution
- Hypercare support

The following is an example of an allocation spreadsheet, showing how roles, durations, services costs, and expenses were calculated:

		2024																							
		March			April				May				June				July				August				
		18	25	01	08	15	22	29	06	13	20	27	03	10	17	24	01	08	15	22	29	05	12	19	26
Project Planning			Implementation Sprints												Factory and User Acceptance Testing				Deployment and Support						
Program Manager	90%	90%	90%	90%	90%	90%	90%	90%	90%	90%	90%	90%	90%	90%	90%	90%	90%	90%	90%	90%	90%	10%	10%		
Project Coordinator	25%	25%	25%	25%	25%	25%	25%	25%	25%	25%	25%	25%	25%	25%	25%	25%	25%	25%	25%	25%	25%	15%	15%		
Functional Analyst Lead	50%	50%	50%	50%	50%	25%	0%	0%	25%	25%	0%	0%	25%	25%	0%	0%	50%	50%	100%	100%	100%	25%	25%		
OCM Lead	50%	50%	50%	50%	25%	25%	25%	25%	25%	25%	50%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	25%	25%		
OCM Resource	50%	50%	50%	50%	25%	25%	25%	25%	25%	25%	50%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%		
MES Architect	100%	100%	75%	75%	50%	50%	50%	50%	50%	50%	50%	50%	50%	25%	25%	25%	10%	10%	10%	10%	10%	10%	10%		
MES Lead	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	50%	50%			
MES Developer				100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	50%		
MES Developer				100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	50%	50%		
MES Developer				50%	50%	50%	50%	50%	50%	50%	50%	50%	50%	50%	50%	50%	50%	50%	50%	50%	50%	50%	50%		
Master Data Lead	50%	100%	100%	80%	80%	80%	80%	70%	70%	70%	70%	50%	50%	50%	100%	100%	50%	25%	25%	25%	10%	10%	10%	10%	
		Totals: \$1,119,242 6590																							
Number trips / drivers		12																							
Round trip miles		275																							
Mileage rate		0.67																							
Mileage cost		\$2,211																							
Hotel nights		192																							
Hotel rate		\$175																							
Hotel total:		\$33,600																							
Rental Car Days		56																							
Rental Car Cost		\$75																							
Rental Car Total		\$4,200																							
Food days		162																							
Food cost per day		\$55																							
Food total		\$8,910																							
Travel Hours		144																							
Travel rate		\$125																							
Total		\$18,000																							
Total Expenses		\$81,921																							

(Published) MES RIS 102.9 - Deployment Roadmap Formulation

Wednesday, February 7, 2024 12:58 PM

Please Reference:

[Deployment Roadmap Formulation](#)

The deployment roadmap details how the solution stack is envisioned to be deployed across the area, site, enterprise based on the overall scope of the initiative. This plan helps to establish the required implementation project phases upon which the quotation effort can be based. As a result, this activity naturally progresses to the Spend Allocation Phase.

There is no prescribed 'one size fits all' methodology for solution deployment, as a variety of factors both internal to the selected solution stack and external to the initiative must be considered. However, this document will help to highlight key decision points that can shape the deployment vision.

A goal at this stage is to keep the roadmap simple in nature so that it can be easily understood by various stakeholders within the customer organization. It is not meant to be a detailed project schedule containing all tasks and dependencies. Rather, it is used to establish consensus and foster discussion concerning when various process scenarios will be realized, and when supporting efforts must be executed to ensure success.

Time Horizons:

Often, a solution roadmap is thought of in terms of short-, mid-, and long-term time horizons.

Short-term scope items are the most thoroughly defined and can usually be quoted in a fixed-price or time & materials manner. Short-term items support platform establishment and 'quick-wins' to anchor the solution, begin to deliver value, and lay the foundation for longer-term initiative. Short-term scope items have a clear ROI in terms of replacement of legacy systems and/or tangible ROI with a reasonable payback period.

Mid-term scope items may be more complex in nature with longer lead times and value propositions that are less clear. They likely require prerequisite aspects of the solution to be established and adopted before they can be implemented. They may have longer development lead times due to custom development requirements (extension of a platform or custom solution development). The ROI payback period of these items may be longer in duration. Mid-term scope items may only be able to be quoted in a budgetary manner with a +/- range due to requirement unclarity or unknown dependencies.

Long-term scope items are the most complex in nature and/or have the longest lead times. The value proposition of these items may be difficult to define but extremely impactful if achieved. These items are often considered 'stretch goals' utilizing advanced technological concepts that may not yet be fully accessible (new features not yet incorporated into a platform, technology not yet commercially available at a reasonable cost, etc.). Examples of long-term scope items may include leveraging of AI technologies, incorporation of advanced functionality such as vision systems, robotics, etc., digital twin simulation, virtual reality, and other technologies.

The long term time horizon helps to establish a vision that can help build momentum for the building blocks required to achieve such a state. These items typically can only be quoted by providing a rough

order of magnitude (ROM) estimate establishing a rough budgetary ballpark.

Mid- and long-term scope items can be fluid in nature as the initiative progresses. Often, unforeseen value is provided in early phases that negates the need for the envisioned functionality, or new priorities come to light that were not thought of at the time of roadmap development. As a result, the roadmap should be revisited periodically throughout the initiative, and should be a review topic during periodic governance reviews.

Internal Dependencies:

The roadmap must allocate proper time to designing, implementing, testing, and deploying the functionality in question. A detailed schedule is not required to determine this, but high-level timing estimates will need to be developed based upon the complexity of the envisioned scope items.

Prerequisite tasks will need to be considered. It may not be possible to provide accurate OEE visualizations until schedule execution functionality is deployed, so that target production counts and product information is known for the run for example. Automating quality procedures may require development of an external interface to quality instrumentation.

Hardware lead times may also be a factor. This includes servers, scanners, tablets, sensors, vision systems, etc.

External Dependencies:

The digitalization roadmap cannot be considered in isolation - external dependencies must be factored into the plan. For example, often MES and ERP initiatives are executed in parallel. The ERP schedule must align to the MES schedule so that needed ERP functions are available when testing / deploying the MES, and vice versa. This also applies to any other parallel initiatives (control system upgrades, quality system deployment, etc.).

Customer team availability should also be considered. If the customer has a small support team or needs time to scale the team, this could impact the deployment timeline. Other external activities such as training time may also impact the timeline.

Decoupling the Magnitude of Change:

A useful strategy when formulating a digitalization roadmap is decoupling large system deployments to reduce the amount of change that must be managed by operations at one time, as well as reduce the size of the team that will be required to support deployment and hypercare activities. In some instances employing these strategies may increase the scope of the implementation or require later re-work, but this is outweighed by the change management benefits.

An example of deploying this strategy is employing a gradual deployment model where functionality is deployed to subsets of the site footprint sequentially. For example, the solution may be deployed to the batching area first, then the packaging area. Or the solution could be deployed across production lines in a staggered manner. However, care must be taken to ensure that these sorts of methodologies do not require operators, supervisors, etc. to 'bounce' between systems as they move from line to line or area to area, which could cause undue confusion and lead to mistakes being made.

Another strategy involves gradual deployment of functionality. Production management may be deployed first, then quality, then efficiency for example. This strategy can be especially beneficial if for example the enterprise relies upon delineated 'point solutions' that can be gradually decommissioned. For example, the production execution functionality could be deployed while the customer continues to

leverage a legacy quality system, then eventually that quality system be replaced by a deployed module within the MES platform.

Yet another strategy with integrated MES and ERP deployments is to decouple the MES deployment from the ERP deployment. For example, it may be possible to extend the MES to communicate with legacy ERP systems, deploy the MES first and anchor it, then deploy the ERP and cut over the MES interfaces. This strategy will require coordination with other customer teams to ensure such a model is achievable, and may require extra work to create additional interfaces to legacy systems, but can have huge benefits in terms of site acceptance.

Greenfield Deployment Considerations:

A greenfield deployment presents unique challenges to roadmap formulation, as there will typically be a base level of functionality that *must* be delivered to support the startup of the facility. It may not be possible to deliver the full scope of the solution on day 1 due to timing constraints, so functionality may need to be prioritized based on immediate vs. longer-term business impact. In these scenarios, the hierarchy of required functions often aligns to the following:

- Work instructions and procedures: New employees and new equipment will require proper work instructions / workflows to be established early so that plant processes are quickly learned and understood. With a greenfield deployment, procedures may change frequently so it is important to make this functionality data-driven in nature with the ability to be quickly updated and re-deployed.
- Production: functions supporting the execution of the production schedule, tracking of order progress, materials used (genealogy), and finished goods output. This information may be received from and provided to the ERP layer to support customer order tracking and material requirements planning (MRP) activities.
- Quality: functions supporting the confirmation of product quality, including sample and test execution, non-conformance workflows, and product disposition. There may be manual processes that can be enacted in the short term to track quality as a stop-gap, but digitalization of this functionality will still be a high-priority. In regulated industries such as life sciences, quality may actually be a higher-priority than production tracking.
- Efficiency: functions supporting efficiency tracking and reporting, such as equipment utilization, overall equipment effectiveness (OEE), etc. Greenfield plants usually must be well established and a basis of data captured before meaningful efficiency insights can be acted upon.
- Maintenance Optimization: A new facility with new equipment will generally perform in a more reliable manner, reducing the need for maintenance management and maintenance process optimization in the short term.
- Energy Management: A new facility with new equipment will likely utilize energy more efficiently than legacy sites. As a result, insights from energy utilization information may not be as important in the short term.

Roadmap Visualization:

The roadmap should be presented in a manner that is easily digestible to our customer contacts, their peers, and those at higher levels in the organization. A PowerPoint slide presenting a timeline view is a good communication vehicle.

Chevron bars can be leveraged to present the duration of a given phase so that it is clear that phases may blend and we are not committing to specific start or end dates, for example the following:



Explicit details must also be included after presentation of the high-level visual.

Enterprise Multiple Site Sequencing:

There are certain factors to balance when defining a multi-site deployment roadmap. Customers will often want our recommendation as to the sequence in which site deployments should occur. The pros and cons of deploying to certain sites early or late in the roadmap should be captured. Some factors for consideration include the following:

- Some sites are more open/tolerant to change than others. This can be due to the site culture, management, workforce age, history of the site, location, and other factors. It may be beneficial to deploy to sites more open to change first, to gain momentum, secure quicker ROI, and groom advocates for the solution that can help win over more reluctant sites.
- Some sites are more ready for the solution than others. Sites possessing modern infrastructure and control system automation may more easily be able to support deployment of the solution without costly infrastructure / controls upgrades. It may be beneficial to deploy to these sites first while in parallel performing upgrades to less modern facilities.
 - o However, this should be balanced with the fact that more ROI may be achievable in less modern sites.
- Smaller sites may support faster deployments with less overall risk to the organization. Deployment to smaller sites early in the roadmap can build momentum and prove out solution functionality with minimal negative impact. However, the ROI for small site deployments may not be as great as that for larger sites, so these two factors should be balanced. It may make sense to deploy to a small site, and then a large site to take advantage of both derisking and ROI attainment factors early in the roadmap.
- It may be beneficial to deploy to sites near an established RoviSys office and/or customer HQ early in the roadmap. Sites early in the deployment roadmap will likely require a larger support structure, so the ability for RoviSys and customer resources to travel to the site expediently could pay dividends.
- Sites with similar equipment and/or work processes could be clustered together to support a more aggressive deployment schedule. For example, all of the type 'A' plants could be deployed sequentially while providing time to develop functionality for all the type 'B' plants as these deployments progress.
- Some sites may have a stronger need for the solution than others. Maybe a given site has a legacy system that is no longer maintainable or is utilizing inefficient paper-based processes. Or, the site manufactures a high-cost product with a high-scrap rate than needs to be lowered. Sites yielding the greatest potential for ROI should be considered early in the roadmap to help justify the initiative.

Manpower Loading:

The distribution of manpower across both the internal RoviSys and external customer project teams must also be considered during roadmap formulation. For example, it may not be feasible to deploy to several sites concurrently if the customer possesses a small training or support team and has not contracted a larger RoviSys team to compensate. In another scenario, a small RoviSys team consisting of a couple individuals may not be able to cover an entire site deployment, requiring an area-by-area deployment model to be put forth instead.

(Published) MES RIS 102.10 - Executive Summary

Wednesday, February 7, 2024 12:58 PM

Please Reference: [Executive Summary](#)

The executive summary presentation distills the activities, key findings, and recommendations into a format that can be digested by broader customer stakeholders including an executive leadership team or steering committee. It should present a compelling plan that highlights the benefits of the initiative, roadmap, and keys to success.

Presentation Components:

The following are the topics typical of an executive summary presentation

Agenda:

The agenda should be presented at the beginning of the presentation so that the group at hand understands what will be covered during the discussion.

Introduction:

Key RoviSys assessment team members, their roles, and credentials should be captured and presented.

RoviSys Background:

Customer stakeholders unfamiliar with RoviSys will likely benefit from a brief overview of our size, locations, and capabilities. This introduction should be quick and to the point, allowing for ample time to cover the logistics and findings within the assessment itself.

DxArchitecture Activity Overview:

This section should present an overview of the assessment activities and overall timeline involved.

Key Individuals and Sites:

This section should present the key individuals from which information was solicited or who reviewed and approved various aspects of the assessment findings. In the case of a global enterprise-centric assessment, the representative site(s) of focus and reasons for selection of those sites as focuses should be discussed.

Capabilities of Focus:

Summarization of a standard capability map and presentation of the capabilities of focus is useful to lay the foundation for the focus of the assessment, including functional capabilities in scope and out of scope.

Key Findings:

The customer stakeholders for whom the presented is targeted likely understands the manufacturing and associated operational processes at a high-level. However, key findings should be summarized, including gaps, concerns, and challenges pertaining to current operational workflows.

Business Process Scenario Summary:

The executive summary review session will likely not allow enough time for each business process scenario to be reviewed in full detail. However, a summary of the process scenarios documented can be presented to provide the audience a sense of the scope of the investigation and functionality to be

provided by the solution.

Technology Selection Summary:

This section presents a summary of the technology options shortlisted, key pros / cons of each, and describes the reasoning for the selection of the technology solution(s) ultimately decided upon.

ROI Summary:

This section summarizes key ways in which the solution is anticipated to provide a return on investment (ROI). RoviSys will likely rely upon the customer project team to aid in providing specific numbers to be included within the presentation, using the guidance provided by RoviSys during the ROI guidance formulation phase.

Deployment Roadmap:

The deployment roadmap should be presented at a high-level so that the team understands the rough timing of the initiative, sequence of facility go-lives, and other logistics. The team may also want a summary of capital expenditure and manpower needs over the timeline for high-level planning purposes.

Next Steps:

The expected next steps should be clearly articulated. Next steps may include a proposal for the initial roadmap phase(s), capital allocation for the initiative, release of an RFP, or other potential paths forward. RoviSys should coordinate with the project team to recommend next steps that are believed to be agreeable to the presentation audience prior to giving the presentation, so that there is little chance of surprise.

Questions:

Time should be allocated at the end of the presentation for questions and follow-up actions.

Presentation Roles:

In addition to the primary presenter, a note taker should be assigned who can document any questions, key points raised, or follow-up actions. The note taker should be prepared to summarize these points verbally at the end of the presentation session, so that it can be assured that RoviSys captured all follow-ups and nothing was misinterpreted.

Recording:

Permission should be requested at the beginning of the presentation to record the session. This is useful so that the presentation can be shared with any stakeholders who were unable to attend, as well as so that RoviSys can review any points made that require clarification or re-examination.

Pre-Review:

A pre-review of the executive summary presentation should be conducted with the customer project champion or a small subset of the customer stakeholder team prior to wider distribution of the presentation and conducting the presentation in front of a larger customer team. This will ensure alignment with the core team and reduce the chance of any surprises concerning our findings and recommendations. The customer project champion or project team may also be able to call out sensitive topics or better ways of communicating our findings and recommendations to ensure the messaging is better received by the broader customer stakeholder community.

MES RAP 103 - DxExecution

Tuesday, January 9, 2024 9:03 AM

Audience

Management Sponsors, Project Managers, Technical Leads, and Solution Architects

Procedure Overview

The **DxExecution Procedure** outlines our approach for executing digital transformation projects. Its primary goal is to streamline project execution, thereby increasing our ability to consistently deliver high-quality results. By seamlessly blending flexibility with structure, DxExecution enables us to adapt to changes while maintaining defined milestones, budgets, and schedules. Our commitment to transparency and regular communication ensures that every stage of the project is successful and that end users receive the tools they need to work more effectively.

The DxExecution RAP consists of three key phases:

1. Mobilization:

This initial phase lays the foundation for the project. During Mobilization, we develop a detailed work breakdown structure and compile a comprehensive project backlog. This backlog prioritizes features and user stories, which are essential for defining future sprints.

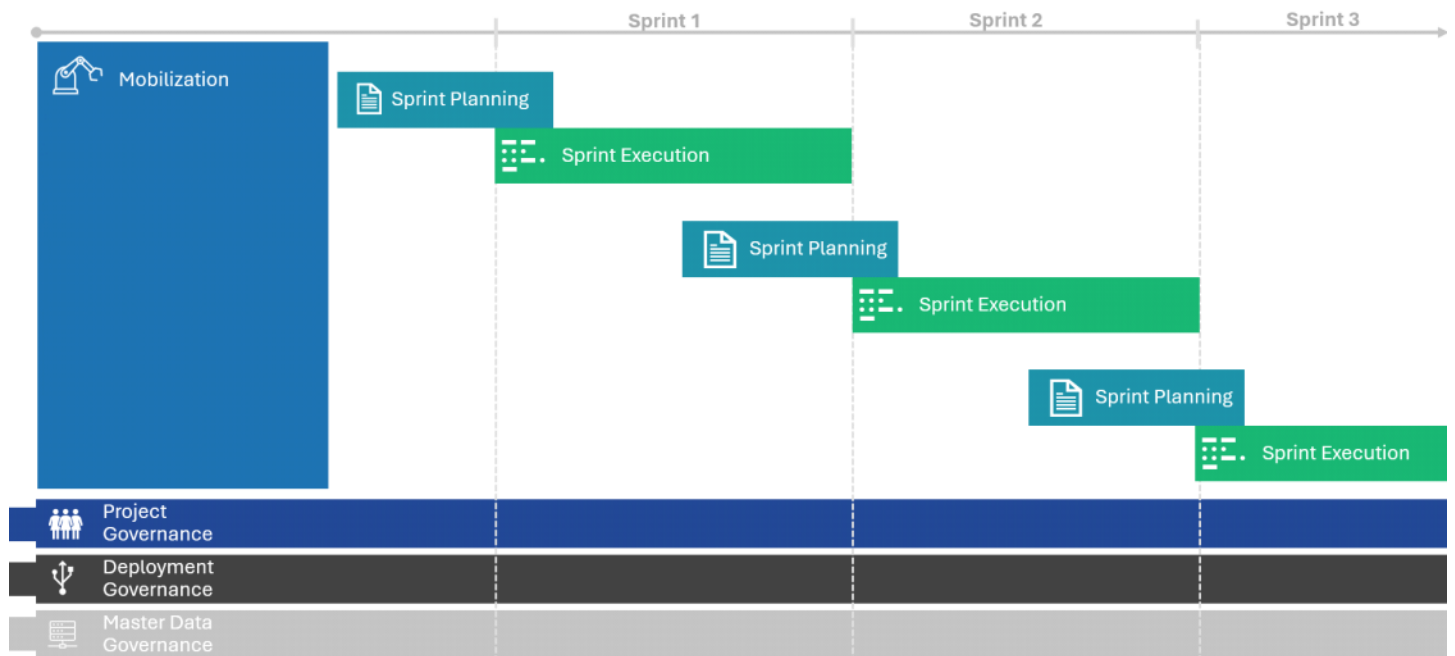
2. Planning:

In the Planning phase, the project backlog is thoroughly evaluated. The team then identifies and organizes the components of each sprint, ensuring that every user story is well-defined and that acceptance criteria are in place for successful implementation.

3. Execution:

During the Execution phase, the development team works on the selected tasks from the sprint backlog to meet the defined goals. Although the Planning and Execution phases are presented sequentially, they are in practice iterative processes that continue until the entire project scope is addressed.





Note: The image above illustrates the sequencing of activities. Please take note of any overlaps.

Procedure Detail

In this section, we offer a comprehensive overview of the RISs featured within each sub-RAP. While the details provided are not exhaustive, our goal is to introduce the purpose and significance of each RIS and to guide you to additional resources for more in-depth information.

Mobilization Sub-RAP

To be a better partner for our customers, our Mobilization procedure is essential for ensuring transparency and providing a clear view of the project scope and deliverables. This approach improves communication between the project team and customers, fostering a better understanding of progress and milestones. By following our procedures and standards, we build customer confidence in our organization and planning, helping to set and manage realistic expectations for timelines and deliverables.

[\[More Detail\]](#)

RIS 103.1.1 – Developer Onboarding

This standard details the creation of an Implementation Team Onboarding Plan. The purpose of the Onboarding Plan is to identify and organize the detailed activities individuals need to complete in order to quickly, efficiently, and sufficiently onboard to a project.

[\[More Detail\]](#)

RIS 103.1.2 – Feature User Stories

This standard is intended to help individuals with a base understanding of Project Management and Agile/SCRUM to understand how we, in the MES division, utilize and create user stories when planning/performing an Agile for Manufacturing project

[\[More Detail\]](#)

RIS 103.1.3 – Work Breakdown

This standard establishes a structured approach to developing an initial Work Breakdown Schedule (WBS) for the project scope. By systematically breaking work down from Features → User Stories → Tasks, teams can enhance the accuracy of delivery timelines, optimize resource allocation, and identify critical interdependencies.

[\[More Detail\]](#)

RIS 103.1.4 – Execution Architecture

The execution architecture is the technical framework used to execute the project. The Technical Lead should outline the technological components required for execution and document them. Documenting the details of how we will execute the project helps in communicating our needs to the customer and assists in onboarding our team.

[\[More Detail\]](#)

RIS 103.1.5 – Execution Schedule

The Execution Schedule standard defines the expectations for creating a structured and reliable project schedule to guide execution. A key aspect of this standard is leveraging the Work Breakdown Schedule (WBS) to sequence tasks, establish priorities, allocate resources, and develop a comprehensive timeline. This process ensures that the critical path is clearly identified and actively managed.

[\[More Detail\]](#)

RIS 103.1.6 – Execution Budget

The Execution Budget standard serves several critical purposes. It ensures that we fulfill our commitments within the agreed-upon budget, meeting customer expectations while maintaining financial discipline. Additionally, it establishes a clear budget baseline against which the project team is measured, based on the resources and costs necessary for successful execution.

[\[More Detail\]](#)

RIS 103.1.7 – Test Plan

The Test Strategy Standard defines the approach for testing throughout the project, ensuring a structured and consistent validation process. It provides an opportunity to refine the baseline established during the proposal phase, aligning expectations with the evolving project scope.

[\[More Detail\]](#)

RIS 103.1.8 – DevOps Plan

The DevOps Plan standard defines the development, deployment, and infrastructure management approach for the project. It establishes the development tools, environment configurations, and deployment strategy, ensuring a structured and efficient workflow.

[\[More Detail\]](#)

RIS 103.1.9 – Traceability Matrix

[\[More Detail\]](#)

RIS 103.1.10 – Risk and Issue Plan

[\[More Detail\]](#)

RIS 103.1.11 – Training Plan

[\[More Detail\]](#)

Planning Sub-RAP

RIS 103.2.1 – Sprint Refinement

RIS 103.2.2 – Developer Assignments

Execution Sub-RAP

RIS 103.3.1 – User Story Development

RIS 103.3.2 – Automated Test Development

RIS 103.3.3 – Test Execution

RIS 103.3.4 – Code Reviews

RIS 103.3.5 – Sprint Progress Reviews

RIS 103.3.6 – Sprint Retrospective

RIS 103.3.7 – “As Built” Design Specification

RIS 103.4 – Project Governance

RIS 103.5 – Deployment Governance

RIS 103.6 – Master Data Governance

Smart Manufacturing Audit Template

Wednesday, September 18, 2024 10:13 AM

DxAssessment

- **Business Drivers**

- ☐ Execution Plan
- ☐ Project Kickoff
 - ☐ Internal Kickoff
 - ☐ Customer Kickoff
- ☐ RACI Matrix
- ☐ Stakeholder Analysis
- ☐ Communication Matrix
- ☐ Budget Definition
- ☐ Current Process State
- ☐ Stakeholder Group Interviews
- ☐ Business Drivers

- **Business Scenarios**

- ☐ Business Process Scenario
- ☐ Key Design Decisions

- **Solution Architecture**

- ☐ Technology Assessment
- ☐ Generic Assessment
- ☐ Hardware Specification
- ☐ ROI Analysis
- ☐ Deployment Roadmap

DxExecution

- **Mobilization**

- ☐ Developer Onboarding
- ☐ Feature User Stories
- ☐ Work Breakdown (WBS)
- ☐ Traceability Matrix
- ☐ Sprint Schedule Plan
- ☐ Test Plan
- ☐ CI/CD Plan
- ☐ Risk & Issue Plan
- ☐ Training Plan

- **Sprint Planning**

- ☐ Sprint Refinement
 - ☐ User Story Acceptance
 - ☐ Customer Kickoff
- ☐ Developer Assignments

- **Sprint Execution**

- ☐ User Story Development
- ☐ Automated Test Development
- ☐ Test Execution (As Applicable):
 - ☐ Unit Testing
 - ☐ Factory Acceptance Test
 - ☐ User Acceptance Testing
- ☐ Code Reviews
- ☐ Sprint Progress Review

- ☐ Sprint Retrospective
- ☐ Design Specification

DxAssessment

- **User Adoption**

- ☐ Training Build
- ☐ Standard Operating Procedures
- ☐ User Guides
- ☐ Deployment Checklists

- **Startup & Anchor**

- ☐ Training Execution
- ☐ Deployment Summary Reports
- ☐ Observation Log
- ☐ Lessons Learned
- ☐ Project Closeout Report

KB - Executing without Prior Assessment (RAP 101 and 102)

Friday, July 12, 2024 8:58 AM

Overview

Every project we undertake is influenced by whether the customer has purchased a prior Assessment package. This document is designed to guide the implementation team when the customer has not purchased a formal Assessment engagement and is beginning the project following our Smart Manufacturing Delivery process. Certain components of our DxDiscovery and DxArchitecture services offerings must be completed at the project's start as a part of Smart Manufacturing Delivery. This additional work should have been included in the proposal, but even if it was not, the effort must be accounted for to yield project success. It is also possible that the customer purchased aspects of the DxDiscovery and/or DxArchitecture offerings, but gaps exist that must be completed as part of project implementation.

Gap Analysis

The following sections review each deliverable from the DxDiscovery and DxArchitecture offerings constituting the Assessment package to indicate whether it is applicable. This document serves as a guideline. The **Project Manager** is responsible for ensuring successful project delivery and for determining which features and deliverables from the Assessment package should be applied, and to what extent, if the full Assessment package was not purchased prior to the engagement.

Miscellaneous Artifacts

MES RIS 101.1 - Assessment Quotation

This is not applicable. The quote has already been submitted and accepted by the customer.

DxDiscovery Features

MES RIS 101.2 - RACI

In this scenario, if the customer has not provided a RACI (Responsible, Accountable, Consulted, Informed) matrix, it is the responsibility of the **Project Manager** to create one. For detailed guidance, please reference MES RAP 101.2. This ensures that roles and responsibilities are clearly defined and understood, facilitating smooth project execution.

MES RIS 101.3 - Infrastructure Evaluation

As a part of the Sales effort, an infrastructure evaluation should have been completed - at least a cursory review of current state / constraints and any infrastructure-related consultation required to support the desired end-state solution to be able to build the quote. Therefore, it is the responsibility of the **Technical Lead** to review the Infrastructure Evaluation and to fill in any details relevant to the implementation of our commissioned scope.

MES RIS 101.4 - Legacy System Review

As part of the discovery phase, reviewing any legacy systems used in manufacturing is crucial for understanding current plant processes, constraints, and potential opportunities for replacement or enhancement. While a full assessment typically involves an in-depth analysis to deliver robust recommendations, in cases where an assessment is not included, a high-level review should be conducted instead. This review focuses on understanding the legacy system's capabilities and how they compare to the system being implemented.

Such insights provide valuable perspective on the customer's existing environment, which can inform the strategy for change management and enhance communication with the customer. The responsibility for conducting this cursory review lies with the **Technical Lead**.

MES RIS 101.5 - Legacy Documentation Review

The above analysis from the Legacy System Review is applicable here, as well. The responsibility for conducting this cursory review lies with the **Technical Lead**.

MES RIS 101.6 - Site Walk Evaluation

It should be assumed that a Site Walk Evaluation was completed as a part of the Sales effort. However, regardless, it is a valuable onboarding tool to have the team visit onsite to develop a deeper comprehension of the site's operations. The **Project Manager** is responsible for organizing any site visits.

MES RIS 101.7 - Panel Walk Down

Legend

-  Applicable
-  Partially Applicable
-  Not Applicable

Management Sponsor

Project Manager

Technical Lead

Functional Analyst

Solution Architect

Customer Success Manager

This is not typically applicable to an MES or MOM software engagement.

MES RIS 101.8 - Operations Management Capability Review

The capabilities being delivered by the MES initiative should already be well established in the scope of work. However, various customer stakeholders may require further education concerning how the MES fits into a broader systems landscape, as well as boundary establishment concerning what capabilities are included in the current scope, what capabilities could be included in a future phase, and what capabilities are typically managed within other applications. As a result, the **Project Manager** and **Technical Lead** should judge the scope of the project and the customer's maturity, and conduct an operations management capability review and any ancillary tasks such as a demo to aid customer stakeholders in envisioning the possibilities of the solution as warranted.

MES RIS 101.9 - Current Process State Definition

This is applicable as the effort helps to baseline the current state of the facility or facilities under the scope of the project. The **Project Manager** is responsible for organizing the visit.

MES RAP 101.10 - Stakeholder Group Interview Process

This is applicable to gain insights facilitating formulation of the 'to-be' business process scenarios. The **Technical Lead** is responsible for Stakeholder Group Interview Process.

MES RIS 101.11 - Perceived Benefit

This is not typically applicable as the customer has already made the decision to move forward with the execution of a system implementation effort. However, it is valuable for the **Project Manager** to capture the expected benefit from the customer.

MES RIS 101.12 - Executive Summary

This is not applicable as it is intended only for summarizing the discovery efforts to build support for the path forward.

DxArchitecture Artifacts

MES RIS 102.1 - RACI

In this scenario, if the customer has not provided a RACI (Responsible, Accountable, Consulted, Informed) matrix, it is the responsibility of the **Project Manager** to create one. For detailed guidance, please reference MES RAP 101.2. This ensures that roles and responsibilities are clearly defined and understood, facilitating smooth project execution.

MES RAP 102.2 - Business Intention Definition

In an ideal scenario, the customer would have already conducted an exercise resulting in known business intentions. Typically, this information is included in an RFP package, though its quality can vary. The **Project Manager** is responsible for understanding the customer's business intentions that are relevant to our scope of work. The **Technical Lead** should be informed and consulted about the business intentions.

If the customer did not complete or share the Business Intentions, the **Project Manager** should complete this exercise and may include the **Management Sponsor** to support the effort.

MES RAP 102.3 - Process Activity Diagram

The process activity diagram documents the various production processes in a 'swim lane' format, showing the system responsible for each activity, as well as system-to-system interactions or user-to-system interactions. This activity is especially useful to facilitate a technical team's understanding of the role of various systems and how they fit together. The process activity diagram highlights areas where detailed system interface design will be required, as well as cases where user interfaces will need to be made available. This is a valuable exercise to complete in order to understand interface constraints on the new system. The **Technical Lead** is responsible for delivering this activity.

MES RAP 102.4 - Business Process Scenario Definition

It is unlikely the customers completed Business Process Scenarios, and, in such cases, the **Project Manager** should include this exercise in our WBS (Work Breakdown Structure), ensuring that our scope of work is clear and

comprehensive. This *effort should* have been included in the quote but, if not, it is up to the **Project Manager** to determine the scale of this exercise to balance the schedule and budget. The **Technical Lead** and **Functional Analyst** are responsible for completing this activity.

WBS	Business Scenario	DevOps
WBS L1	n/a	Epic
WBS L2	SIPOC Business Scenario	Feature
WBS L3	SIPOC Step	User Story
WBS L4	n/a	Task

MES RAP 102.5 - Generic Architecture

The generic architecture is typically leveraged to portray architectural touchpoints and constraints envisioned independent of selection of a specific technology stack to facilitate technology selection. In this case, the technology to be leveraged should have already been selected, minimizing the need for this intermediate activity.

MES RAP 102.6 - Technology Selection Process

In this case, our involvement typically begins after the customer has already selected the technology. Our primary role is to support the implementation of a solution that achieves the customer's business objectives using the chosen technology. Our focus is on ensuring that the selected technology is effectively integrated and meets the desired outcomes as defined by the customer.

However, there may be nuances concerning the technology selected, including integration layers, certain modules to leverage or not to leverage, etc. that have not yet been determined prior to the commencement of the project. In these cases, an abridged technology selection effort per MES RAP 101.8 may be required to help close these decision gaps. The **Technical Lead** is responsible for engaging a **Solution Architect** to drive this task, if necessary, to address the gap.

MES RAP 102.7 - Solution Architecture Definition

During the quotation process, the platform-specific solution architecture should have been provided either by the customer or by RoviSys. If this was not completed at that stage, the **Solution Architect** or **Technical Lead**, if no **Solution Architect** is assigned to the project, should take the initiative to build a platform-specific solution architecture. This architecture should be detailed enough to cover the scope of our work. If further details are necessary but fall outside our scope, the customer should be asked to provide those additional details. The **Technical Lead** will be responsible for facilitating this effort, ensuring that all necessary information is obtained for a successful implementation.

Additionally, aspects related to system sizing are often not covered by the solution architecture, but are needed by the customer to begin procurement and configuration activities. For example, the number of servers, software versions, resource allocation (CPU, RAM, disk), etc. may need to be further defined or refined by RoviSys as part of solution architecture definition.

MES RAP 102.8 - ROI Guidance Formulation

Before proceeding, it is crucial that the customer has completed an ROI evaluation (i.e., the project justification). If this information was not provided during the quotation process, the **Management Sponsor** or **Customer Success Manager** should reach out to the customer to obtain the necessary details at the onset of the project so it can be included in the necessary documentation and meetings. Understanding the customer's ROI targets is essential for us to meet their expectations effectively. Without clear knowledge of these targets, our efforts would be misdirected, akin to aiming blindly without a specific goal in sight.

Once the ROI targets are known, the **Project Manager** should communicate these targets clearly and broadly to the entire team (e.g., via Project Execution Plan, IKO, status meetings, etc.). This ensures that everyone is aligned with the ultimate goal and can work cohesively towards achieving the desired outcomes for the customer. By having a well-informed and unified team, we can better meet the customer's expectations and successfully achieve their ROI targets.

MES RAP 102.9 - Deployment Roadmap Formulation

In this scenario, the customer should have already completed the deployment roadmap that we will be executing. If this information was not provided during the quotation process, the **Project Manager** should request it from the customer at the onset of the project so it can be reviewed during the IKO. Having the deployment roadmap is essential as it will be consulted when building the project schedule, ensuring that our execution aligns with the customer's planned timeline and objectives, where possible or we work with the customer to reset the

expectations.

 MES RAP 101.110 - Executive Summary

This is not applicable as it is intended only for summarizing the DxArchitecture efforts in a summary to gain support for moving forward with the broader implementation initiative.

MES RAP 103.1 - Mobilization Step

Wednesday, March 20, 2024 10:11 AM

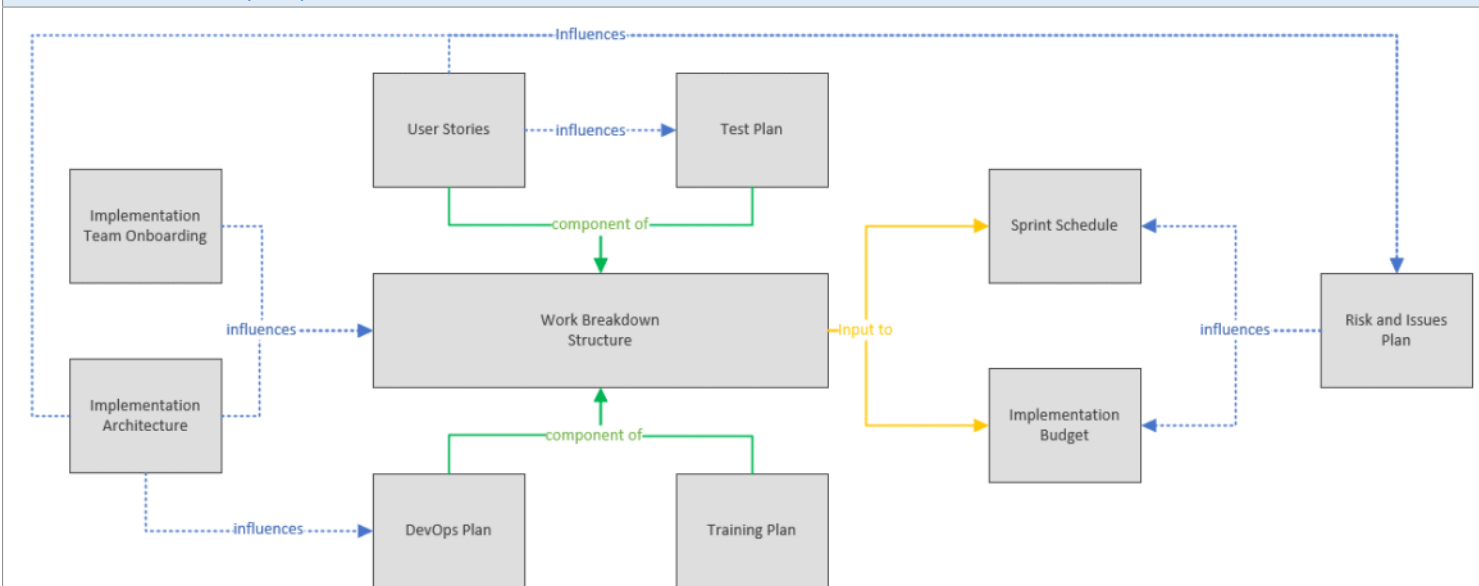
Overview

To be a better partner for our customers, our Mobilization procedure is essential for ensuring transparency and providing a clear view of the project scope and deliverables. This approach improves communication between the project team and customers, fostering a better understanding of progress and milestones. By following our procedures and standards, we build customer confidence in our organization and planning, helping to set and manage realistic expectations for timelines and deliverables.

This procedure encompasses the following deliverables. The standards for each of these deliverables are RIS' as children to this RAP. Our experience shows that by following this procedure, we significantly increase our chances of successfully delivering the project to our customers.

Deliverable Standard	
Implementation Team Onboarding	The standard by which an implementation team should be onboarded.
User Stories	The standard for how Feature User Stories should be created.
Work Breakdown	The standard for how work should be deconstructed, organized, sequenced, etc.
Implementation Architecture	The standard of the architecture that will be implemented.
Project Schedule	The standard of how to build a Project Schedule.
Implementation Budget	The standard of how to build an implementation budget.
Test Plan	The standard of how to build a testing plan to support the testing of the solution.
DevOps Plan	The standard of how the project will leverage DevOps
Risk and Issue	The standard for risks and issues will be logged, managed, and monitored on the project.
Training Plan	The standard for how to build a training plan to empower the customer to own the solution.

Deliverable Relationship Map



<https://rovi.sharepoint.com/sites/mes/Shared%20Documents/3.Strategy/PMO/Mobilization%20Deliverable%20Relationship%20Map.vsd?web=1>

Improvement Notes:

- Before we submit a WOR, we need to have the Dev Hints established and estimates agreed upon; Talk to Aaron

RIS 103.1.1 - Execution Team Onboarding

Wednesday, July 17, 2024 9:13 AM

Purpose

This standard details the creation of an Implementation Team Onboarding Plan. The purpose of the Onboarding Plan is to identify and organize the detailed activities individuals need to complete in order to quickly, efficiently, and sufficiently onboard to a project.

Overview

Onboarding a project implementation team in an organized, repeatable manner is crucial for ensuring consistency, efficiency, clarity, and quality assurance. A structured process guarantees that all team members receive the same information and training, leading to a uniform understanding of project goals, processes, and expectations. This approach minimizes the time and resources spent on onboarding, enabling team members to become productive more quickly. Clearly defined procedures help new members understand their roles and responsibilities, reducing confusion and ensuring everyone knows what is expected of them. Additionally, a repeatable onboarding process helps maintain high-quality standards by providing a reliable framework for introducing new team members to the project.

Accepted Standard Files

- Onboarding Checklist - Word
- Onboarding Checklist - OneNote

Definitions

"RACI" A tool used to clarify roles and responsibilities within a project or process.
See the reference table/ legend in the right margin -->

RACI

- Management Sponsor - Accountable
- Project Manager - Responsible
- Technical Lead - Consulted
- Solution Architect - Consulted
- Team Member - Informed

Standard Use

Checklist Setup

- Project Manager is assigned to the project.
- The Project Manager downloads the Onboarding Checklist and follows the checklist for Project Manager.
- Project Manager works with the Technical Lead and Solution Architect to tailor the checklist for their project.
- When individuals onboard to the project, the Project Manager must ensure that the individual(s) complete the onboarding checklist.

Checklist Execution

- A Team Member is assigned to the project.
- The Project Manager assigned the Team Member the Onboarding Checklist.
- The Team Member completes the items in the checklist document.
- The Team Member informs the Project Manager that the onboarding is completed.
- The Team Member begins work on the project.

Checklist Maintenance

- Project Manager and Technical Lead maintain the checklist as a living document throughout the project.

<- AJW - I think it's worth including some reference to CKO/IKO here. For resources staffed on the project from the beginning, what checklist items must be completed before CKO/IKO?

Abbr.	Title	Description
A	Accountable	Person who is the "owner" of that line item (assumes Informed, as well).
R	Responsible	Person who will deliver/ do the line item.
C	Consulted	Person is consulted for activities in this area (assumes Informed, as well)
I	Informed	Person is kept informed of activities in this area

>> [TEMPLATE] Onboarding Checklist OneNote

Wednesday, July 17, 2024 12:54 PM

Jump To Role

[Project Manager](#)

[Technical Lead](#)

[Solution Architect](#)

[Functional Analyst](#)

[Engineer](#)

[Customer Success Manager](#)

Project Manager

General Onboarding

- ☐ Meet with Sales Resource/ Assessment Team (if applicable)
- ☐ Meet with RM and/or Director
- ☐ Read Proposal (send to Technical Lead (& Solutions Architect if applicable))
- ☐ Confirm the opportunity in Comet

Note: Make sure all information is accurate (adjust as needed)

Grandview

Set up the project in Grandview (use the list below to assist in this task)

- ☐ Project Information
- ☐ Order Information
- ☐ Company Team
- ☐ Project Contacts
- ☐ Shipping Address
- ☐ Invoicing Address
- ☐ Project Tasks
- ☐ Employee Task Assignments (Task Codes)
- ☐ Project Status Tab View (Use Grandview Templates)
- ☐ Project Execution
- ☐ Custom Project Status Logs
- ☐ Schedule
- ☐ Files (Review RIS 537 – Configuration Management)

Teams Group Configuration

Configure the Teams group for the project team

- ☐ Set up OneNote & Team Site for the project
- Note: The initial Team Site should be auto-generated by Grandview*

Staffing Configuration

Configure the Teams group for the project team

- ☐ Set up the project in Staffing
- ☐ Staff all resources on the project

RAP 302 Initial Documentation

Set up the project in Grandview (use the list below to assist in this task)

- ☐ Complete Initial Draft of Project Management & Execution Plan
- ☐ Meet with Technical Lead (& Solutions Architect if applicable)
- ☐ New task to do

Misc.

- ☐ Schedule IKO
- ☐ Schedule CKO
- ☐ Setup DevOps for the project

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Technical Lead

- ☐ Read Proposal
- ☐ Meet with Project Manager (& Solutions Architect if applicable)
- ☐ Meet with Sales Resource (if additional clarification on Proposal is needed)
- ☐ Meet with RM (w/ Solutions Architect if applicable)

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Solution Architect

- ☐ Read Proposal
- ☐ Meet with Project Manager (& Solutions Architect if applicable)
- ☐ Meet with Sales Resource (if additional clarification on Proposal is needed)
- ☐ Meet with RM (w/ Solutions Architect if applicable)

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Functional Analyst

- ☐ Read Proposal
- ☐ Meet with Project Manager (& Solutions Architect if applicable)
- ☐ Meet with Sales Resource (if additional clarification on Proposal is needed)
- ☐ Meet with RM (w/ Solutions Architect if applicable)

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Engineer

- ☐ Read Proposal
- ☐ Meet with Project Manager (& Solutions Architect if applicable)
- ☐ Meet with Sales Resource (if additional clarification on Proposal is needed)
- ☐ Meet with RM (w/ Solutions Architect if applicable)

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Customer Success Manager

- ☐ Read Proposal
- ☐ Meet with Project Manager (& Solutions Architect if applicable)

- ☐ Meet with Sales Resource (if additional clarification on Proposal is needed)
- ☐ Meet with RM (w/ Solutions Architect if applicable)

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RIS 103.1.2 - Feature User Stories

Wednesday, July 17, 2024 9:15 AM

Deliverable: Complete backlog of Feature User Stories

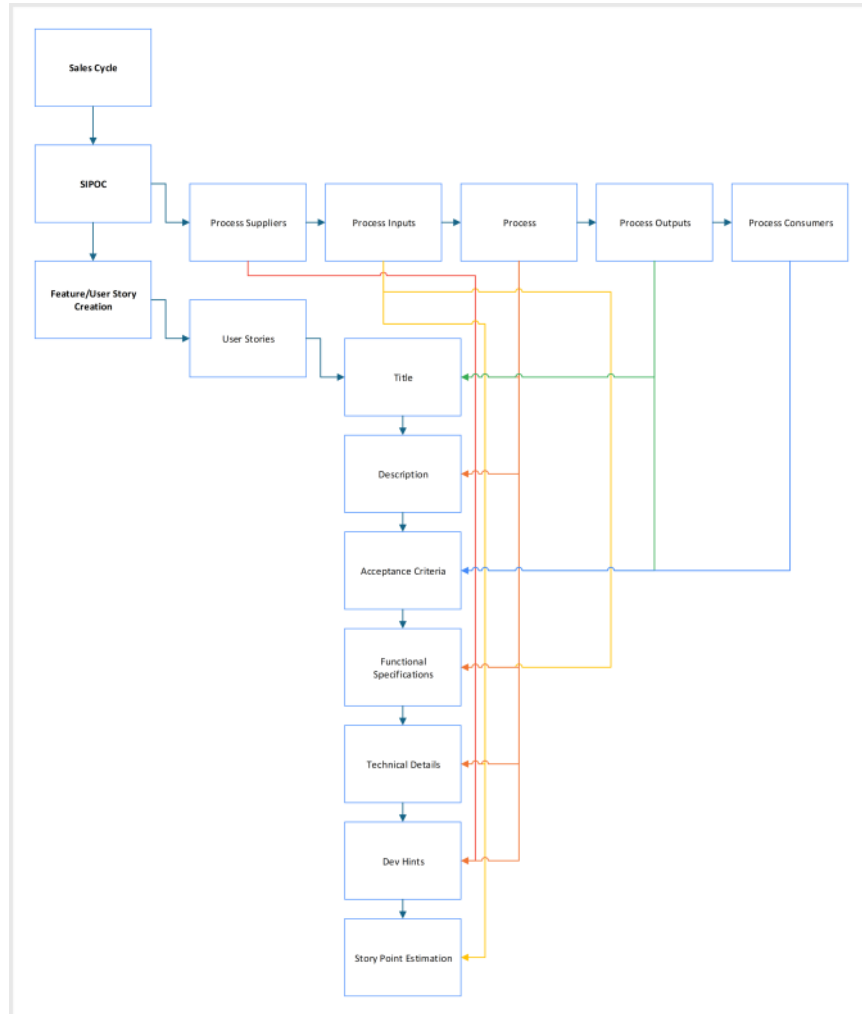
In Progress - Jacques (Last Updated: 2024.09.24)

User Stories

• Purpose

- This standard is intended to help individuals with a base understanding of Project Management and Agile/SCRUM to understand how we, in the MES division, utilize and create user stories when planning/performing an Agile for Manufacturing project
- Definition:** A **User Story** is a brief, self-contained unit of work that's written from the perspective of the end user to explain how a software feature will provide value to the customer. The purpose of a user story is to help developers understand the context behind what they're building and to help QA get into the mindset of the user while testing.

• Workflow



• Key Terms

Term	Definition
Epic	Large project or goal.
Feature	Smaller, tangible functionality contributing to the epic.
Title	A brief and descriptive name for the user story that summarizes the functionality or feature being implemented.
Description	A detailed explanation of the user story, including the problem it addresses, the intended users, and the value it provides. This is often written in the following format (e.g., "As a [user], I want to [action] so that [benefit].").
Acceptance Criteria	Specific conditions or requirements that must be met for the user story to be considered complete. These criteria ensure that the functionality works as expected and aligns with the user's needs.
Functional Specifications	A more detailed description of what the user story should do, focusing on the behavior and interactions from the user's perspective. It outlines the exact functionality, including user interfaces, inputs, outputs, and any rules or constraints.
Technical	Information on how the feature should be implemented from a technical perspective. This can

Linh's Notes:

I would include a chart on how Epics, Features, User Stories, and Tasks tie together. Either DevOps or just a chart that we build. Someone who doesn't know the hierarchy would be confused.

We should include screenshots of a Epic, Feature, User Story, and Task and highlight which fields should be filled out. What do you think?

Need examples of what ADO looks like. Someone who has never seen it before will be very confused.

Story Points = estimate in terms of hours?

Do meetings and refinement sessions fall under Analysis or Design tasks?

Is the point of this RIS to inform others what a User Story is, or how to build one?

AJW Notes:

It may be useful to include that the chart below (or something similar) for those familiar with RVS processes who are new to MES. From [KB - Executing without Prior Assessment \(RAP 101\)](#) :

WBS	Business Scenario	DevOps
WBS L1	n/a	Epic
WBS L2	SIPOC Business Scenario	Feature
WBS L3	SIPOC Step	User Story
WBS L4	n/a	Task

I think it would pay to include some examples of good User Stories here (or references of where you can find them)

Same question as Linh, Is this intended only to be introductory knowledge about User Stories & Agile/Scrum as a concept, or should this also serve as a guide for creation? As an introduction, this give you the information you need to know, but I worry that it will be read and then forgotten. We may have more success driving consistency if this document is meant to be actively used while performing the relevant tasks (I suspect an actual guide may be more appropriate as a child 103.1.2.1

I think it is worth adding some material in the Roles & Responsibilities section to correlate the scrum roles you have defined to typical roles in a RoviSys project, e.g., Scrum Master :: RoviSys PM

I think the CRM23B example in the "what information is required..." section is very helpful to those unfamiliar, but I would like to see more detail on what each feature there is or means. E.g., Operator Cockpit - Create the User interface for Operators

SIPOC materials should provide a reference/link to [MES RIS 102.4 - Business Process Scenario Definition \(SIPOC\)](#)

Nitpicking, but one yellow arrow on the workflow diagram is not like the others.

Details	include architectural considerations, data models, APIs, and other technical requirements.
Dev Hints	A numerical estimate that reflects the complexity, effort, and time required to complete the user story. This is typically done using a relative scale, such as Fibonacci numbers, and helps in planning and prioritizing work.
Tags	Keywords or labels that help categorize and organize the user story. Tags can be used for filtering, searching, or grouping stories by features, components, teams, or other criteria.

SIPOC	Definition
Suppliers	The entities (people, organizations, or systems) that provide the necessary inputs for the process.
Inputs	The resources, materials, or information needed to perform the process.
Process	The series of steps or activities carried out to convert inputs into outputs.
Outputs	The products, services, or results produced by the process.
Customers	The end-users or beneficiaries of the process outputs.

• Roles & Responsibilities

Roles	Responsibilities
Scrum Master	Facilitates the Scrum process, ensuring the team adheres to Agile principles and practices. Removes impediments and fosters collaboration among team members. Prioritizes managing the work items in the current sprint.
Product Owner	Represents the stakeholders and is responsible for defining the product backlog, prioritizing features, and ensuring the team delivers value to the customer. Prioritizes preparing work items for future sprints.
Solution Architect	Designs the overall system architecture, ensuring that the technical structure aligns with the business goals and is scalable, reliable, and maintainable.
Technical Lead	Provides technical direction and oversight to the development team, ensuring best practices are followed and the technical implementation aligns with the project's goals.
Functional Analyst	Analyzes business requirements and translates them into functional specifications for the development team, ensuring that the software meets the needs of the users.
Senior Developer/Engineer	An experienced developer responsible for implementing complex features, mentoring junior developers, and contributing to the overall code quality and technical decisions.

• What Information is Required in a User Story

- To start you need an epic and, in most cases, features.
- What we include in Features:
 - Title - A clear, concise title that describes what the feature is about.
 - Description - A brief summary of the feature, explaining its purpose and the problem it solves. You can use the SIPOC to answer the following questions if you're having trouble formulating this:
 - Objective?
 - Who?
 - What?
 - When?
 - Definition of Done/Success Criteria
 - Related Documents
- So for example, the CRM23B epic would be "The Implementation of Critical Manufacturing" and the features for that project would have been:
 - PLM
 - Operator Cockpit
 - Supplier Integration
 - SAP
- Meaning that every user story that was created can be placed in one of the 4 categories listed above.

• Utilizing SIPOC

- In user story creation you care about suppliers and inputs because of how they affect what you're creating, you care about the process because that is essentially the user story, and you care about the outputs and customers because that's the end result of your work, and who will be evaluating whether or not the work that was done is correct.
- You can read more about SIPOC and how we utilize it here:
 - [MES RIS 101.7 - Business Process Scenario Definition](#)
- The fields you should fill out in ADO using the information from your SIPOC:

ADO Field	Definition
Title	• SIPOC Focus: Process • This should be a concise summary of the process you are improving or feature you are working on. Think about the core process or goal this story targets.
Description	• SIPOC Focus: Process & Outputs • Provide a detailed description of what the user story covers. This should outline the steps in the process (from the SIPOC) and describe the expected outputs. Think about how the inputs get transformed by the process.
Acceptance Criteria	• SIPOC Focus: Outputs & Customers • Define what success looks like from the perspective of the customers in your SIPOC. This is where you articulate what needs to be achieved for the outputs to meet customer expectations.
Functional Specifications	• SIPOC Focus: Inputs & Process • Break down the functional requirements that detail how the process transforms the inputs. Functional specifications should specify what technical or business inputs are necessary and how they flow through the process.
Technical Details	• SIPOC Focus: Inputs & Process • Highlight the technical inputs and details involved. You might refer to specific systems, APIs, or components needed to execute the process.
Dev Hints	• SIPOC Focus: Suppliers & Process

	• These can include helpful insights about potential challenges in working with the suppliers (teams, systems, or external tools) and process improvements. This field can include information on dependencies or specific implementation tips.
Story Point Estimation	• SIPOC Focus: Process & Inputs • Estimate the effort required to transform the inputs into the desired outputs. This estimation takes into account the complexity of the process as defined in the SIPOC, including technical details and any constraints.
Tags	• SIPOC Focus: Customers & Suppliers • Use tags to categorize the story. Consider your stakeholders (customers) and the teams (suppliers) responsible for delivering on this user story. These tags could help in filtering or identifying key stories that are customer-specific or related to certain systems or processes.

- **Using ADO**
 - [Link to "Creating an item in Azure Dev Ops" Document](#)
 - [Link to "Importing into Azure Dev Ops" Document](#)
- **MES Accepted Standards/Tools**
 - Upload Document
 - [Link to Template A](#)
 - [Link to Template B](#)
 - ADO
 - [Link to Template A](#)
 - [Link to Template B](#)
- **Things to Consider**
 - Who on the project is capable of being a "Reviewer"
 - Who owns the responsibility of approving PRs and merging branches back into Main

List of Potential Tasks

Note that not all of the tasks listed below are applicable for every user story. This is a list for you, as the SCRUM Master, to pick and choose from (in otherwise, don't put all 17 of these tasks on every User Story you create - just the ones you need)

- **Analysis Tasks**
 - Developer Review (Includes WBS Creation)
 - Requirement Clarification - Meeting with stakeholders or the product owner to clarify any ambiguous aspects of the user story.
- **Deployment and Configuration Tasks**
 - Environment Setup - Setting up or configuring development environment to support the new feature.
- **Design Tasks**
 - Architectural Design- Planning the architecture or high-level design, particularly if the user story impacts the system's structure.
 - UI/UX Design - Creating wireframes, mockups, or design assets for any user interface changes required by the story.
 - Database Design - Designing any necessary database schema changes, such as new tables, columns, or relationships.
- **Development Tasks**
 - Front-End Development
 - Back-End Development
 - Refactoring
 - Unit Testing - Writing unit tests for individual components or functions to ensure they work as expected.
- **Testing Tasks**
 - Integration Testing - Ensuring that the front-end and back-end components work together correctly.
 - End-to-End Testing - Writing and executing tests that simulate real user interactions to verify that the story's functionality works as intended
- **Documentation Tasks**
 - User Documentation - Updating user manuals, help files, or online documentation to reflect the new functionality.
 - Technical Documentation - Documenting any architectural changes, API updates, or configuration changes for future reference.
 - Release Notes - Preparing notes for the release that describe the new features and any impact on users.
- **Review and Approval Tasks**
 - Code Review - Conducting peer reviews of the code to ensure quality and adherence to coding standards.
 - Acceptance Review - Final review of the code by the Technical Lead before it is merged into the main branch.

Example Breakdown:

User Story "XYZ" would have the following tasks listed under it:

1. Developer Review (WBS for development task written here)
2. Requirement Clarification (So speaking to customer to clarify things)
3. Environment Setup (if needed)
4. Back-End Development (Completed by Developer A)
5. Unit Testing (Completed by Developer A)
6. Integration Testing (Completed by Developer B)
7. End-to-End Testing (Completed by Developer B or C)
8. User Documentation (Completed by Developer A)
9. Code Review (By more Sr. Developer/Technical Lead)
10. Acceptance Review (By Whoever Approves Published User Stories & Merges Into Main Branch)

RIS 103.1.3 - Work Breakdown

Wednesday, July 17, 2024 9:15 AM

Overview

This standard establishes a structured approach to developing an initial **Work Breakdown Schedule (WBS)** for the project scope. By systematically breaking work down from **Features → User Stories → Tasks**, teams can enhance the accuracy of delivery timelines, optimize resource allocation, and identify critical interdependencies.

In this standard, we outline the expectations for how each project should effectively leverage the WBS to drive execution, maintain alignment, and ensure successful delivery. By adhering to these guidelines, teams can improve project predictability and efficiency while maintaining a clear roadmap for progress.

RIS 103.1.4 - Execution Architecture

Wednesday, July 17, 2024 9:15 AM

Overview

When executing a project, the team responsible must make the final decision about how it will be carried out, even if it was previously architected during the assessment step or by the customer. It's crucial to consider earlier architectures as they build the execution architecture.

No matter how a project was architected during the assessment step or by the customer, when it comes to executing the project the team responsible for executing it need to make the final decision about how it will be executed. Certainly they should take the earlier architectures into consideration as they build the execution architecture.

What is the Execution Architecture?

The execution architecture is the technical framework used to execute the project. The Technical Lead should outline the technological components required for execution and document them. Documenting the details of how we will execute the project helps in communicating our needs to the customer and assists in onboarding our team.

The architecture should include, but is not limited to, the following aspects:

- Server Structure
 - o What are the solution environments required?
 - Do you have a DEV instance for software development?
 - Do you have a QA instance for software testing?
 - What will your PROF instance be once you cutover?
 - Sometimes customers will promote their QA instance to PROD at Go-Live to mitigate risks of deployment errors and to reduce the cut-over effort.
 - Will we be developing on the customer instances, our own, or a hybrid?
- Code Management
 - o Where are you managing your source code?
 - Is the customer requiring it on their environment?
 - o What pipeline(s) need to be set up for code migration/ deployments?
- System Integration
 - o What systems are we integrating to in this project?
 - What are the details of those environments? (e.g., IP address, instance type, etc.)
 - What type of data is required on those instances to support our integration testing?
- Validation
 - o Will we be doing automated testing?
 - What tools will you use?
 - Where will you store tests?
 - Can you leverage any of our existing frameworks?
- RoviSys Accelerators
 - o Will you be using a RoviSys Accelerator as the foundation?

Accepted Standard Files

- TBD - Word
- TBD - Visio

Definitions

TBD TBD

RACI

- Management Sponsor - Informed
- Project Manager - Accountable
- Technical Lead - Responsible
- Solution Architect - Consulted
- Team Member - Informed
- Customer - Informed

Standard Use

Server Structure

1. Technical Lead reviews existing documentation (e.g., Proposal, Solution Architecture Definition, etc.) to understand the architectural decisions made by the assessment team and/ or Customer.
2. Technical Lead creates or edits a server diagram of the recommended Execution Architecture.
3. Technical Lead meets with the Customer to review the recommended architecture
4. Once the architecture is accepted by the Customer, the Technical Lead is responsible for ensuring the Customer delivers on the architecture requirements and is responsible for ensuring RoviSys delivers on our architectural requirements.
5. The Technical Lead informs the project team of the Server Structure. This should be documented in the Project Execution Plan to be used during onboarding of Team Member(s) and communicated during the IKO and CKO meetings.

Code Management

1. Technical Lead reviews existing documentation (e.g., Proposal, Solution Architecture Definition, etc.) to understand the architectural decisions made by the assessment team and/ or Customer.
2. Technical Lead creates a recommendation for code management standards to be used on this project.
3. The Technical Lead informs the project team of the code management strategy. This should be documented in the Project Execution Plan to be used during onboarding of Team Member(s) and communicated during the IKO and CKO meetings.

Checklist Maintenance

1. Project Manager and Technical Lead maintain the checklist as a living document throughout the project.

RIS 103.1.5 - Project Schedule

Wednesday, July 17, 2024 9:15 AM

Overview

The Execution Schedule standard defines the expectations for creating a structured and reliable project schedule to guide execution. A key aspect of this standard is leveraging the Work Breakdown Schedule (WBS) to sequence tasks, establish priorities, allocate resources, and develop a comprehensive timeline. This process ensures that the critical path is clearly identified and actively managed.

At this stage of the project, we have a well-defined scope and a clear understanding of the approach. With this foundation, we can create a schedule with a high degree of accuracy—one that provides confidence in both planning and execution.

RIS 103.1.6 - Budget

Wednesday, July 17, 2024 9:15 AM

Overview

The Execution Budget standard serves several critical purposes. It ensures that we fulfill our commitments within the agreed-upon budget, meeting customer expectations while maintaining financial discipline. Additionally, it establishes a clear budget baseline against which the project team is measured, based on the resources and costs necessary for successful execution.

This standard provides essential guidance on Earned Value Management (EVM), stakeholder expectation management (both customer and RoviSys), and handling financial adjustments throughout the project lifecycle. By following these principles, teams can proactively manage costs, track performance, and drive project success with confidence.

RIS 103.1.7 - Test Plan

Wednesday, July 17, 2024 9:15 AM

Overview

The Test Strategy Standard defines the approach for testing throughout the project, ensuring a structured and consistent validation process. It provides an opportunity to refine the baseline established during the proposal phase, aligning expectations with the evolving project scope.

This standard outlines the test strategy deliverable, ensuring all stakeholders have a clear understanding of what will be tested, how testing will be conducted, and who is responsible for each testing activity. By establishing these expectations upfront, teams can drive efficiency, improve quality assurance, and ensure a smooth testing process from start to finish.

RIS 103.1.8 - DevOps Plan

Wednesday, July 17, 2024 9:15 AM

Overview

The DevOps Plan standard defines the development, deployment, and infrastructure management approach for the project. It establishes the development tools, environment configurations, and deployment strategy, ensuring a structured and efficient workflow.

By setting a clear baseline for both the project team and customer stakeholders, this standard fosters a shared understanding of processes, responsibilities, and expectations. This alignment enhances collaboration, streamlines development, and ensures smooth deployment throughout the project lifecycle.

RIS 103.1.9 - Traceability Matrix

Friday, February 7, 2025 4:12 PM

Overview

The Traceability Matrix standard establishes how projects will map requirements

RIS 103.1.10 - Risk and Issues

Wednesday, July 17, 2024 9:15 AM

Overview

RIS 103.1.11 - Training Plan

Wednesday, July 17, 2024 9:15 AM

TBD

MES RAP 103.2 - Design Architecture Strategy

Wednesday, March 20, 2024 10:12 AM

MES RAP 103.3 - Sprint Pre-Work

Wednesday, March 20, 2024 10:16 AM

In Progress - Jacques (Last Updated: 2024.09.24)

Sprint Planning

Evaluating the project backlog, the team determines the components of each Sprint - ensuring User Story and Acceptance Criteria are ready for implementation.

- **Sprint Refinement**
 - Acceptance Criteria
 - User Stories Approved
- **Developer Assignments**
 - Text

MES RAP 103.4 - Sprint Execution

Wednesday, March 20, 2024 10:17 AM

In Progress - Jacques (Last Updated: 2024.09.24)

Sprint Execution

This involves the development team working on the selected tasks from the sprint backlog, aiming to complete the defined goals.

- **Design Specification**
 - Text
- **(User Story) Development**
 - Text
- **Automated Test Development**
 - Text
- **Test Execution (As Applicable)**
 - Unit Testing
 - Factory Acceptance Testing
 - User Acceptance Testing
- **Code Reviews**
 - Text
- **Sprint Progress Reviews**
 - Text
- **Sprint Retrospective**
 - Text

MES RAP 104 - DxDeployment

Friday, August 23, 2024 9:16 AM

MES RAP 105 - DxLean

Friday, August 23, 2024 9:17 AM

MES RAP 106 - DxEvolution

Friday, August 23, 2024 9:17 AM

(Published) MES RAP 107 - DxSustainment

Friday, August 23, 2024 9:11 AM

Please Reference:

[DxSustainment](#)

The goal of the DxSustainment effort is to ensure that the customer's technology investment remains healthy and viable. A solution that is out-of-date, hard to support, buggy, or slow-performing will be poorly received by the system's user base, and is more likely to be poorly used or abandoned. The DxSustainment effort applies both to systems implemented and deployed by RoviSys and systems deployed by others. In the latter case, the customer may have become frustrated by the previous vendor's lack of ongoing support and reached out to RoviSys to help salvage the system so that the customer can recoup their technology investment.

In the case of a system installed by RoviSys, we may be able to draw upon the experience of the project team who implemented the system, as well as archived source code, configuration, and documentation. In the case of a system installed by others, we will need to quickly triage the system state. In both cases, detailed planning and execution of the sustainment strategy is key to resolving ongoing issues and ensuring proactive steps are taken to maintain system value.

The following features are captured as part of our DxSustainment productized services offering:

[System Evaluation:](#)

The system evaluation effort allows the team to gather information concerning the system that will be leveraged to formulate a sustainment strategy. Infrastructure, system versions, log files, server resources, and other metrics will be captured and analyzed.

[Knowledge Transfer:](#)

Knowledge transfer allows RoviSys to gain information from various user roles concerning their usage of the system, and specific challenges and pain points. This may include bugs that must be worked around, ideas for enhancements, issues accessing the system, and performance concerns.

Additional knowledge transfer may also be required between the current RoviSys project support team and new team members who may be brought into the project engagement in a support capability. This includes not only knowledge transfer concerning the system software platform, but also nuances concerning the implementation and common support issues.

[Support Desk Establishment:](#)

The support desk provides a means for the customer to submit a support incident (such as an issue or enhancement) and provides notification to the support team of this submittal. The support desk also facilitates the ability to track issues and enhancements reported by the customer, including resolution status, individual assigned as responsible for the resolution of the support incident, and a trail of correspondence that can be consulted should future similar issues arise.

[Knowledge Base Creation:](#)

A knowledge base concerning previous project issues and resolutions greatly aids the project team by

reducing duplication of troubleshooting efforts. Common support issues, including abnormal occurrences, error messages, and system state can be searched and returned to the support resource, providing a means of issue resolution.

[Periodic Health Audit:](#)

This proactive exercise allows the overall health and wellbeing of the software application to be captured, preventing potential issues from causing bugs or system outages that could have wide-ranging impacts to the customer's manufacturing operations. The health audits must be periodic in nature, with a consistent checklist to be followed and means of escalating found issues so that they are rectified before they cause production impact.

(Published) MES RIS 107.1 - DxSustainment Quotation

Tuesday, February 18, 2025 12:22 PM

Please Reference:

[DxSustainment Quotation](#)

When quoting a DxSustainment extended support agreement for an MES project, it is crucial to structure the agreement to include necessary activities, define clear support tiers, and establish key assumptions that will impact service delivery. A well-defined agreement ensures that both RoviSys and the customer have a clear understanding of expectations, available resources, and the cost structure for ongoing support.

Key Activities to be Quoted in an MES Extended Support Agreement:

Note that the number of hours allocated to each key activity should be scaled based on the scope, size, and age of the system being supported. Additionally, if the agreement encompasses multiple sites with substantial differences, the activity durations must be scaled according to the number of sites.

1. Project Management

Effective project management is essential for ensuring the smooth execution of support activities. The proposal should include time allocation for:

- Coordination of support tasks
- Scheduling and tracking of activities
- Regular status meetings with stakeholders
- Documentation and reporting of progress

2. Knowledge Transfer Activities

A critical component of an effective support agreement is allocating time for knowledge transfer. This ensures that the support team has the necessary insights into the MES implementation, reducing resolution time for future issues. The proposal should include quoting for the following activities:

- Documentation handover sessions
- Training on customizations and specific configurations
- Walkthroughs of application processes

3. Code and Configuration Review

A comprehensive review of the MES codebase and system configuration should be conducted at the beginning of the support engagement by a lead or architect-level resource. This review provides:

- Understanding of custom scripts and automation logic
- Validation of system parameters and configurations
- Identification of potential technical debt that may affect supportability

Time should be allocated in the proposal for this review to ensure efficient future support.

3. System Evaluation

Before beginning an extended support agreement, an initial system evaluation should be performed to:

- Assess system performance and stability
- Identify known issues and their current resolutions
- Confirm the availability of logs, monitoring tools, and documentation

The proposal should specify the required hours for this activity to ensure a smooth transition into the support phase.

4. Periodic Health Audits

Ongoing system health checks help proactively identify issues before they escalate. The proposal should

include allocated hours for:

- Performance analysis
- Log review and error analysis
- Security and compliance checks
- Optimization recommendations

Including periodic health audits in the agreement ensures system reliability and continued efficiency.

5. Minor Enhancements

A dedicated bucket of hours can be allocated for minor enhancements to improve usability and functionality. The proposal should include:

- Small configuration changes
- User interface improvements
- Performance optimizations
- Minor automation or workflow adjustments

By setting aside time for minor enhancements, the MES can continuously evolve to better support operational needs.

Support Tier Structure and Rate Allocation

This support engagement is typically proposed on a time and materials basis, not to exceed the amount indicated in the agreement. Any work performed beyond the allocated hours would require prior approval and an additional proposal.

To ensure clarity in cost and service expectations, MES extended support agreements should define structured support tiers and a base fee that scales based on the level of support required:

Base Fee: A tiered base fee will be established based on the level of support provided, covering essential administrative and standby costs. To ensure clarity in cost and service expectations, MES extended support agreements should define structured support tiers:

1. Normal Business Hours - Non-Critical Issues & Enhancements
 - Covers minor enhancements, usability improvements, and non-urgent issues
 - Standard hourly rate applies
2. Business Hours - Critical Issues
 - Covers system downtime, data integrity issues, and production-impacting problems
 - Higher priority response with an elevated hourly rate
3. After-Hours 24x7 Support
 - Covers emergencies requiring immediate attention outside of normal business hours
 - Premium hourly rate due to resource availability and response time commitment

By clearly defining support buckets, organizations can better forecast costs while ensuring that urgent issues receive the attention they require.

Key Assumptions in the Agreement

To deliver effective support, the following assumptions should be included within the DxSustainment proposal:

- **RoviSys Contact Information:** Upon contract award, RoviSys will provide contact details, including a support hotline number for after-hours emergencies and an email for the RoviSys support helpdesk, which will be routed to the appropriate project team. To deliver effective support, the following assumptions must be met:

- **Project Management Burndown Reporting:** The project manager will provide periodic burndown reporting. If the number of remaining hours falls below an established threshold, the customer will be alerted, and a supplementary support proposal will be provided to ensure there is no lapse in coverage. To deliver effective support, the following assumptions must be met:
- **Customer Maintains Vendor Support Agreement:** The customer must maintain an active support agreement with the MES vendor to ensure continued access to vendor updates, patches, and assistance.
- **Authority to Communicate with Vendor:** The support team must be permitted to act on the customer's behalf in communications with the vendor to facilitate issue resolution and necessary escalations.
- **Remote Access:** The support team must have secure remote access to MES servers, databases, and logs.
- **Availability of External Resources:** Key personnel, such as MES administrators and IT support, must be available for knowledge transfer and collaboration.
- **Access to the Latest Code and Configuration:** All MES configurations, scripts, and system documentation must be maintained in a version control repository and made accessible to the support team.
- **System Stability Prior to Support Engagement:** The MES should be in a stable and operational state before the support team assumes responsibility. Any major issues or incomplete implementations should be resolved beforehand.
- **Systems Within the Scope of Support:** The proposal should clearly delineate systems that are under the scope of the DxSustainment support engagement and those that are excluded. For example, while RoviSys may be responsible for the MES, we may not be equipped to support external systems such as the ERP or WMS (warehouse management solution). We may also not be able to support associated middleware enabling communication to these applications that may have been deployed by those business teams.

Proposal Template:

The following proposal template may be leveraged when quoting a DxSustainment support initiative.

[P24XXX01a - DxSustainment Platform Support.pptx](#)

(Published) MES RIS 107.2 - System Evaluation

Friday, August 23, 2024 9:13 AM

Please Reference: [System Evaluation](#)

The system evaluation effort allows RoviSys to formulate a baseline of the customer's environment, gaining an understanding that will be leveraged when formulating the sustainment strategy. This is of particular importance in cases where RoviSys did not perform the initial system development and installation, but may also be important in cases where RoviSys did deploy the system but has not been exposed to the system for some time. During this time period the customer may have modified the system or contracted others to provide support and implementation.

Prerequisite Information:

Prior to the system evaluation activity, a request should be made to the customer to provide any system documentation that may be available. This includes the following potential items:

- System architecture diagram
- System functional or design documentation
- System integration specifications
- System testing documentation
- System user manuals / user guides
- Export of incidents related to the system from customer's ticket management system, including any incident notes recorded

This information should be reviewed prior to formulation of the system evaluation checklist and execution of the system evaluation and knowledge transfer productized services features.

System Evaluation Checklist Formulation:

The customer environment should be evaluated in a deliberate manner according to a pre-defined plan, maximizing the time of both RoviSys and customer resources. This checklist should be tailored to the specifics of the environment, based upon the known software platforms / solutions. For example, specific file locations and system configuration items to be confirmed should be noted in detail, rather than 'fumbling around' while on a shared web conference call with the customer.

Specific aspects / activities that should be incorporated to confirm when generating the system evaluation checklist include the following:

- Application installation locations
- Server infrastructure
 - Server names and associated resources
- Product version information, including last patch / hotfix installation
- Installed system modules
- Any custom-developed modules or reports integrated within the solution
- External system connections
- Control system connectivity (via OPC / MQTT layer for example)
 - Any tag data errors or misconfigurations
- System log file locations and log message review

- Capture of any errors or warnings
- Windows system log review
 - Capture of any errors or warnings
- SQL databases leveraged
 - Presence of any custom tables, stored procedures, or functions
 - SQL server health
 - SQL log size, SQL backup evidence, index fragmentation, running maintenance plans
 - SQL Agent jobs configured and running status (successfully running, disabled, running with errors?)
- Client hardware leveraged
 - Terminals, workstations, thin clients, mobile devices, scanners, etc.
 - Brand, version, and form factor
- System configuration aspects
 - Backend plant model structure
 - Backend event / logic configuration
 - SQL database schema review
 - Configured frontend displays
 - Configured frontend reports
 - Etc.

System Access:

At this stage, the customer may be reluctant to provide RoviSys direct access to the production environment. Even if this access is granted, it is recommended that we perform any system evaluation activities while being shadowed by a customer technical contact, to ensure that we do not take an action that may inadvertently impact production or are tied to a coincidental issue impacting production.

Either an in-person site visit or web conference call are effective ways to conduct this evaluation, with the customer sharing their screen and navigating through the various aspects to be evaluated on the environment.

At this stage remote access for the RoviSys support team may be requested to ensure that we have proper access to troubleshoot future issues or perform system enhancements.

Post System Evaluation:

Leveraging the noted software version information, RoviSys can consult with vendor documentation to determine if the version in question is the latest offered by the vendor, or if major / minor revisions exist. Should upgrades exist, RoviSys must be prepared to discuss upgrade options with the customer, including the benefits of upgrading and possible risks were the customer not to upgrade.

Please consult MES RIS 107.6 - Technology Upgrade, for details concerning execution of a technology upgrade for the customer.

Outside of the potential for software update, all noted concerns, observations, and opportunities must be organized, categorized, and prioritized according to severity. RoviSys can then prepare both a written summary and review of these findings with the customer. The following details should be captured for each item, in a concise manner able to be understood by a non-technical stakeholder:

- Description of item
- Consequence if not addressed
- Benefit if addressed

- Effort to address (either effort expended against the current DxSustainment contract or a required subsequent proposal)
 - A longer-term enhancement may require creation of a separate implementation proposal, but a high-level overview can be provided as part of this summarization

The vendor may need to be consulted in the case of a system issue / error that is not documented and for which no patch exists. If the customer has no active support or maintenance contract in place with the vendor directly, the customer will likely need to renew their product support contract with the vendor to allow RoviSys to act on their behalf to engage the vendor to resolve the unknown error / issue.

(Published) MES RIS 107.3 - Knowledge Transfer

Friday, August 23, 2024 9:13 AM

Please Reference:

[Knowledge Transfer](#)

Knowledge transfer may be required in two different scenarios. In the case of a system that was not originally implemented by RoviSys, the knowledge transfer process allows RoviSys to gain information from various user roles concerning their usage of the system, and specific challenges and pain points. This may include bugs that must be worked around, ideas for enhancements, issues accessing the system, and performance concerns.

In the case of a system that was installed by RoviSys, knowledge transfer may be required between the current RoviSys project support team and new team members who may be brought into the project engagement in a support capability. This includes not only knowledge transfer concerning the system software platform, but also nuances concerning the implementation and common support issues.

Prior to the Knowledge Transfer Session:

As with system evaluation, creation of a knowledge transfer checklist is important to provide structure to the knowledge transfer effort and ensure that it is conducted efficiently. An initial step in the knowledge transfer process is identification of the specific user roles who will participate, particularly in the case of knowledge transfer being provided to RoviSys by the customer. End user roles are most adept at how the software functions and how it is utilized on a day-to-day basis.

It may first be necessary to obtain a customer process overview by a customer SME so that the RoviSys team understands the manufacturing process and can put system operations within the context of these processes.

Based upon the system evaluation effort, RoviSys should be able to map out displays and functions of focus. As these are presented by relevant customer SMEs, RoviSys can check off discovered displays / functions to ensure that none are missed.

During the Knowledge Transfer Session:

Leveraging the created checklist, RoviSys personnel can execute a session with each identified customer super user to capture how the software is leveraged. The customer should demonstrate common functions, data input, data output, and decisions made leveraging the software. Any functions with which the user struggles or bugs / workarounds captured should be noted.

An attempt should be made to record these sessions whenever possible so that redundant sessions are not required should the RoviSys support team change.

The RoviSys team should at a minimum consist of a lead who is driving the session and asking prompting questions, as well as a designated notetaker.

Post Knowledge Transfer Session:

Notes should be sanitized, condensed, and made available to the customer for reference / review. These notes should also be stored in a shared repository accessible to the broader RoviSys DxSustainment team, such as a OneNote notebook.

(Published) MES RIS 107.4 - Support Desk Establishment

Friday, August 23, 2024 9:13 AM

Please Reference:

[Support Desk Establishment](#)

The support help desk system provides a means for the customer to submit a support incident (such as an issue or enhancement) and provides notification to the support team of this submittal. The support help desk also facilitates the ability to track issues and enhancements reported by the customer, including resolution status, individual assigned as responsible for the resolution of the support incident, and a trail of correspondence that can be consulted should future similar issues arise.

There are generally two options for the support help desk. A customer may have an IT help desk solution that they wish to leverage for the engagement with RoviSys. In this case, RoviSys may work with the customer to add the RoviSys team to this help desk solution, as well as periodically add/remove team members who may be added to the RoviSys project or removed from the project. If the customer desires to utilize their help desk software solution, requirements that must be available via this solution include the following abilities:

- Can add all RoviSys team members who will be supporting the implementation
- Provide notification to RoviSys team via email and text (as well as phone if a 24x7 real-time response is required) when a support incident is logged
- Allow an individual to assign themselves to the support incident as the owner / responder
- Track the resolution status of the support incident
- Capture the history of the incident, including resolution
- Allow search of incidents / resolutions specific to the sustainment project

Should the customer not possess a support helpdesk solution, or if the solution possessed cannot meet the above requirements, the RoviSys customer support helpdesk can be leveraged (JitBit). This help desk allows the customer to submit a help desk request via email, which will then send email (and optionally text) notifications to the assigned support team.

A JitBit customer support helpdesk project can be established by a JitBit administrator. The following RoviPedia article provides details concerning usage of the customer support helpdesk:

<https://grandview.rovisys.com/Rovipedia/2773/JitBit-RoviSys-Customer-Support-Helpdesk-Users-Guide?s=JitBit%20customer%20support>

(Published) MES RIS 107.5 - Knowledge Base Creation

Friday, August 23, 2024 9:14 AM

Please Reference:

[Knowledge Base Creation](#)

Maintaining a comprehensive knowledge base for a digitalization initiative is crucial to ensuring long-term success and efficiency of the DxSustainment effort. A well-documented repository of system configurations, processes, best practices, and troubleshooting guides significantly accelerates onboarding for new team members, reducing the learning curve and enabling them to contribute effectively. Additionally, having readily available documentation streamlines troubleshooting efforts by providing quick access to historical issues, resolutions, and system insights, minimizing downtime and improving operational continuity. Furthermore, as the MES evolves, a structured knowledge base serves as a foundation for future enhancements, ensuring that past decisions, lessons learned, and system dependencies are well understood, thereby facilitating informed decision-making and seamless system improvements.

There are two primary conduits for knowledge base assistance. They are described as follows:

RoviSys Knowledge Hub:

The RoviSys Knowledge Hub repository, hosted in Microsoft Teams, may be utilized for general platform-specific information and assistance. This Teams instance possesses channels for the common MES and digitalization platforms utilized within RoviSys, such as GE, Critical Manufacturing, Parsec TrakSYS, and others. Questions may be posted to these channels concerning best practice, platform errors, or how to seek assistance from the vendor.

The channel can also be searched using Microsoft Teams' search functionality to find issues that may have been posted by others.

Additionally, file structures behind each channel store training and demonstration materials in certain cases.

Relevant insights from the RoviSys Knowledge Hub should be subsequently incorporated into the project-specific knowledge base for easy reference.

Project-Specific Knowledge Base:

It is recommended that the DxSustainment initiative also establish a Knowledge Hub repository for internal project usage, ensuring that information is accessible to both current and future team members. Although there is currently no global RoviSys standard for a Knowledge Base application, recommended options include Microsoft DevOps or Microsoft OneNote. Key functions that should be supported by the knowledge base platform selected include the following:

- Creation of a knowledge base entry, with the ability to input text, images, videos
 - o Assignment of a title for the entry
 - o Capture of the person creating the entry
 - o Capture of a date for the entry
 - Date entry allows a subsequent user who queries the article to judge the relevance of the entry based upon age
 - o Usage of tagging to categorize the entry according to function or module

- Ability to efficiently modify an existing knowledge base entry should there be a platform change or should new information come to light
- Ability to search for a word or phrase
 - Ability to sort based upon date
 - Ability to search based upon keyword / tag

Enforcement of Knowledge Base Management:

The importance of maintaining the knowledge base must be emphasized with the project team from the start of the DxSustainment engagement. The team must create knowledge base entries as issues or best practices are encountered, and keep these up-to-date should system changes occur. The responsibility for enforcement of these practices with the team ultimately lies with the project manager.

A well-maintained, searchable knowledge base helps reduce unnecessary inquiries, particularly after hours, by enabling team members to find solutions independently. A proper knowledge base allows a less experienced team member or one who may not be familiar with the issue in question to find a resolution via the knowledge base before reaching out to others on the project team.

(Published) MES RIS 107.6 - Periodic Health Audit

Friday, August 23, 2024 9:14 AM

Please Reference: [Periodic Health Audit](#)

A periodic health audit allows RoviSys to verify the overall well-being of the customer's digitalization stack, and recommend actions to ensure system performance, reliability, and robustness. The health audit must follow a pre-defined checklist to ensure that no required steps are missed. This checklist should be executed by an assigned DxSustainment project team member and archived in a project repository as completed.

The initial audit (or any audit involving a new auditor who is brought onto the DxSustainment engagement) should be overseen by the technical lead to ensure that the auditor properly completes the audit checklist steps.

The checklist will be formulated by the project technical lead (or a designee) and tailored specifically to the project, including the software systems in place, software hosting model (on-premises vs. cloud), and application functionality. The checklist must be very specific in nature, such that a resource unfamiliar with the overall build and architecture of the solution could execute it if required. This includes calling out specific server names, disks utilized, SQL server locations, log file locations, application service / module names, and other critical information to facilitate expedient and complete completion of the checklist.

Frequency:

An audit frequency should be established that balances cost with the risk to the system should an unnoticed error go undetected. A mission critical system responsible for production execution should generally be audited more frequently than a passive reporting system. A monthly audit is often a recommended cadence.

Checklist Items:

The following are typical items to be verified as part of the periodic health audit:

- Server resources
 - Average CPU usage
 - RAM utilization
 - Free hard disk and hard disk consumption since last audit period
- Application status
 - All required Windows services are in a valid running state
 - All application modules required are enabled and in a valid running state
- Windows event logs
- Application log files
 - Errors or warnings that may be indicative of an issue
 - Size of log files, including any buildup that may pose risk of filling up the hard disk
- SQL Server performance
 - Long-running queries
 - Index fragmentation
 - Size of SQL log file
 - Confirmation of regular SQL server backup execution

- Shop floor connectivity
 - Invalid control system addresses or communication errors
- External system connectivity
 - Errors generated by system interfaces
 - Unprocessed data
- Application client performance
 - Responsiveness of client displays in production environment
- Application vulnerability
 - Examination of any recent patches / updates released by the software vendor that should be applied to address any application bugs or security holes

Automated Verifications:

Some items in the checklist above may be monitored in an automated manner, using IT tools at the customer's disposal. Some examples of items that can be monitored in an automated manner include application status, server resources, and event logs / log files. However, a periodic manual audit must still verify these items, in the event that the automated tools are not working or potential issues reported by the automated tools are ignored.

Root Cause Resolution:

In cases where a potential issue is discovered, particularly concerning matters of resource utilization, the RoviSys auditor must examine the root cause of the issue rather than recommend a resolution that simply masks the issue. As a common example, in the cases of poor system performance or disk utilization approaching 100%, RoviSys may recommend a system resource increase (addition of CPU cores, RAM, or hard disk to the server). However, there may be an underlying issue, such as a repetitive error being generated by the system, SQL fragmentation, data collection triggered too frequently, etc. In these cases the underlying cause of the issue must be resolved to avoid unnecessary system resource cost or the potential that the issue becomes worse over time.

Checklist Template:

While the RoviSys DxSustainment project team is free to leverage a template that is formulated based upon the specific nuances of the project and system in question, the following is a linkage to a general template in Excel format. This template consists of the following sheets:

- Title page, including project information, checklist version, execution date, and RoviSys resource responsible for execution
- Issue summary, including reference to checklist item, comments, suggested resolution, and urgency
- Detailed system health audit checklist

Checklist Summary Report:

Upon completion of the periodic health audit, the audit checklist must be reviewed by the DxSustainment technical lead and then submitted to the customer for review. While the full checklist should be submitted, a summary of concerns / issues requiring follow-up should be formulated and presented as the first section of the audit, including a description of the issue, the potential consequences of inaction, and urgency of the issue. The technical lead or project PM may also need to schedule a follow-up review of issues found with the customer, including the recommended resolution. This follow-up should be scheduled based upon the urgency of the identified issues.

Issue resolutions generally fall into three broad categories:

- Issues that are able to be resolved under the scope of the DxSustainment initiative
- Issues that may require additional project scope, necessitating a proposal for RoviSys services to be provided to the customer
 - If a system outage can be avoided via immediate RoviSys action even if commercial details are not yet established, it is recommended that these actions be taken in as urgent a manner as possible as a show of good will to the customer.
- Issues that must be addressed by the customer's IT team

(Published) MES RIS 107.7 - Technology Upgrade

Friday, August 23, 2024 9:14 AM

Please Reference:

[Technology Upgrade](#)

Either the system evaluation or periodic health audit may have uncovered the need to perform a technology upgrade. Upgrades typically arise due to a combination of the following three factors:

- New application version that provides some benefit
- Hardware obsolesce that necessitates a hardware refresh (server, client, or network)
- Supporting software obsolesce that necessitates upgrading this supporting software, such as operating system, SQL server, or other supporting applications
 - In some cases due to compatibility issues, the MES/MOM/digitalization platform in question may also need to be upgraded to ensure compatibility with the upgraded OS, SQL server, or supporting software. Vendor documentation should be consulted to confirm this.

When preparing for a discussion concerning platform upgrade options, RoviSys should be prepared to answer the following questions:

- What critical bugs or security issues are addressed by the latest version of the software?
 - Is there an intermediate version or patch that could be applied without performing a full upgrade?
- What lower-priority bugs or security issues are addressed by the latest version of the software?
- What other benefits exist to upgrading, including improved user experience or additional functionality that could be leveraged by the customer?
- Are there any software licensing costs required to upgrade to a later version?
 - Often customers may not have a current support contract with their vendor, requiring the vendor to pay for updated licensing.
- What upgraded infrastructure will be required as part of migrating to the latest version of the software?
 - This includes servers, OS version, SQL server version, and client hardware
- What user experience changes and corresponding change management concerns will need to be addressed as part of upgrading the software
 - This includes re-training, documentation changes, etc.
- What re-validation effort will be required as part of the software upgrade (specific to life sciences customers)
- What re-development effort will be required as part of the software upgrade (including re-configuration due to application changes as well as redevelopment of custom functionality that may have been implemented at some point in the past).
 - What is the associated cost of this re-development effort?
- How long will the system be down / unavailable as part of the upgrade process?

(Published) MES RIS 107.8 - Infrastructure Refresh

Friday, August 23, 2024 9:15 AM

Please Reference:

[Infrastructure Refresh](#)

Over the lifecycle of an MES application, it may be necessary to refresh on-premises infrastructure due to normal system aging and corresponding software advancement. Refreshing infrastructure is a complex process with multiple considerations. This article outlines key factors to include in an MES infrastructure refresh..

The customer's IT organization must formulate and approve a plan detailing the activities involved in the infrastructure refresh.

This plan must first consider formulation of an architecture to present the future state infrastructure. Considerations in this architecture include the following:

- Will the infrastructure hosting the MES platform remain on-premises or is a cloud migration planned? Migration to the cloud requires confirmation of software support, network implementation to ensure data movement between on-premises and cloud, and the potential for edge devices to aid in bridging data transfer to/from cloud-hosted applications.
- Server sizing, including CPU, RAM, hard disk, etc.: Servers should be sized based on both current and future need, considering increased data volume and additional system demands should the MES platform be upgraded.
 - o Application stacks are often deployed to virtual infrastructure, enabling dynamic allocation of resources when needed. However, IT organizations often limit resource allocation for individual applications and take a conservative approach. It is important to express the recommended application requirements to ensure no impact to the production environment after the refresh.
- Server OS and software versioning: It must be confirmed that the versions of the various components in the digitalization software stack are compatible with the operating system and associated software (such as SQL server, web server, frameworks (such as .NET, python, JavaScript, etc.) to be deployed to the refreshed infrastructure. It may be necessary to also consider a technology upgrade should there be a desire to upgrade to an operating system or SQL version that is incompatible with the current version of the MES or other digitalization software components.
- Custom integrations and APIs, including scripts, third-party tools, and middleware, must be reviewed for compatibility with the upgraded infrastructure. Any changes to these integrations required must be documented in the architecture and tested as part of the upgrade plan.
- High availability and Redundancy: an infrastructure refresh is a good time to consider improvements to the infrastructure to improve reliability and redundancy. This includes failover mechanisms such as clustered servers, load balancers, and RAID storage.
- Cybersecurity compliance: deployment of upgraded firewalls, endpoint protection, and security appliances should all be considered to mitigate cyber security risks.
 - o This deployment may require configuration to ensure the digitalization platforms continue to function properly. For example, ports leveraged by the application may need to be opened or certain point to point communication exceptions be configured to ensure data flows to/from external application and devices.
 - o Incorporation of enhanced security protections, such as multi-factor authentication (MFA) should also be considered.
- Many digitalization platforms are beginning to move to a containerized environment.

Containerization is advantageous in that it permits more rapid deployment and many better support redundancy, but may present unique architectural constraints such as requirement for a specific OS and installation of the container hosting application. Should containerization be supported, migration to a containerized environment must be factored into the architecture and migration plan.

The plan must then consider the migration steps. As part of migration, the digitalization software stack may need to be transitioned from the existing to new infrastructure. Depending on the magnitude of the refresh, this may entail application re-installation, database migration, configuration and code re-deployment, and other factors. The plan should look to minimize both risk and system downtime wherever possible.

Considerations to be documented as part of the migration plan include:

- Benchmark key performance indicators before and after the refresh, such as database query times, application latency, and server resource utilization, to ensure no performance degradation and ideally a performance improvement that helps to justify the refresh effort.
- A contingency strategy must be incorporated to ensure any production disruption is minimized. A common aspect of such strategy is maintaining the legacy infrastructure throughout the process so that as a contingency production can fall back to the existing infrastructure if required.
 - o Usage of paper-based records or other manual contingencies may also factor into the infrastructure refresh plan. This data will need to be synchronized post-cutover and a plan for doing so, including required roles and resources, must be established.
- Will server names change or remain the same? A change to server names may require configuration changes within the MES, Historian, and/or other middleware components. Testing should be conducted to confirm that there are no hardcoded server names within the code/configuration or missed server name references.
- Can the new servers be stood up and application installation occur prior to the cutover, reducing the cutover time? Doing this may require the new servers to be on an isolated network or certain data conduits to be disabled so that there are not conflicts with the production environments.
 - o Data migration likely cannot occur until the actual cutover. In some cases, a majority of the data may be able to be migrated, and then a partial migration of data captured in production subsequent to the bulk migration may be feasible, but this is generally not supported by most MES applications. As a result, customized tools may need to be developed to facilitate this if minimization of downtime is of sufficient importance to justify this. More commonly, the database will need to be migrated in its entirety during the cutover and restored to the new environment, then any needed configuration changes (such as server renames) made.
 - o An infrastructure refresh is an ideal opportunity to purge non-critical data according to an established retention policy. This will ensure both data retention compliance as well as ensure that the database is as small as possible, increasing the efficiency of the database migration.
 - It may be necessary to shrink the database and associated data files to realize the size reduction.
 - This is also a good time to ensure that periodic database maintenance is performed prior to the refresh, such as addressing index fragmentation.
- Capture of key backup steps, including database backups, configuration backups, code backups, and environment snapshots to ensure contingency plans can be executed should they be required.
- Reestablishment of any data conduits that may have been disabled to prevent accidental interference with production data conduits.
- Timing of when the refresh will take place. It may be feasible to perform the upgrade during a time in which the plant is down (not scheduled, holiday, shift change, etc.) or when data load is minimized (night shift for example).

- Communication plan to inform operations of any system outages, needed workarounds (such as paper-based data capture) and expected durations.
- Coordination channels during the infrastructure refresh. An onsite presence may be required, or an open web conference call established to ensure proper communications between the various teams. A chain-of-command hierarchy may be needed to ensure that should quick decisions need to be made approvals can be granted.
- Manage licensing migration, whether transferring existing licenses or acquiring new ones from vendors.
 - o Some platform vendors leverage licensing dependent on the specific hardware (such as MAC address or physical server keys), requiring a license refresh from the vendor prior to enablement of the new environment.
 - o Licensing for ancillary components, such as OPC servers or other middleware, must also be considered.
- Redirection of client workstations, tablets, etc. to the new infrastructure, particularly in the event that there is a server name change or migration from on-premises to containerized or cloud infrastructure.
- Testing the various digitalization applications on the upgraded infrastructure in a quarantined manner prior to the upgrade. It is recommended to setup the new infrastructure and install the applications, taking care to isolate them from production systems and data connections, to confirm the upgrade steps and verify proper application function prior to the cutover.
- A system outage associated with the refresh may cause inaccurate OEE / performance metrics. Manual corrections may be required to ensure this impact is not factored into periodic performance reporting.
- There may be an opportunity to introduce or leverage automated test scripts to quickly ensure system integrity post-refresh without extensive manual testing steps. When considering implementation of such test scripts, areas with the greatest probability of impact post-refresh should be considered, such modules that may be sensitive to data conduit interruption, server name changes, or other similar aspects.
- Enhanced system logging may need to be enabled during the refresh process to allow for closer monitoring of potential errors or issues. Ensure that any enhanced tracking that was enabled is disabled after normal operation is confirmed, to prevent performance impact or excessive system resource consumption by logging activities.
- Any system documentation that must be changed must be identified and edited as part of the refresh activities.
- Disaster recovery testing should be planned to ensure backup restoration and failover mechanisms work as expected.

After the refresh, increased attention must be given to the application and associated infrastructure, allocating support time and resources to respond to any issues. In particular, the following factors should be closely monitored:

- Server performance, including CPU, RAM, and hard disk usage
- Inability of certain clients to connect
- Any reports of system slowness, screens or functions that will not execute, or system bugs
- Failed data transactions, or external system communication errors (for example to-from the ERP, shop floor equipment, instrumentation, etc.)

After the upgrade, document lessons learned and improvement opportunities to streamline future infrastructure refreshes, capturing any pain points or strategies employed to ensure that any future infrastructure refreshes are conducted as smoothly as possible.

Based on the above description, the following is a concise checklist that can be considered:

Infrastructure Architecture Planning

- ☐ Decide whether to keep MES on-premises or migrate to the cloud.
- ☐ Confirm software compatibility for cloud migration and edge device integration.
- ☐ Size servers appropriately (CPU, RAM, storage) for current and future needs.
- ☐ Ensure high availability and redundancy (failover clusters, RAID, load balancers).
- ☐ Verify operating system and software stack compatibility (SQL Server, .NET, middleware, etc.).
- ☐ Review and document custom integrations, APIs, and third-party middleware compatibility.
- ☐ Assess and implement cybersecurity measures (firewalls, endpoint protection, MFA, network security).
- ☐ Consider containerization for improved scalability and deployment flexibility.

Migration Planning & Execution

- ☐ Benchmark performance metrics (database query times, latency, server utilization) pre- and post-refresh.
- ☐ Develop a contingency plan with rollback procedures and legacy infrastructure availability.
- ☐ Define a clear **data migration strategy**, including full migration or incremental data synchronization.
- ☐ Determine if **server names will change**, and update configurations accordingly.
- ☐ Establish an **isolated test environment** for pre-cutover installation and validation.
- ☐ Purge non-critical data based on **retention policies** to optimize database performance.
- ☐ Perform **database maintenance** (index optimization, shrinking, and cleanup).
- ☐ Ensure comprehensive **backup strategy** (database, configuration, code, and snapshots).
- ☐ Plan the **timing of the refresh** to minimize downtime (e.g., night shift, scheduled maintenance).
- ☐ Develop a **communication plan** for downtime alerts, workarounds, and affected teams.
- ☐ Coordinate **onsite or virtual presence** for real-time issue resolution.
- ☐ Manage **software licensing migration**, ensuring vendor compliance and hardware-based dependencies.
- ☐ Redirect **client devices** (workstations, tablets) to the new infrastructure.

Testing & Validation

- ☐ Test **all digitalization applications** in a **quarantined environment** before production deployment.
- ☐ Verify **data conduits** (SCADA, ERP, IIoT, OPC) are restored and operational.
- ☐ Validate **OEE and performance metrics** accuracy post-upgrade.
- ☐ Utilize **automated test scripts** to streamline system validation and detect potential issues.
- ☐ Enable **enhanced system logging** for real-time issue tracking during the refresh.
- ☐ Identify and update **system documentation** to reflect infrastructure changes.
- ☐ Conduct **disaster recovery testing** to verify backup restoration and failover procedures.

Post-Refresh Monitoring & Optimization

- ☐ Monitor **server performance** (CPU, RAM, disk usage).
- ☐ Check **client connectivity** and address any issues.
- ☐ Investigate **system slowness, unresponsive functions, or application failures**.
- ☐ Resolve **failed data transactions or integration errors** with ERP, shop floor systems, and other applications.
- ☐ Document **lessons learned** and improvement opportunities for future refreshes.

(Published) MES RIS 107.9 - System Impact Notification and Updates

Friday, August 23, 2024 9:15 AM

~~Please Reference:~~

[System Impact Notifications and Updates](#)

This procedure establishes a structured process for notifying shop floor personnel about changes, upgrades, or enhancements to digitalization applications and/or infrastructure. This procedure ensures that all affected personnel are informed about the change, potential impacts, workarounds, and contingency plans to maintain operational continuity. While proper communication is ultimately the customer's responsibility, the customer may be unfamiliar with such strategies and rely upon RoviSys to aid in proper execution of a system impact notification and update plan.

This procedure applies to all planned digitalization application or infrastructure changes, including but not limited to:

- System upgrades and migrations
- Software patches and enhancements
- Server, database, and network changes
- Integration updates with ERP, SCADA, and shop floor equipment
- Security and compliance updates affecting user access

The following roles and responsibilities must be involved in the notification process:

- **IT / OT Departments:** Responsible for planning, executing, and communicating changes
- **Operations / Production Managers:** Ensure shop floor personnel understand the impact and workarounds
- **Shift Supervisors:** Act as liaisons between IT and shop floor employees
- **MES Administrators:** Validate system changes and provide troubleshooting support
- **End Users (Operators, Technicians, Engineers):** Follow the instructions and report any issues

The following is a recommended notification process workflow:

Step 1: Change Assessment and Approval:

1. Identify the scope and impact of the digitalization change
2. Assess potential risks and determine contingency measures
3. Obtain approval from IT/OT leadership and operations management

Step 2: Notification Preparation:

1. Draft a notification message covering:
 - What is changing (e.g., "MES will be upgraded to version X.X")
 - When the change will take place (e.g., "Scheduled for March 12, 2025, from 8 PM to 2 AM")
 - Expected impact (e.g., "MES will be unavailable during this period; certain functionalities may be limited post-upgrade")
 - Workarounds (if any) (e.g., "Use paper-based reporting for downtime tracking")
 - Contingency plan (e.g., "If the upgrade fails, we will roll back to the previous version within 2 hours")
 - Support contact information (e.g., "For urgent issues, contact the MES Helpdesk at ext. 1234")
2. Format the message for different communication channels (email, posters, announcements, digital signage, etc.)

Step 3: Notification Distribution

1. Send initial notification (in ample time prior to the change, such as 1 week) via:
 - Email to shop floor supervisors, production leads, and affected personnel
 - Bulletin boards and break room posters
 - Digital factory signage or announcements on MES terminals
 - Operations team meetings and shift huddles
 - Internal messaging platforms (e.g., Microsoft Teams, Slack, WhatsApp for Business)
2. Send a reminder (such as 1 day before the change)
3. Send a final reminder (such as 1 hour before the implementation starts)

Step 4: Implementation and Monitoring

1. Execute the planned MES infrastructure change
2. Have IT/OT and MES administrators on standby for immediate issue resolution
3. Instruct shift supervisors to monitor the impact on production and escalate issues as needed

Step 5: Post-Implementation Communication

1. Notify personnel once the change is successfully completed
2. Provide an update on system performance and any lingering issues
3. Offer refresher training (if applicable) on new features or processes
4. Gather feedback from operators and supervisors

Step 6: Lessons Learned and Continuous Improvement

1. Conduct a **post-change review** with IT/OT and operations to discuss:
 - Issues encountered and resolutions
 - Effectiveness of the notification process
 - Any required improvements for future infrastructure changes
2. Update any SOPs/procedures if necessary based on feedback

5. Escalation Procedure

If unexpected issues arise, ensure that stakeholders understand the escalation conduit. For example:

1. **Operators** report issues to Shift Supervisors.
2. **Shift Supervisors** escalate urgent concerns to MES Administrators.
3. **MES Administrators** troubleshoot and involve the IT/OT teams if needed.

A hotline may need to be established and captured in the notifications so that users have a direct line to help should an issue occur during or after the system impact.

(Published) MES RIS 107.10 - Root Cause Analysis

Friday, August 23, 2024 9:15 AM

Please Reference:

[Root Cause Analysis](#)

Performing a Root Cause Analysis (RCA) for an MES (Manufacturing Execution System) application issue requires a structured approach to identifying and resolving the problem effectively. While the specific troubleshooting steps depend greatly on the MES platform, data connections, and infrastructure, the following are general strategies:

1. Define the Problem Clearly

- Gather details on what the issue is, when it started, who reported it, and how it affects operations.
- Categorize the issue (e.g., performance degradation, incorrect data, connectivity issues, UI errors).

2. Reproduce & Validate the Issue

- Check if the issue is consistent or intermittent.
- Identify the affected modules, transactions, or processes.
- Determine whether it occurs in production, test, or both environments.

3. Collect and Analyze Logs & System Data

- Check MES application logs for errors or warnings.
- Review database logs for slow queries, deadlocks, or data inconsistencies.
- Utilize a database query profiling tool (such as SQL Server Profiler) to analyze long-running queries.
- Monitor server performance (CPU, memory, disk I/O).
- Examine network logs for latency or packet loss.
- Check browser console logs (for web-based MES systems).

4. Isolate Contributing Factors

- Recent software updates or configuration changes.
- Database schema modifications or incorrect data inputs.
- External system integrations (ERP, SCADA, IoT).
- User actions that may trigger the issue.
- System resource constraints or hardware failures.

5. Use the "5 Whys" or Fishbone Diagram

- Apply the 5 Whys method to drill down into the root cause.
- Use a fishbone (Ishikawa) diagram to categorize possible causes (software, hardware, network, process, user actions).

6. Test Hypotheses & Validate the Root Cause

- Replicate the issue in a controlled test environment.
- Apply temporary fixes (rollback changes, restart services, etc.).
- Use debugging tools to trace errors and bottlenecks.

7. Implement & Verify the Fix

- Deploy a fix in a staging environment before production.
- Ensure that the solution does not create new issues.
- Monitor system behavior post-fix.

8. Prevent Recurrence

- Implement automated monitoring for similar failures.
- Improve error handling & logging.
- Update documentation with RCA findings.
- Train users or update SOPs if needed.

While MES troubleshooting varies by platform, infrastructure, and integrations, these general strategies provide a structured approach to identifying and resolving issues efficiently.

Usage of 5 Whys Strategy:

The **5 Whys** is a root cause analysis (RCA) technique used to identify the underlying cause of a problem by repeatedly asking "Why?" until the fundamental issue is uncovered. It was developed as part of **Toyota's Lean Manufacturing System** but is widely used in other areas, including MES troubleshooting.

How to Perform the 5 Whys Analysis (example):

- 1. State the Problem Clearly**
 - Example: "Production reports in the MES are showing incorrect data."
- 2. Ask 'Why?' (First Level)**
 - **Why is the production report incorrect?**
→ Because the MES is pulling incorrect values from the database.
- 3. Ask 'Why?' Again (Second Level)**
 - **Why is the MES pulling incorrect values?**
→ Because the database has duplicate and outdated records.
- 4. Ask 'Why?' Again (Third Level)**
 - **Why does the database have duplicate and outdated records?**
→ Because the data synchronization process failed.
- 5. Ask 'Why?' Again (Fourth Level)**
 - **Why did the data synchronization process fail?**
→ Because a scheduled job responsible for syncing data was not executed.
- 6. Ask 'Why?' Again (Fifth Level - Root Cause)**
 - **Why was the scheduled job not executed?**
→ Because the job scheduler crashed due to a lack of memory on the server.

Root Cause Identified:

- The MES reporting issue originated from insufficient server memory, leading to job failures and, eventually, incorrect production reports.

Usage of a Fishbone Diagram to Perform Systematic Evaluation:

A Fishbone Diagram, also known as an Ishikawa Diagram or Cause-and-Effect Diagram, is a tool used to identify the root causes of a problem systematically. It allows the troubleshooter to visually map out potential factors contributing to an issue, organizing them into major categories.

Structure of a Fishbone Diagram:

- The "head" of the fish represents the problem statement (effect).

- The "spine" is the main horizontal line extending from the head.
- Major categories (causes) branch off the spine as bones.
- Sub-causes are further detailed under each category.

Common Categories in MES Troubleshooting:

For MES troubleshooting, common cause categories may include:

1. Software – Bugs, application crashes, configuration errors.
2. Hardware – Server failures, disk I/O, memory issues.
3. Database – Slow queries, data corruption, connection timeouts.
4. Network – Latency, firewall issues, dropped packets.
5. User Actions – Incorrect input, unauthorized access, training gaps.
6. External Systems – ERP, SCADA, IoT device failures, API issues.

Example Use Case

Problem: MES system reports incorrect production data

Potential Causes (Categorized):

- Software: Bug in the report generation module
- Hardware: Database server running low on disk space
- Database: Data synchronization failure
- Network: High latency between MES and SCADA
- User Actions: Incorrect manual data entry
- External Systems: ERP sending duplicate data records

Of increased benefit is once formulated, a created fishbone diagram can be saved to the knowledge base established for the platform, shortcutting future potential troubleshooting efforts.

Increasing Root Cause Analysis Effectiveness:

As part of the DxSustainment offering, RoviSys should aid the customer in establishing practices that will aid in future root cause analysis. Such practices include the following:

- Maintain detailed version control of MES configurations and database changes
- Ensure all deployed updates have a rollback procedure documented in case a recent update introduces issues
- Ensure that development and sandbox / staging environments are leveraged to replicate production scenarios
- Establish application performance monitoring (APM) tools for real-time system insights and ensure notifications are configured to key individuals should issues be detected by these tools
- Leveraging infrastructure refresh and technology upgrade strategies, ensure that the infrastructure and software applications are kept reasonably up-to-date
- Evaluate the inclusion of additional logging and monitoring, particularly around risk areas such as external system transactions
- Capture of root cause issue and associated resolution in a searchable knowledge base

(Published) MES RIS 107.11 - Minor Enhancements

Friday, August 23, 2024 9:16 AM

Please Reference:

[Minor Enhancements](#)

As part of a DxSustainment contract, performing minor enhancements can improve system functionality and usability without necessitating a full-scale change request (CR). However, clear guidelines must be in place to determine when an enhancement qualifies as a minor update rather than a major change requiring formal approval. Additionally, a structured approach to risk assessment, testing, and migration is essential to ensure system stability and maintainability.

Establishing an Enhancement Threshold:

A key aspect of MES sustainment is defining what constitutes a minor enhancement versus a significant change requiring a formal change request (CR). A well-defined threshold should consider factors such as:

- Impact on existing functionality – Minor enhancements should not disrupt critical processes or introduce significant workflow changes.
- Complexity and effort required – Small configuration changes, UI tweaks, and minor automation improvements generally fall within sustainment scope, while extensive logic modifications may require a CR.
- Risk level – The enhancement should not introduce potential vulnerabilities or require substantial regression testing across the system.

By establishing clear criteria, MES sustainment teams can ensure that minor enhancements are handled efficiently while more substantial modifications undergo proper review and approval processes.

Risk Assessment for Minor Enhancements:

Even minor enhancements carry some level of risk, making risk assessment a crucial step before implementation. The risk assessment process should include:

- Impact analysis – Evaluating which areas of the MES may be affected by the enhancement, including dependencies on external systems such as ERP or LIMS, or shop-floor equipment.
- Regression testing scope – Identifying the extent of testing required to verify that the enhancement does not introduce unintended consequences.
- Rollback strategy – Ensuring that changes can be reversed if issues arise during implementation.

By proactively assessing risks, sustainment teams can prevent disruptions and ensure a seamless transition from development to production.

Prerequisites for Implementing Minor Enhancements:

Before making any MES enhancement, several prerequisites should be met to maintain system integrity and minimize risks:

1. Development-Test-Production Environment – A structured environment for deploying changes is essential. All enhancements should be developed and tested in a dedicated development environment before being validated in a test environment and finally deployed to production.

2. Source Code and Configuration Management – The latest version of all MES configurations, scripts, and custom code must be maintained in a version control system to prevent conflicts and ensure recoverability.
3. Stakeholder Approval and Communication – Even minor enhancements should be communicated with relevant stakeholders, including operations, IT, and MES administrators, to ensure alignment with business needs.
4. Validation documentation revisions - In a validated environment common in industries such as life sciences, even small changes must follow proper change controls, and be coordinated with the customer's validation team.

Additional Best Practices:

Beyond risk assessment and environmental prerequisites, the following best practices can further enhance the sustainment process:

- Change Documentation – Even for minor enhancements, maintaining documentation of the changes, rationale, and expected impact helps with future troubleshooting and audits.
- Automated Testing Where Possible – Automating regression tests can improve validation efficiency and reduce the risk of unintended consequences.
- Monitoring Post-Deployment – Implementing monitoring mechanisms to track system behavior after deployment can help quickly identify and address any unexpected issues.

(Published) MES RIS 107.12 - Vendor Support Engagement

Friday, August 23, 2024 9:16 AM

Please Reference:

[Vendor Support Engagement](#)

As an independent systems integrator, we are uniquely positioned to act as a liaison between our customers and their vendors. Our role is to bridge communication gaps, ensure smooth coordination of support activities, and advise on best practices for system maintenance and upgrades. This document outlines the procedures we should follow to assist our customers in managing vendor relationships, troubleshooting application support issues, and planning for infrastructure refreshes and updates.

RoviSys can provide more effective support when we have an established relationship with a prioritized vendor. If we are unfamiliar with a vendor or lack an existing relationship, we may still assist, but our effectiveness may be limited. Customers should be made aware of this potential limitation.

Proposal Assumptions:

As part of the DxSustainment contract establishment, roles and responsibilities should be clearly delineated. RoviSys should be positioned in an assist role, but ultimately the vendor must be positioned as responsible for the proper function of their software platform.

Additionally, any prerequisites to be established to ensure RoviSys access to vendor support must be clearly stated in the assumptions. For example, the customer may be required to ensure they maintain a support contract with the vendor, maintain a current supported version of the software, and devote recommended server resources to the application per the vendor's guidelines. Customers often assume that having a support contract with RoviSys eliminates the need for a separate vendor support contract, but this is incorrect.

Assumptions must also be clearly stated in the proposal around response times / SLAs. RoviSys cannot be expected to meet a certain response or resolution time in the event that an issue requires vendor engagement, as RoviSys has little control over the timing of the vendor response. This must be clarified in the proposal to minimize the risk to RoviSys.

Vendor Support Engagement Execution:

The following are ways in which RoviSys can aid in vendor engagement that should be considered throughout the duration of our DxSustainment contract:

Establishing Vendor Contact Points

- Maintain an up-to-date database of customer vendors, including contact persons, escalation paths, and support processes
- Ensure that customers are aware of the appropriate channels for contacting vendors
- Act as a primary or secondary point of contact for vendor communications when required by the customer

Streamlining Issue Reporting

- Triage issues by distinguishing:
 - Platform-related issues requiring vendor engagement
 - Application or customization issues that the vendor cannot resolve

- Training-related concerns that can be handled internally
- Consider that if an issue is improperly triaged, the vendor may be more reluctant to support the customer in the future, and RoviSys may be perceived by the vendor in a negative light*
- Assist our customers in submitting support tickets with vendors, ensuring that all necessary technical details and logs are included.
 - Obtain additional information from the customer's environment as requested by the vendor.
- Facilitate remote or on-site vendor support sessions as needed
- Follow up on reported issues and track progress until resolution

Best Practices and Optimization

- Offer guidance on industry best practices related to system performance, security, and usability via coordination with the software vendor
- Share knowledge on vendor-recommended configurations and optimizations to enhance system performance
- Act as a trusted advisor to help customers navigate technological advancements and strategic decisions related to vendor platforms
- Advocate for customer feature requests or necessary fixes based upon operational challenges via consultation with vendor contacts

Vendor Support Watchouts:

As the customer's trusted advisor, RoviSys can help guard against unrealistic vendor expectations or requests. Vendor support resources often lack perspective concerning management of their platform in a production environment, leading to asks that may not be practical. These include the following:

- A vendor may request a customer deploy a patch or upgrade to the latest version of the software as an initial step before providing support. This may be impractical in a production environment due to resulting system downtime or risk of outage should a problem arise. RoviSys can help plan for such an update, or independently evaluate if the request makes sense. The request may be unreasonable if the customer is running a supported version of the software, even if the customer is not running the latest revision.
- A vendor may request a database backup without realizing that it may be too large to transmit efficiently or contain proprietary information that cannot be shared with a third party. RoviSys can help communicate this impracticality to the vendor, as well as scrub the database or export configuration relevant to issue troubleshooting.

SaaP Reference Docs

Monday, June 10, 2024 2:28 PM

Reference docs

[RoviSys Agile for Manufacturing Approach - 2024.pptx](#)
[P23FSR01e - Remote Operations Center - Assessment and Design.pptx](#)
[Smart Manufacturing Delivery Process.pptx](#)
[MES Mock-Up April 24.pptx](#)
[SaaP.xlsx](#)
[Jessica Brainstorming \(Web view\)](#)
[SaaP Product Development Brainstorming \(Web view\)](#)
[DxOps Transformation feature definitions.docx](#)